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Reverse engineering the NMEA 2000 standardized and manufacturer proprietary data

In the late 90s a number of marine electronics manufacturers realised that there was a need for a better protocol than NMEA 0183, which did not suit larger networks. The result was the NMEA 2000 standard. This is based on a CAN bus running at 250.000 kbit/s. It allows up to 250 devices on the same bus, although this will probably be overloaded.

Unfortunately, the NMEA 2000 standard is using the same "closed" mentality as ISO and other industry standards. This is completely different from the world of computing, where all such protocols are open for use by anyone. In 2008, as I wanted to interface to my brand new NMEA 2000 interface and show the data on this newfangled device called an iPhone, I started reverse engineering the protocol. At the time there was limited info available. This has improved over time; interestingly even the NMEA leaks a lot of information that is publicly available. For instance the list of PGNs and the field *names* are available at http://www.nmea.org/Assets/july%202010%20nmea2000_v1-301_app_b_pgn_field_list.pdf.

This document will provide a human readable explanation of everything we (as there are now over 20 contributors) have learned. Right now some basic concepts are not yet explained, but you will find all PGNs, lookup lists, and data types here.

If you have data (even if it is just a logfile for a new device) to contribute, please open an issue at <https://github.com/canboat/canboat/issues> or send a pull request at <https://github.com/canboat/canboat/pulls>.

Note: It should be obvious that all data below is *not authoritative*; it is just our (my) interpretation. If you are a manufacturer and want to create a NMEA 2000 device, become a member and buy the standard!

ISO-11783 and NMEA2000 header

An "Extended 29 bit identifier" CAN frame, as mandated by NMEA 2000, consists of a 39 bit header followed by (up to) 64 data bits. The header bits are:

Bit	CAN	ISO 11783	Notes
1	SOF	SOF	Start of Frame
2	ID 28	P 3	Priority bit 3
3	ID 27	P 2	Priority bit 2
4	ID 26	P 1	Priority bit 1
5	ID 25	R 1	Reserved bit 1
6	ID 24	DP	Data page
7	ID 23	PF 8	PDU format bit 8
8	ID 22	PF 7	PDU format bit 7
9	ID 21	PF 6	PDU format bit 6
10	ID 20	PF 5	PDU format bit 5
11	ID 19	PF 4	PDU format bit 4
12	ID 18	PF 3	PDU format bit 3
13	SRR	SRR	Substitute Remote Request bit
14	IDE	IDE	Identifier Extension bit
15	ID 17	PF 2	PDU format bit 2
16	ID 16	PF 1	PDU format bit 1
17	ID 15	PS 8	PDU specific bit 8
18	ID 14	PS 7	PDU specific bit 7
19	ID 13	PS 6	PDU specific bit 6



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Bit	CAN	ISO 11783	Notes
20	ID 12	PS 5	PDU specific bit 5
21	ID 11	PS 4	PDU specific bit 4
22	ID 10	PS 3	PDU specific bit 3
23	ID 9	PS 2	PDU specific bit 2
24	ID 8	PS 1	PDU specific bit 1
25	ID 7	SA 8	Source Address bit 8
26	ID 6	SA 7	Source Address bit 7
27	ID 5	SA 6	Source Address bit 6
28	ID 4	SA 5	Source Address bit 5
29	ID 3	SA 4	Source Address bit 4
30	ID 2	SA 3	Source Address bit 3
31	ID 1	SA 2	Source Address bit 2
32	ID 0	SA 1	Source Address bit 1
33	RTR	RTR	Remote Transmission Request bit
34	r 1	r 1	CAN reserved bit 1
35	r 0	r 0	CAN reserved bit 0
36	DLC 4	DLC 4	Data Length Code bit 4
37	DLC 3	DLC 3	Data Length Code bit 3
38	DLC 2	DLC 2	Data Length Code bit 2
39	DLC 1	DLC 1	Data Length Code bit 1

For NMEA2000 the "R 1" bit is always 0, but in SAE J1939 it is not. J1939 calls this the "Extended Data Page" (EDP). In general, we don't care about these and the interfaces will generally only produce the 29 bits named "ID xx" in the above table. This is called the *CAN Id*.

The source address and priority are easily derived from the above table: they form contiguous bits in the header. But the PGN and destination address are formed from the remaining $(29 - 3 - 8) = 18$ bits in a convoluted fashion. The algorithm is as follows:

- Generate the values for PF (8 bits) and PS (8 bits) from their contiguous bits in the header.
- Generate a value RDP (2 bits) from the "R 1" and "DP" bits.
- If the value for PF is less than 240 (0xF0) this is called "PDU1" format. PS contains the destination address and the PGN is formed by `[RDP][PF]00000000`.
- Otherwise if PF is in range 240 - 255 this is called "PDU2" format. The destination is always 'global' (i.e. address 255) and the PGN is formed by `[RDP][PF][PS]`.

Note that this means that all PDU1 PGNs are 256 apart, as their lower 8 bits are always zero. Thus there are much fewer of them.

PGN ranges

There are a number of PGN (Parameter Group Number) ranges. When expressed in decimal the ranges do not make much sense, but in hexadecimal they do -- which is explained above. The PGN ranges are divided up by PDU1/PDU2 but also by the fact that there is the need to have manufacture private PGNs, so some blocks have been carved out for this purpose. The ranges are:

Range Hex	Range Dec	PDU	Step	Number of possible PGNs	Use	Framing
0xE800-0xEE00	59392 - 60928	PDU1	256	7	ISO 11783 (protocol)	Single frame
0xEF00	61184	PDU1		1	Manufacturer proprietary	Single frame
0xF000-0xFFFF	61440 - 65279	PDU2	1	3840	Standardized	Single frame
0xFF00-0xFFFF	65280 - 65535	PDU2	1	256	Manufacturer proprietary	Single frame
0x1ED00-0x1EE00	126208 - 126464	PDU1	256	2	Standardized (protocol)	Fast packet
0x1EF00	126720	PDU1		1	Manufacturer proprietary	Fast packet
0x1F000-0x1FEFF	126976 - 130815	PDU2	1	3840	Standardized	Mixed single/fast



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Range Hex	Range Dec	PDU	Step	Number of possible PGNs	Use	Framing
0x1FF00-0x1FFF	130816 - 131071	PDU2	1	256	Manufacturer proprietary	Fast packet

Note: There are some missing ranges in the above table: Apparently no PGN is used in range 0x0000-0xE700 or 0x10000-0x1EC00. The reason is not obvious to me. Maybe someone can enlighten us?

Packet framing

NMEA 2000 messages that are 8 bytes or less can be transmitted in a single CAN frame. For messages of 9 or more bytes there is an ISO 11783 defined method called Transport Protocol that can be used to transmit up to 1785 bytes. See [PGN 60416](#). This is *not* generally used though. What is used is an alternative method with less overhead and less complexity for the sender. This is called *fast* packet framing.

In fast packet framing, the first packet contains two protocol bytes and six data bytes. Following packets contain one protocol byte and seven data bytes. Up to 32 packets can be used for a single message so the total maxing data length is $6 + 31 * 7 = 223$ bytes. The first byte in all frames contains a sequence counter in the high 3 bits and a frame counter in the lower 5 bits. The second byte in the first frame contains the total number of bytes in all packets that will be sent (excluding the single header byte in each of the following packets). As there is no way to acknowledge or deny reception, if a message is missed the receiver will have to wait for the next transmission of the message.

It is unclear to us whether the standard mandates that frames are transmitted in order. According to [Yacht Devices](#) there are (older?) devices that transmit out-of-order as this is easier to program with certain CAN chips, and various new MFDs ignore out-of-order packet transmissions. As of v4.0.0 the CANBoat analyzer program will analyze out-of-order frames, although it means it has to heuristically determine when a packet is complete.

PGN list

0xE800-0xEE00: Standardized single-frame addressed

Standardized PGNs in PDU1 (addressed) single-frame PGN range 0xE800 to 0xEE00 (59392 - 60928). When this is shown during analysis it means the PGN is not reverse engineered yet.

0xE800: PGN 59392 - ISO Acknowledgement

This message is provided by ISO 11783 for a handshake mechanism between transmitting and receiving devices. This message is the possible response to acknowledge the reception of a 'normal broadcast' message or the response to a specific command to indicate compliance or failure.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Control		0 .. 252	8 bits lookup ISO_CONTROL	
2	Group Function		0 .. 252	8 bits unsigned NUMBER	
3	Reserved			24 bits RESERVED	
4	PGN	Parameter Group Number of requested information	0 .. 262143	24 bits PGN	

0xEA00: PGN 59904 - ISO Request

As defined by ISO, this message has a data length of 3 bytes with no padding added to complete the single frame. The appropriate response to this message is based on the PGN being requested, and whether the receiver supports the requested PGN.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 3 bytes long and contains 1 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	PGN		0 .. 262143	24 bits PGN	

0xEB00: PGN 60160 - ISO Transport Protocol, Data Transfer

ISO 11783 defines this PGN as part of the Transport Protocol method used for transmitting messages that have 9 or more data bytes. This PGN represents a single packet of a multipacket message.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 2 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID	Sequence number of multi-packet frame	0 .. 252	8 bits unsigned NUMBER	
2	Data	Multi-packet packetized data - 56 bits organized as seven 8-bit bytes, each of the bytes is transmitted in the same order as it would in a standard (i.e., single frame) message. Unused bytes shall be filled with 0xFF.		56 bits BINARY	

0xEC00: PGN 60416 - ISO Transport Protocol, Connection Management - Request To Send

This PGN description applies when the following field(s) match:

Group Function Code	16
---------------------	----

ISO 11783 defines this group function PGN as part of the Transport Protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is to prepare the receiver for the fact that this sender wants to transmit a long message. The receiver will respond with CTS.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Function Code	16: RTS	0 .. 255	8 bits lookup ISO_COMMAND	
2	Message size	bytes	0 .. 65532	16 bits unsigned NUMBER	
3	Packets	packets	0 .. 252	8 bits unsigned NUMBER	
4	Packets reply	packets sent in response to CTS	0 .. 252	8 bits unsigned NUMBER	
5	PGN		0 .. 262143	24 bits PGN	

Source: <https://embeddedflakes.com/j1939-transport-protocol/>

0xEC00: PGN 60416 - ISO Transport Protocol, Connection Management - Clear To Send

This PGN description applies when the following field(s) match:

Group Function Code	17
---------------------	----

ISO 11783 defines this group function PGN as part of the Transport Protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is to signal to the sender that the receive is ready to receive a number of frames.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. It is not transmitted at a regular interval, but when data is available or it is requested.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Function Code	17: CTS	0 .. 255	8 bits lookup ISO_COMMAND	
2	Max packets	Number of frames that can be sent before another CTS is required	0 .. 252	8 bits unsigned NUMBER	
3	Next SID	Number of next frame to be transmitted	0 .. 252	8 bits unsigned NUMBER	
4	Reserved			16 bits RESERVED	
5	PGN		0 .. 262143	24 bits PGN	

Source: <https://embeddedflakes.com/j1939-transport-protocol/>

0xEC00: PGN 60416 - ISO Transport Protocol, Connection Management - End Of Message

This PGN description applies when the following field(s) match:

Group Function Code	19
---------------------	----

ISO 11783 defines this group function PGN as part of the Transport Protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is to mark the end of the message.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Function Code	19: EOM	0 .. 255	8 bits lookup ISO_COMMAND	
2	Total message size	bytes	0 .. 65532	16 bits unsigned NUMBER	
3	Total number of frames received	Total number of of frames received	0 .. 252	8 bits unsigned NUMBER	
4	Reserved			8 bits RESERVED	
5	PGN		0 .. 262143	24 bits PGN	

Source: <https://embeddedflakes.com/j1939-transport-protocol/>

0xEC00: PGN 60416 - ISO Transport Protocol, Connection Management - Broadcast Announce

This PGN description applies when the following field(s) match:

Group Function Code	32
---------------------	----

ISO 11783 defines this group function PGN as part of the Transport Protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is to announce a broadcast of a long message spanning multiple frames.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Function Code	32: BAM	0 .. 255	8 bits lookup ISO_COMMAND	
2	Message size	bytes	0 .. 65532	16 bits unsigned NUMBER	
3	Packets	frames	0 .. 252	8 bits unsigned NUMBER	
4	Reserved			8 bits RESERVED	
5	PGN		0 .. 262143	24 bits PGN	

Source: <https://embeddedflakes.com/j1939-transport-protocol/>

0xEC00: PGN 60416 - ISO Transport Protocol, Connection Management - Abort

This PGN description applies when the following field(s) match:

Group Function Code	255
---------------------	-----

ISO 11783 defines this group function PGN as part of the Transport Protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is to announce an abort of a long message spanning multiple frames.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 4 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Function Code	255: Abort	0 .. 255	8 bits lookup ISO_COMMAND	
2	Reason			8 bits BINARY	
3	Reserved			24 bits RESERVED	
4	PGN		0 .. 262143	24 bits PGN	

Source: <https://embeddedflakes.com/j1939-transport-protocol/>

0xEE00: PGN 60928 - ISO Address Claim

This network management message is used to claim network address, reply to devices requesting the claimed address, and to respond with device information (NAME) requested by the ISO Request or Complex Request Group Function. This PGN contains several fields that are requestable, either independently or in any combination. Note that there are several places where this 64-bit data also occurs, these are named ISO_NAME and can be recursively explained.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 10 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Unique Number	ISO Identity Number	0 .. 2097148	21 bits unsigned NUMBER	
2	Manufacturer Code		0 .. 2044	11 bits lookup MANUFACTURER_CODE	
3	Device Instance Lower	ISO ECU Instance	0 .. 6	3 bits unsigned NUMBER	
4	Device Instance Upper	ISO Function Instance	0 .. 29	5 bits unsigned NUMBER	
5	Device Function	ISO Function	0 .. 252	8 bits indirect lookup DEVICE_FUNCTION where the first column is the value from field 7 ("Device Class")	
6	Spare			1 bits SPARE	
7	Device Class		0 .. 125	7 bits lookup DEVICE_CLASS	
8	System Instance	ISO Device Class Instance	0 .. 13	4 bits unsigned NUMBER	
9	Industry Group		0 .. 6	3 bits lookup INDUSTRY_CODE	
10	Arbitrary address capable	Field indicates whether the device is capable to claim arbitrary source address. Value is 1 for NMEA200 devices. Could be 0 for J1939 device claims	0 .. 1	1 bits lookup YES_NO	



Source: <https://web.archive.org/web/20150910070107/http://www.nmea.org/Assets/20140710%20nmea-2000-060928%20iso%20address%20claim%20pgn%20corrigendum.pdf>

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0xEF00: Manufacturer Proprietary single-frame addressed

Manufacturer proprietary PGNs in PDU1 (addressed) single-frame PGN 0xEF00 (61184). When this is shown during analysis it means the PGN is not reverse engineered yet.

0xEF00: PGN 61184 - Seataalk: Wireless Keypad Light Control

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	1

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	1: Wireless Keypad Light Control	0 .. 252	8 bits unsigned NUMBER	true
5	Variant		0 .. 252	8 bits unsigned NUMBER	
6	Wireless Setting		0 .. 252	8 bits unsigned NUMBER	
7	Wired Setting		0 .. 252	8 bits unsigned NUMBER	
8	Reserved			16 bits RESERVED	

0xEF00: PGN 61184 - Seataalk: Wireless Keypad Control

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID		0 .. 252	8 bits unsigned NUMBER	true
5	Variant		0 .. 252	8 bits unsigned NUMBER	



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6	Beep Control		0 .. 252	8 bits unsigned NUMBER	
7	Reserved			24 bits RESERVED	

0xEF00: PGN 61184 - Victron Battery Register

This PGN description applies when the following field(s) match:

Manufacturer Code	358
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	358: Victron Energy	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Register Id		0 .. 65532	16 bits unsigned NUMBER	
5	Payload		0 .. 4294967292	32 bits unsigned NUMBER	

0xF000-0xFEFF: Standardized single-frame non-addressed

PGNs in PDU2 (non-addressed) single-frame PGN range 0xF000 to 0xFEFF (61440 - 65279). When this is shown during analysis it means the PGN is not reverse engineered yet.

0xFDE9: PGN 65001 - Bus #1 Phase C Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	Reserved			16 bits RESERVED	

0xFDEA: PGN 65002 - Bus #1 Phase B Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	



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3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	Reserved			16 bits RESERVED	

0xFDEB: PGN 65003 - Bus #1 Phase A Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	Reserved			16 bits RESERVED	

0xFDEC: PGN 65004 - Bus #1 Average Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	Reserved			16 bits RESERVED	

0xFDED: PGN 65005 - Utility Total AC Energy

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Total Energy Export		kWh 0 .. 4294967292	32 bits unsigned NUMBER	
2	Total Energy Import		kWh 0 .. 4294967292	32 bits unsigned NUMBER	

0xFDEE: PGN 65006 - Utility Phase C AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR 0 .. 65532	16 bits unsigned NUMBER	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER_FACTOR	
4	Reserved			30 bits RESERVED	

0xFDEF: PGN 65007 - Utility Phase C AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFDF0: PGN 65008 - Utility Phase C Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		A 0 .. 65532	16 bits unsigned NUMBER	

0xFDF1: PGN 65009 - Utility Phase B AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR 0 .. 65532	16 bits unsigned NUMBER	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER_FACTOR	
4	Reserved			30 bits RESERVED	

0xFDF2: PGN 65010 - Utility Phase B AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known



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This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFDF3: PGN 65011 - Utility Phase B Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		A 0 .. 65532	16 bits unsigned NUMBER	

0xFDF4: PGN 65012 - Utility Phase A AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER_FACTOR	
4	Reserved			14 bits RESERVED	

0xFDF5: PGN 65013 - Utility Phase A AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFDF6: PGN 65014 - Utility Phase A Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known



This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		$\sqrt{}$ 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		$\sqrt{}$ 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		Δ 0 .. 65532	16 bits unsigned NUMBER	

0xFDF7: PGN 65015 - Utility Total AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER FACTOR	
4	Reserved			14 bits RESERVED	

0xFDF8: PGN 65016 - Utility Total AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFDF9: PGN 65017 - Utility Average Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		$\sqrt{}$ 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		$\sqrt{}$ 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		Δ 0 .. 65532	16 bits unsigned NUMBER	

0xFDFA: PGN 65018 - Generator Total AC Energy



This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Total Energy Export		kWh 0 .. 4294967292	32 bits unsigned NUMBER	
2	Total Energy Import		kWh 0 .. 4294967292	32 bits unsigned NUMBER	

0xFDFB: PGN 65019 - Generator Phase C AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER FACTOR	
4	Reserved			14 bits RESERVED	

0xFDFC: PGN 65020 - Generator Phase C AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFDFD: PGN 65021 - Generator Phase C Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		A 0 .. 65532	16 bits unsigned NUMBER	

0xFDFE: PGN 65022 - Generator Phase B AC Reactive Power



This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER_FACTOR	
4	Reserved			14 bits RESERVED	

0xFDF: PGN 65023 - Generator Phase B AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFE00: PGN 65024 - Generator Phase B Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		A 0 .. 65532	16 bits unsigned NUMBER	

0xFE01: PGN 65025 - Generator Phase A AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER_FACTOR	
4	Reserved			14 bits RESERVED	

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0xFE02: PGN 65026 - Generator Phase A AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

0xFE03: PGN 65027 - Generator Phase A Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		A 0 .. 65532	16 bits unsigned NUMBER	

0xFE04: PGN 65028 - Generator Total AC Reactive Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Reactive Power		VAR -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Power factor		6.10352e-05 Cos Phi 0 .. 3.999755859375	16 bits unsigned NUMBER	
3	Power Factor Lagging		0 .. 2	2 bits lookup POWER_FACTOR	
4	Reserved			14 bits RESERVED	

0xFE05: PGN 65029 - Generator Total AC Power

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Real Power		W -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	
2	Apparent Power		VA -2000000000 .. 2294967292	32 bits signed NUMBER stored with offset -2000000000	

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0xFE06: PGN 65030 - Generator Average Basic AC Quantities

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Line-Line AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
2	Line-Neutral AC RMS Voltage		V 0 .. 65532	16 bits unsigned NUMBER	
3	AC Frequency		0.0078125 Hz 0 .. 511.96875	16 bits unsigned NUMBER	
4	AC RMS Current		A 0 .. 65532	16 bits unsigned NUMBER	

0xFED8: PGN 65240 - ISO Commanded Address

ISO 11783 defined this message to provide a mechanism for assigning a network address to a node. The NAME information in the data portion of the message must match the name information of the node whose network address is to be set. ISO 11783-5 requires that this message to be sent using the BAM Transport Protocol method. The appropriate response to this message is defined in section 5.2.3 of 11783-5.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This PGN is 9 bytes long and contains 11 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Unique Number	ISO Identity Number		21 bits BINARY	
2	Manufacturer Code		0 .. 2044	11 bits lookup MANUFACTURER_CODE	
3	Device Instance Lower	ISO ECU Instance	0 .. 6	3 bits unsigned NUMBER	
4	Device Instance Upper	ISO Function Instance	0 .. 29	5 bits unsigned NUMBER	
5	Device Function	ISO Function	0 .. 252	8 bits indirect lookup DEVICE_FUNCTION where the first column is the value from field 7 ("Device Class")	
6	Reserved			1 bits RESERVED	
7	Device Class		0 .. 125	7 bits lookup DEVICE_CLASS	
8	System Instance	ISO Device Class Instance	0 .. 13	4 bits unsigned NUMBER	
9	Industry Code		0 .. 6	3 bits lookup INDUSTRY_CODE	
10	Reserved			1 bits RESERVED	
11	New Source Address		0 .. 252	8 bits unsigned NUMBER	

0xFF00-0xFFFF: Manufacturer Proprietary single-frame non-addressed

Manufacturer proprietary PGNs in PDU2 (non-addressed) single-frame PGN range 0xFF00 to 0xFFFF (65280 - 65535). When this is shown during analysis it means the PGN is not reverse engineered yet.

0xFF00: PGN 65280 - Furuno: Heave

This PGN description applies when the following field(s) match:

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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Heave		0.001 m 2147483.647 .. 2147483.644	32 bits signed NUMBER	
5	Reserved			16 bits RESERVED	

0xFF04: PGN 65284 - Maretron: Proprietary DC Breaker Current

This PGN description applies when the following field(s) match:

Manufacturer Code	137
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 6.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Bank Instance		0 .. 252	8 bits unsigned NUMBER	true
5	Indicator Number		0 .. 252	8 bits unsigned NUMBER	
6	Breaker Current		0.1 A 0 .. 6553.2	16 bits unsigned NUMBER	
7	Reserved			16 bits RESERVED	

0xFF05: PGN 65285 - Airmar: Boot State Acknowledgment

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 5.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	



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2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Boot State		0..6	3 bits lookup BOOT_STATE	
5	Reserved			45 bits RESERVED	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0xFF05: PGN 65285 - Lowrance: Temperature

This PGN description applies when the following field(s) match:

Manufacturer Code	140
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	140: Lowrance	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Temperature Source		0..252	8 bits lookup TEMPERATURE_SOURCE	true
5	Actual Temperature		0.01 K 0..655.32	16 bits unsigned NUMBER	
6	Reserved			24 bits RESERVED	

0xFF06: PGN 65286 - Chetco: Dimmer

This PGN description applies when the following field(s) match:

Manufacturer Code	409
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	409: Chetco	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Instance		0..252	8 bits unsigned NUMBER	true
5	Dimmer1		0..252	8 bits unsigned NUMBER	
6	Dimmer2		0..252	8 bits unsigned NUMBER	
7	Dimmer3		0..252	8 bits unsigned NUMBER	
8	Dimmer4		0..252	8 bits unsigned NUMBER	
9	Control		0..252	8 bits unsigned NUMBER	



0xFF06: PGN 65286 - Airmar: Boot State Request

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0xFF07: PGN 65287 - Airmar: Access Level

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Format Code		0 .. 252	8 bits unsigned NUMBER	
5	Access Level		0 .. 6	3 bits lookup ACCESS LEVEL	
6	Reserved			5 bits RESERVED	
7	Access Seed/Key	When transmitted, it provides a seed for an unlock operation. It is used to provide the key during PGN 126208.	0 .. 4294967292	32 bits unsigned NUMBER	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0xFF07: PGN 65287 - Simnet: Configure Temperature Sensor

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect

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- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

0xFF08: PGN 65288 - Seataalk: Alarm

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	SID			8 bits BINARY	
5	Alarm Status		0 .. 252	8 bits lookup SEATALK ALARM STATUS	
6	Alarm ID		0 .. 252	8 bits lookup SEATALK ALARM ID	
7	Alarm Group		0 .. 252	8 bits lookup SEATALK ALARM GROUP	
8	Alarm Priority			16 bits BINARY	

0xFF09: PGN 65289 - Simnet: Trim Tab Sensor Calibration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	



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3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

0xFF0A: PGN 65290 - Simnet: Paddle Wheel Speed Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

0xFF0C: PGN 65292 - Simnet: Clear Fluid Level Warnings

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

0xFF0D: PGN 65293 - Simnet: LGC-2000 Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known



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This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

0xFF0D: PGN 65293 - Diverse Yacht Services: Load Cell

This PGN description applies when the following field(s) match:

Manufacturer Code	641
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	641: Diverse Yacht Services	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Instance		0 .. 252	8 bits unsigned NUMBER	true
5	Reserved			8 bits RESERVED	
6	Load Cell		0 .. 4294967292	32 bits unsigned NUMBER	

0xFF16: PGN 65302 - Simnet: AP Unknown 1

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

Seen as sent by AC-42 only so far.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This single-frame PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 7. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 65532	16 bits unsigned NUMBER	
7	D		0 .. 252	8 bits unsigned NUMBER	



8	Reserved			8 bits RESERVED	
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0xFF19: PGN 65305 - Simnet: Device Status

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Report	2

This PGN is reported by an Autopilot Computer (AC/NAC)

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- One or more of the lookup fields contain missing or incorrect values

This single-frame PGN is 8 bytes long and contains 7 fields. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Model		0 .. 252	8 bits lookup SIMNET_DEVICE_MODEL	
5	Report	2: Status	0 .. 252	8 bits lookup SIMNET_DEVICE_REPORT	
6	Status		0 .. 252	8 bits lookup SIMNET_AP_STATUS	
7	Spare			24 bits SPARE	

0xFF19: PGN 65305 - Simnet: Device Status Request

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Report	3

This PGN is sent by an active AutoPilot head controller (AP, MFD, Triton2). It is used by the AC (AutoPilot Controller) to verify that there is an active controller. If this PGN is not sent regularly the AC may report an error and go to standby.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Model		0 .. 252	8 bits lookup SIMNET_DEVICE_MODEL	
5	Report	3: Send Status	0 .. 252	8 bits lookup SIMNET_DEVICE_REPORT	
6	Spare			32 bits SPARE	

0xFF19: PGN 65305 - Simnet: Pilot Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4



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This PGN is reported by an Autopilot Computer (AC/NAC)

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- One or more of the lookup fields contain missing or incorrect values

This single-frame PGN is 8 bytes long and contains 7 fields. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Model		0 .. 252	8 bits lookup SIMNET_DEVICE_MODEL	
5	Report	10: Mode	0 .. 252	8 bits lookup SIMNET_DEVICE_REPORT	
6	Mode		0 .. 65535	16 bits bitfield SIMNET_AP_MODE_BITFIELD	
7	Spare			16 bits SPARE	

0xFF19: PGN 65305 - Simnet: Device Mode Request

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Report	11

This PGN is sent by an active AutoPilot head controller (AP, MFD, Triton2). It is used by the AC (AutoPilot Controller) to verify that there is an active controller. If this PGN is not sent regularly the AC may report an error and go to standby.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Model		0 .. 252	8 bits lookup SIMNET_DEVICE_MODEL	
5	Report	11: Send Mode	0 .. 252	8 bits lookup SIMNET_DEVICE_REPORT	
6	Spare			32 bits SPARE	

0xFF19: PGN 65305 - Simnet: Sailing Processor Status

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Report	23

This PGN has been seen to be reported by a Sailing Processor.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect



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This single-frame PGN is 8 bytes long and contains 6 fields. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Model		0 .. 252	8 bits lookup SIMNET_DEVICE_MODEL	
5	Report	23: Sailing Processor Status	0 .. 252	8 bits lookup SIMNET_DEVICE_REPORT	
6	Data			32 bits BINARY	

0xFF1D: PGN 65309 - Navico: Wireless Battery Status

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Status		0 .. 252	8 bits unsigned NUMBER	
5	Battery Status		% 0 .. 252	8 bits unsigned NUMBER	
6	Battery Charge Status		% 0 .. 252	8 bits unsigned NUMBER	
7	Reserved			24 bits RESERVED	

0xFF20: PGN 65312 - Navico: Wireless Signal Status

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Unknown		0 .. 252	8 bits unsigned NUMBER	



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5	Signal Strength		% 0 .. 252	8 bits unsigned NUMBER	
6	Reserved			32 bits RESERVED	

0xFF3C: PGN 65340 - Simnet: AP Unknown 2

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

Seen as sent by AC-42 only so far.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This single-frame PGN is 8 bytes long and contains 9 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 252	8 bits unsigned NUMBER	
7	D		0 .. 252	8 bits unsigned NUMBER	
8	E		0 .. 252	8 bits unsigned NUMBER	
9	Reserved			8 bits RESERVED	

0xFF3D: PGN 65341 - Simnet: Autopilot Angle

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 6.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			16 bits RESERVED	
5	Mode		0 .. 252	8 bits lookup SIMNET AP MODE	
6	Reserved			8 bits RESERVED	
7	Angle		0.0001 rad	16 bits unsigned NUMBER	



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		0 .. 6.2831852	
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0xFF41: PGN 65345 - Seataalk: Pilot Wind Datum

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Wind Datum		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
5	Rolling Average Wind Angle		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	Reserved			16 bits RESERVED	

0xFF46: PGN 65350 - Simnet: Magnetic Field

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known
- The manufacturer specific PGN seems to not contain the mandatory Company and Industry fields; this is likely a bug or at least incompatibility in the device

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	A		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
2	B		% 0 .. 252	8 bits unsigned NUMBER	
3	C		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
4	D		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
5	Reserved			8 bits RESERVED	

0xFF4F: PGN 65359 - Seataalk: Pilot Heading

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect



- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	SID			8 bits BINARY	
5	Heading True		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	Heading Magnetic		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
7	Reserved			8 bits RESERVED	

0xFF50: PGN 65360 - Seataalk: Pilot Locked Heading

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	SID			8 bits BINARY	
5	Target Heading True		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	Target Heading Magnetic		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
7	Reserved			8 bits RESERVED	

0xFF51: PGN 65361 - Seataalk: Silence Alarm

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	



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4	Alarm ID		0 .. 252	8 bits lookup SEATALK_ALARM_ID	
5	Alarm Group		0 .. 252	8 bits lookup SEATALK_ALARM_GROUP	
6	Reserved			32 bits RESERVED	

0xFF5B: PGN 65371 - Seataalk: Keypad Message

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 11 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID		0 .. 252	8 bits unsigned NUMBER	true
5	First key		0 .. 252	8 bits unsigned NUMBER	
6	Second key		0 .. 252	8 bits unsigned NUMBER	
7	First key state		0 .. 2	2 bits unsigned NUMBER	
8	Second key state		0 .. 2	2 bits unsigned NUMBER	
9	Reserved			4 bits RESERVED	
10	Encoder Position		0 .. 252	8 bits unsigned NUMBER	
11	Reserved			8 bits RESERVED	

0xFF5E: PGN 65374 - SeaTalk: Keypad Heartbeat

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID		0 .. 252	8 bits unsigned NUMBER	true
5	Variant		0 .. 252	8 bits unsigned NUMBER	



6	Status		0 .. 252	8 bits unsigned NUMBER	
7	Reserved			24 bits RESERVED	

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0xFF63: PGN 65379 - Seatalk: Pilot Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Pilot Mode		0 .. 65532	16 bits lookup SEATALK_PILOT_MODE_16	
5	Sub Mode			16 bits BINARY	
6	Pilot Mode Data			8 bits BINARY	
7	Reserved			8 bits RESERVED	

0xFF80: PGN 65408 - Airmar: Depth Quality Factor

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	Depth Quality Factor		0 .. 13	4 bits lookup AIRMAR_DEPTH_QUALITY_FACTOR	
6	Reserved			36 bits RESERVED	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0xFF81: PGN 65409 - Airmar: Speed Pulse Count

This PGN description applies when the following field(s) match:



Manufacturer Code	135
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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	Duration of interval		0.001 S 0 .. 65.532	16 bits unsigned DURATION	
6	Number of pulses received		0 .. 65532	16 bits unsigned NUMBER	
7	Reserved			8 bits RESERVED	

Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>

0xFF82: PGN 65410 - Airmar: Device Information

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	Internal Device Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Supply Voltage		0.01 V 0 .. 655.32	16 bits unsigned NUMBER	
7	Reserved			8 bits RESERVED	

Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>

0xFF8C: PGN 65420 - Simnet: AP Unknown 3

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

Seen as sent by AC-42 only so far.



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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This single-frame PGN is 8 bytes long and contains 9 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 252	8 bits unsigned NUMBER	
7	D		0 .. 252	8 bits unsigned NUMBER	
8	E		0 .. 252	8 bits unsigned NUMBER	
9	Reserved			8 bits RESERVED	

0xFFC8: PGN 65480 - Simnet: Autopilot Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			48 bits RESERVED	

0x1ED00 - 0x1EE00: Standardized fast-packet addressed

Standardized PGNs in PDU1 (addressed) fast-packet PGN range 0x1ED00 to 0x1EE00 (126208 - 126464). When this is shown during analysis it means the PGN is not reverse engineered yet.

0x1ED00: PGN 126208 - NMEA - Request group function

This PGN description applies when the following field(s) match:

Function Code	0
---------------	---

This is the Request variation of this group function PGN. The receiver shall respond by sending the requested PGN, at the desired transmission interval.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

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This fast-packet PGN is at least 11 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 5 (with name "Number of Parameters"). It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	0: Request	0 .. 252	8 bits lookup GROUP FUNCTION	
2	PGN	Requested PGN	0 .. 262143	24 bits PGN	
3	Transmission interval	0x0: Turn off transmission; 0xFFFF FFFE = Restore default interval, 0xFFFF FFFF in this field and 0xFFFF in field 4 = Transmit now without changing timing variables	0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	
4	Transmission interval offset		0.01 s 0 .. 655.32	16 bits unsigned DURATION	
5	Number of Parameters	How many parameter pairs will follow	0 .. 252	8 bits unsigned NUMBER	
6 Set 1	Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD INDEX	
7 Set 1	Value	Parameter value		Variable length VARIABLE	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1ED00: PGN 126208 - NMEA - Command group function

This PGN description applies when the following field(s) match:

Function Code	1
---------------	---

This is the Command variation of this group function PGN. This instructs the receiver to modify its internal state for the passed parameters. The receiver shall reply with an Acknowledge reply.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 6 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 5 (with name "Number of Parameters"). It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	1: Command	0 .. 252	8 bits lookup GROUP FUNCTION	
2	PGN	Commanded PGN	0 .. 262143	24 bits PGN	
3	Priority		0 .. 13	4 bits lookup PRIORITY	
4	Reserved			4 bits RESERVED	
5	Number of Parameters	How many parameter pairs will follow	0 .. 252	8 bits unsigned NUMBER	
6 Set 1	Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD INDEX	
7 Set 1	Value	Parameter value		Variable length VARIABLE	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1ED00: PGN 126208 - NMEA - Acknowledge group function

This PGN description applies when the following field(s) match:

Function Code	2
---------------	---



This is the Acknowledge variation of this group function PGN. When a device receives a Command, it will attempt to perform the command (change its parameters) and reply positively or negatively.

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This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

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This fast-packet PGN is at least 6 bytes long and contains 6 fields. The 1 fields starting at field 6 (with name "Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 5 (with name "Number of Parameters".) It is not transmitted at a regular interval, but when data is available or it is requested.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	2: Acknowledge	0 .. 252	8 bits lookup GROUP FUNCTION	
2	PGN	Commanded PGN	0 .. 262143	24 bits PGN	
3	PGN error code		0 .. 13	4 bits lookup PGN ERROR CODE	
4	Transmission interval/Priority error code		0 .. 13	4 bits lookup TRANSMISSION INTERVAL	
5	Number of Parameters		0 .. 252	8 bits unsigned NUMBER	
6 Set 1	Parameter		0 .. 13	4 bits lookup PARAMETER FIELD	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1ED00: PGN 126208 - NMEA - Read Fields group function

This PGN description applies when the following field(s) match:

Function Code	3
---------------	---

This is the Read Fields variation of this group function PGN. The receiver shall respond by sending a Read Reply variation of this PGN, containing the desired values. This PGN is special as it contains two sets of repeating fields, and the fields that contain the information how many repetitions there are do not have a fixed offset in the PGN as the fields 3 to 5 are only present if field 2 is for a proprietary PGN

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is at least 9 bytes long and contains 11 fields. The 2 fields starting at field 9 (with name "Selection Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 7 (with name "Number of Selection Pairs".) The 1 fields starting at field 11 (with name "Parameter") form repeating set 2. The set is repeated n times, where n is determined by the value of field 8 (with name "Number of Parameters".) It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	3: Read Fields	0 .. 252	8 bits lookup GROUP FUNCTION	
2	PGN	Commanded PGN	0 .. 262143	24 bits PGN	
3	Manufacturer Code	Only in PGN when Commanded PGN is proprietary	0 .. 2044	11 bits lookup MANUFACTURER CODE	
4	Reserved	Only in PGN when Commanded PGN is proprietary		2 bits RESERVED	
5	Industry Code	Only in PGN when Commanded PGN is proprietary	0 .. 6	3 bits lookup INDUSTRY CODE	
6	Unique ID		0 .. 252	8 bits unsigned NUMBER	
7	Number of Selection Pairs		0 .. 252	8 bits unsigned NUMBER	

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8	Number of Parameters		0 .. 252	8 bits unsigned NUMBER	
9 Set 1	Selection Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD_INDEX	
10 Set 1	Selection Value			Variable length VARIABLE	
11 Set 2	Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD_INDEX	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1ED00: PGN 126208 - NMEA - Read Fields reply group function

This PGN description applies when the following field(s) match:

Function Code	4
---------------	---

This is the Read Fields Reply variation of this group function PGN. The receiver is responding to a Read Fields request. This PGN is special as it contains two sets of repeating fields, and the fields that contain the information how many repetitions there are do not have a fixed offset in the PGN as the fields 3 to 5 are only present if field 2 is for a proprietary PGN

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 9 bytes long and contains 12 fields. The 2 fields starting at field 9 (with name "Selection Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 7 (with name "Number of Selection Pairs".) The 2 fields starting at field 11 (with name "Parameter") form repeating set 2. The set is repeated n times, where n is determined by the value of field 8 (with name "Number of Parameters".) It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	4: Read Fields Reply	0 .. 252	8 bits lookup GROUP_FUNCTION	
2	PGN	Commanded PGN	0 .. 262143	24 bits PGN	
3	Manufacturer Code	Only in PGN when Commanded PGN is proprietary	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
4	Reserved	Only in PGN when Commanded PGN is proprietary		2 bits RESERVED	
5	Industry Code	Only in PGN when Commanded PGN is proprietary	0 .. 6	3 bits lookup INDUSTRY_CODE	
6	Unique ID		0 .. 252	8 bits unsigned NUMBER	
7	Number of Selection Pairs		0 .. 252	8 bits unsigned NUMBER	
8	Number of Parameters		0 .. 252	8 bits unsigned NUMBER	
9 Set 1	Selection Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD_INDEX	
10 Set 1	Selection Value			Variable length VARIABLE	
11 Set 2	Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD_INDEX	
12 Set 2	Value			Variable length VARIABLE	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1ED00: PGN 126208 - NMEA - Write Fields group function



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This PGN description applies when the following field(s) match:

Function Code

5

This is the Write Fields variation of this group function PGN. The receiver shall modify internal state and reply with a Write Fields Reply message. This PGN is special as it contains two sets of repeating fields, and the fields that contain the information how many repetitions there are do not have a fixed offset in the PGN as the fields 3 to 5 are only present if field 2 is for a proprietary PGN

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 9 bytes long and contains 12 fields. The 2 fields starting at field 9 (with name "Selection Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 7 (with name "Number of Selection Pairs".) The 2 fields starting at field 11 (with name "Parameter") form repeating set 2. The set is repeated n times, where n is determined by the value of field 8 (with name "Number of Parameters".) It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	5: Write Fields	0 .. 252	8 bits lookup GROUP FUNCTION	
2	PGN	Commanded PGN	0 .. 262143	24 bits PGN	
3	Manufacturer Code	Only in PGN when Commanded PGN is proprietary	0 .. 2044	11 bits lookup MANUFACTURER CODE	
4	Reserved	Only in PGN when Commanded PGN is proprietary		2 bits RESERVED	
5	Industry Code	Only in PGN when Commanded PGN is proprietary	0 .. 6	3 bits lookup INDUSTRY CODE	
6	Unique ID		0 .. 252	8 bits unsigned NUMBER	
7	Number of Selection Pairs		0 .. 252	8 bits unsigned NUMBER	
8	Number of Parameters		0 .. 252	8 bits unsigned NUMBER	
9 Set 1	Selection Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD INDEX	
10 Set 1	Selection Value			Variable length VARIABLE	
11 Set 2	Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD INDEX	
12 Set 2	Value			Variable length VARIABLE	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1ED00: PGN 126208 - NMEA - Write Fields reply group function

This PGN description applies when the following field(s) match:

Function Code

6

This is the Write Fields Reply variation of this group function PGN. The receiver is responding to a Write Fields request. This PGN is special as it contains two sets of repeating fields, and the fields that contain the information how many repetitions there are do not have a fixed offset in the PGN as the fields 3 to 5 are only present if field 2 is for a proprietary PGN

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 9 bytes long and contains 12 fields. The 2 fields starting at field 9 (with name "Selection Parameter") form repeating set 1. The set is repeated n times, where n is determined by the value of field 7 (with name "Number of Selection Pairs".) The 2 fields starting at field 11 (with name "Parameter") form repeating set 2. The set is repeated n times, where n is determined by the value of field 8 (with name "Number of Parameters".) It is not transmitted at a regular interval, but when data is available or it is requested.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code	6: Write Fields Reply	0 .. 252	8 bits lookup GROUP FUNCTION	
2	PGN	Commanded PGN	0 .. 262143	24 bits PGN	
3	Manufacturer Code	Only in PGN when Commanded PGN is proprietary	0 .. 2044	11 bits lookup MANUFACTURER CODE	
4	Reserved	Only in PGN when Commanded PGN is proprietary		2 bits RESERVED	
5	Industry Code	Only in PGN when Commanded PGN is proprietary	0 .. 6	3 bits lookup INDUSTRY CODE	
6	Unique ID		0 .. 252	8 bits unsigned NUMBER	
7	Number of Selection Pairs		0 .. 252	8 bits unsigned NUMBER	
8	Number of Parameters		0 .. 252	8 bits unsigned NUMBER	
9 Set 1	Selection Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD INDEX	
10 Set 1	Selection Value			Variable length VARIABLE	
11 Set 2	Parameter	Parameter index	0 .. 252	8 bits unsigned FIELD INDEX	
12 Set 2	Value			Variable length VARIABLE	

Source: <https://web.archive.org/web/20160408085901/http://www.nmea.org/Assets/20140109%20nmea-2000-corrigendum-tc201401031%20pgn%20126208.pdf>

0x1EE00: PGN 126464 - PGN List (Transmit and Receive)

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 1 bytes long and contains 2 fields. The 1 fields starting at field 2 (with name "PGN") form repeating set 1. The set is repeated until there is no more data in the PGN. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Function Code		0 .. 252	8 bits lookup PGN LIST FUNCTION	
2 Set 1	PGN		0 .. 262143	24 bits PGN	

0x1EF00: Manufacturer Proprietary fast-packet addressed

Manufacturer Proprietary PGNs in PDU1 (addressed) fast-packet PGN 0x1EF00 (126720).When this is shown during analysis it means the PGN is not reverse engineered yet.

0x1EF00: PGN 126720 - Seataalk1: Pilot Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	240
command	129
Seataalk1 Command	132



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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 21 bytes long and contains 11 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	240: Seataalk 1 Encoded	0 .. 252	8 bits lookup SEATAALK MESSAGE ID	true
5	command	129: Seataalk1	0 .. 252	8 bits lookup SEATAALK COMMAND	true
6	Seataalk1 Command	132: Compass heading Autopilot course and Rudder position	0 .. 252	8 bits lookup SEATAALK1 COMMAND	true
7	Unknown 1			24 bits BINARY	
8	Pilot Mode		0 .. 252	8 bits lookup SEATAALK PILOT MODE	
9	Sub Mode		0 .. 252	8 bits unsigned NUMBER	
10	Pilot Mode Data			8 bits BINARY	
11	Unknown 2			80 bits BINARY	

0x1EF00: PGN 126720 - Seataalk1: Pilot Hull Type

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	108
command	22

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 14 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	108: Pilot Configuration	0 .. 252	8 bits lookup SEATAALK MESSAGE ID	true
5	command	22: Hull Type	0 .. 252	8 bits lookup SEATAALK COMMAND	true
6	Unknown			8 bits BINARY	
7	Hull Type		0 .. 252	8 bits lookup SEATAALK PILOT HULL TYPE	
8	Unknown 2			64 bits BINARY	



0x1EF00: PGN 126720 - Seataalk: Pilot Auto Turn

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	108
command	38

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	108: Pilot Configuration	0 .. 252	8 bits lookup SEATALK MESSAGE ID	true
5	command	38: Auto Turn	0 .. 252	8 bits lookup SEATALK COMMAND	true
6	Unknown			8 bits BINARY	
7	Enabled		0 .. 252	8 bits lookup YES_NO	
8	Unknown 2		0 .. 252	8 bits lookup YES_NO	
9	Unknown 3			16 bits BINARY	

0x1EF00: PGN 126720 - Seataalk1: Device Identification

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	240
command	129
Seataalk1 Command	144

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 7 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	240: Seataalk 1 Encoded	0 .. 252	8 bits lookup SEATALK MESSAGE ID	true
5	command	129: Seataalk1	0 .. 252	8 bits lookup SEATALK COMMAND	true
6	Seataalk1 Command	144: Device Indentification	0 .. 252	8 bits lookup SEATALK1 COMMAND	true

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7	Reserved			8 bits RESERVED	
8	device		0 .. 252	8 bits lookup SEATALK_DEVICE_ID	

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0x1EF00: PGN 126720 - Seataalk1: Display Brightness

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	140
command1	12
Command	0

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 10 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	140: Display	0 .. 252	8 bits lookup SEATALK_MESSAGE_ID	true
5	command1	12: Settings	0 .. 252	8 bits lookup SEATALK_COMMAND	true
6	Group		0 .. 252	8 bits lookup SEATALK_NETWORK_GROUP	
7	Shared		0 .. 252	8 bits lookup SEATALK_SHARED	
8	Command	0: Brightness	0 .. 252	8 bits unsigned NUMBER	
9	Brightness		% 0 .. 252	8 bits unsigned NUMBER	
10	Unknown 2			8 bits BINARY	

0x1EF00: PGN 126720 - Seataalk1: Display Color

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	140
command1	12
Command	1

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 10 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	



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2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	140: Display	0 .. 252	8 bits lookup SEATALK MESSAGE ID	true
5	command1	12: Settings	0 .. 252	8 bits lookup SEATALK COMMAND	true
6	Group		0 .. 252	8 bits lookup SEATALK NETWORK GROUP	
7	Unknown 1			8 bits BINARY	
8	Command	1: Color	0 .. 252	8 bits unsigned NUMBER	
9	Color		0 .. 252	8 bits lookup SEATALK DISPLAY COLOR	
10	Unknown 2			8 bits BINARY	

0x1EF00: PGN 126720 - Seataalk1: Keystroke

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4
Proprietary ID	240
command	129
Seataalk1 Command	134

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 22 bytes long and contains 10 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	240: Seataalk 1 Encoded	0 .. 252	8 bits lookup SEATALK MESSAGE ID	true
5	command	129: Seataalk1	0 .. 252	8 bits lookup SEATALK COMMAND	true
6	Seataalk1 Command	134: Keystroke	0 .. 252	8 bits lookup SEATALK1 COMMAND	true
7	device		0 .. 252	8 bits unsigned NUMBER	
8	key		0 .. 252	8 bits lookup SEATALK KEYSTROKE	
9	keyInverted	Bit negated version of key	0 .. 252	8 bits unsigned NUMBER	
10	Unknown data			112 bits BINARY	

0x1EF00: PGN 126720 - Fusion: Media Control

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	3

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	3: Media Command	0..65532	16 bits lookup FUSION_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	
6	Command		0..252	8 bits lookup FUSION_COMMAND	

0x1EF00: PGN 126720 - Fusion: Sirius Control

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	30

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	30: Send Sirius Command	0..65532	16 bits lookup FUSION_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	
6	Command		0..252	8 bits lookup FUSION_SIRIUS_COMMAND	
7	Data		0..65532	16 bits unsigned NUMBER	

0x1EF00: PGN 126720 - Fusion: Request Status

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	1

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 4 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	1: Request Status	0..65532	16 bits lookup FUSION_MESSAGE_ID	true

0x1EF00: PGN 126720 - Fusion: Set Source

This PGN description applies when the following field(s) match:



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Manufacturer Code	419
Industry Code	4
Proprietary ID	2

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	2: Set Source	0..65532	16 bits lookup FUSION MESSAGE ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	

0x1EF00: PGN 126720 - Fusion: Set Mute

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	17

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	17: Set Mute	0..65532	16 bits lookup FUSION MESSAGE ID	true
5	Command		0..252	8 bits lookup FUSION MUTE COMMAND	

0x1EF00: PGN 126720 - Fusion: Set Zone Volume

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	24

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	

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4	Proprietary ID	24: Set Zone Volume	0 .. 65532	16 bits lookup FUSION_MESSAGE_ID	true
5	Zone		0 .. 252	8 bits unsigned NUMBER	
6	Volume		0 .. 252	8 bits unsigned NUMBER	

0x1EF00: PGN 126720 - Fusion: Set All Volumes

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	25

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	25: Set All Volumes	0 .. 65532	16 bits lookup FUSION_MESSAGE_ID	true
5	Zone1		0 .. 252	8 bits unsigned NUMBER	
6	Zone2		0 .. 252	8 bits unsigned NUMBER	
7	Zone3		0 .. 252	8 bits unsigned NUMBER	
8	Zone4		0 .. 252	8 bits unsigned NUMBER	

0x1EF00: PGN 126720 - Fusion: Set Power

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Proprietary ID	28

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	28: Power	0 .. 65532	16 bits lookup FUSION_MESSAGE_ID	true
5	Power		0 .. 252	8 bits lookup FUSION_POWER_STATE	

0x1EF00: PGN 126720 - Airmar: Attitude Offset

This PGN description applies when the following field(s) match:



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Manufacturer Code	135
Industry Code	4
Proprietary ID	32

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 9 bytes long and contains 7 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	32: Attitude Offsets	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Azimuth offset	Positive: sensor rotated to port, negative: sensor rotated to starboard	0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
6	Pitch offset	Positive: sensor tilted to bow, negative: sensor tilted to stern	0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
7	Roll offset	Positive: sensor tilted to port, negative: sensor tilted to starboard	0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Calibrate Compass

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	33

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect

This fast-packet PGN is 24 bytes long and contains 16 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	33: Calibrate Compass	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Calibrate Function		0 .. 252	8 bits lookup AIRMAR CALIBRATE FUNCTION	
6	Calibration Status		0 .. 252	8 bits lookup AIRMAR CALIBRATE STATUS	
7	Verify Score	TBD	0 .. 252	8 bits unsigned NUMBER	
8	X-axis gain value	default 100, range 50 to 500	0.01 -327.67 .. 327.64	16 bits signed NUMBER	



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9	Y-axis gain value	default 100, range 50 to 500	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
10	Z-axis gain value	default 100, range 50 to 500	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
11	X-axis linear offset		0.01 I -327.67 .. 327.64	16 bits signed NUMBER	
12	Y-axis linear offset		0.01 I -327.67 .. 327.64	16 bits signed NUMBER	
13	Z-axis linear offset		0.01 I -327.67 .. 327.64	16 bits signed NUMBER	
14	X-axis angular offset		0.1 deg -3276.7 .. 3276.4	16 bits signed NUMBER	
15	Pitch and Roll damping	default 30, range 0 to 200	0.05 s -1638.35 .. 1638.2	16 bits signed DURATION	
16	Compass/Rate gyro damping	default -30, range -2400 to 2400, negative indicates rate gyro is to be used in compass calculations	0.05 s -1638.35 .. 1638.2	16 bits signed DURATION	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: True Wind Options

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	34

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 6 bytes long and contains 6 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	34: True Wind Options	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	COG substitution for HDG	Allow use of COG when HDG not available?	0 .. 2	2 bits lookup YES NO	
6	Reserved			22 bits RESERVED	

Source: <http://www.airmartechology.com/uploads/installguide/PB200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Simulate Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	135
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Industry Code	4
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This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 6 bytes long and contains 6 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	35: Simulate Mode	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Simulate Mode		0 .. 2	2 bits lookup OFF ON	
6	Reserved			22 bits RESERVED	

Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Calibrate Depth

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	40

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 6 bytes long and contains 6 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	40: Calibrate Depth	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Speed of Sound Mode	actual allowed range is 1350.0 to 1650.0 m/s	0.1 m/s 0 .. 6553.2	16 bits unsigned NUMBER	
6	Reserved			8 bits RESERVED	

Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Calibrate Speed

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	41

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 4 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "Input frequency") form repeating set 1. The set is repeated *n* times, where *n* is determined by the value of field 5 (with name "Number of pairs of data points"). It is not transmitted at a regular interval, but when data is available or it is requested.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	41: Calibrate Speed	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Number of pairs of data points	actual range is 0 to 25. 254=restore default speed curve	0 .. 252	8 bits unsigned NUMBER	
6 Set 1	Input frequency		0.1 Hz 0 .. 6553.2	16 bits unsigned NUMBER	
7 Set 1	Output speed		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Calibrate Temperature

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	42

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 6 bytes long and contains 7 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	42: Calibrate Temperature	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Temperature instance		0 .. 2	2 bits lookup AIRMAR TEMPERATURE INSTANCE	true
6	Reserved			6 bits RESERVED	
7	Temperature offset		0.001 K -9.999 .. 9.999	16 bits signed NUMBER	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Speed Filter None

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	43
Filter type	0

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.



This fast-packet PGN is 6 bytes long and contains 7 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	43: Speed Filter	0 .. 252	8 bits lookup AIRMAR_COMMAND	true
5	Filter type	0: No filter	0 .. 13	4 bits unsigned NUMBER	
6	Reserved			4 bits RESERVED	
7	Sample interval	Interval of time between successive samples of the paddlewheel pulse accumulator	0.01 S 0 .. 655.32	16 bits unsigned DURATION	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Speed Filter IIR

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	43
Filter type	1

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 8 bytes long and contains 8 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	43: Speed Filter	0 .. 252	8 bits lookup AIRMAR_COMMAND	true
5	Filter type	1: IIR filter	0 .. 13	4 bits unsigned NUMBER	
6	Reserved			4 bits RESERVED	
7	Sample interval	Interval of time between successive samples of the paddlewheel pulse accumulator	0.01 S 0 .. 655.32	16 bits unsigned DURATION	
8	Filter duration	Duration of filter, must be bigger than the sample interval	0.01 S 0 .. 655.32	16 bits unsigned DURATION	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Temperature Filter None

This PGN description applies when the following field(s) match:



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Filter type	0

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 6 bytes long and contains 7 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	44: Temperature Filter	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Filter type	0: No filter	0 .. 13	4 bits unsigned NUMBER	
6	Reserved			4 bits RESERVED	
7	Sample interval	Interval of time between successive samples of the water temperature thermistor	0.01 S 0 .. 655.32	16 bits unsigned DURATION	

Source: <http://www.airmartechology.com/uploads/installguide/DST200UserManual.pdf>

0x1EF00: PGN 126720 - Airmar: Temperature Filter IIR

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	44
Filter type	1

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 8 bytes long and contains 8 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	44: Temperature Filter	0 .. 252	8 bits lookup AIRMAR COMMAND	true
5	Filter type	1: IIR filter	0 .. 13	4 bits unsigned NUMBER	
6	Reserved			4 bits RESERVED	
7	Sample interval	Interval of time between successive samples of the water temperature thermistor	0.01 S 0 .. 655.32	16 bits unsigned DURATION	



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8	Filter duration	Duration of filter, must be bigger than the sample interval	0.01 S 0 .. 655.32	16 bits unsigned DURATION	
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Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>**0x1EF00: PGN 126720 - Airmar: NMEA 2000 options**

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4
Proprietary ID	46

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 6 bytes long and contains 6 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	46: NMEA 2000 options	0 .. 252	8 bits lookup AIRMAR_COMMAND	true
5	Transmission Interval		0 .. 2	2 bits lookup AIRMAR_TRANSMISSION_INTERVAL	
6	Reserved			22 bits RESERVED	

Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>**0x1EF00: PGN 126720 - Airmar: Addressable Multi-Frame**

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 3 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID		0 .. 252	8 bits unsigned NUMBER	true

0x1EF00: PGN 126720 - Maretron: Deviation Calibration Response

This PGN description applies when the following field(s) match:

Manufacturer Code	137
Industry Code	4
Software code	1



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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Product code		0 .. 65532	16 bits lookup MARETRON PRODUCT CODE	
5	Software code	1: Version 1	0 .. 65532	16 bits lookup MARETRON SOFTWARE CODE	
6	Command	80: Deviation calibration	0 .. 252	8 bits lookup MARETRON COMMAND	
7	Status		0 .. 252	8 bits lookup MARETRON STATUS DEVIATION	

Source: https://web.archive.org/web/20220809223313/https://www.maretron.com/support/manuals/SSC200UM_1.8.html

0x1EF00: PGN 126720 - Maretron: Slave Response

This PGN description applies when the following field(s) match:

Manufacturer Code	137
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- One or more of the lookup fields contain missing or incorrect values
- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Product code		0 .. 65532	16 bits unsigned NUMBER	
5	Software code		0 .. 65532	16 bits unsigned NUMBER	
6	Command		0 .. 252	8 bits unsigned NUMBER	
7	Status		0 .. 252	8 bits unsigned NUMBER	

0x1EF00: PGN 126720 - Garmin: Day Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	229
Industry Code	4
Unknown ID 1	222
Unknown ID 2	5
Unknown ID 3	5
Unknown ID 4	5
Mode	0

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:



- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This fast-packet PGN is 11 bytes long and contains 11 fields.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	229: Garmin	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Unknown ID 1	222: Always 222	0 .. 252	8 bits unsigned NUMBER	
5	Unknown ID 2	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
6	Unknown ID 3	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
7	Unknown ID 4	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
8	Spare			16 bits SPARE	
9	Mode	0: Day	0 .. 252	8 bits lookup GARMIN COLOR MODE	
10	Spare			8 bits SPARE	
11	Backlight		0 .. 252	8 bits lookup GARMIN BACKLIGHT LEVEL	

0x1EF00: PGN 126720 - Garmin: Night Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	229
Industry Code	4
Unknown ID 1	222
Unknown ID 2	5
Unknown ID 3	5
Unknown ID 4	5
Mode	1

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 11 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	229: Garmin	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Unknown ID 1	222: Always 222	0 .. 252	8 bits unsigned NUMBER	
5	Unknown ID 2	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
6	Unknown ID 3	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
7	Unknown ID 4	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
8	Spare			16 bits SPARE	
9	Mode	1: Night	0 .. 252	8 bits lookup GARMIN COLOR MODE	
10	Spare			8 bits SPARE	



11	Backlight		0 .. 252	8 bits lookup GARMIN BACKLIGHT LEVEL	
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0x1EF00: PGN 126720 - Garmin: Color mode

This PGN description applies when the following field(s) match:

Manufacturer Code	229
Industry Code	4
Unknown ID 1	222
Unknown ID 2	5
Unknown ID 3	5
Unknown ID 4	5
Mode	13

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 11 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	229: Garmin	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Unknown ID 1	222: Always 222	0 .. 252	8 bits unsigned NUMBER	
5	Unknown ID 2	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
6	Unknown ID 3	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
7	Unknown ID 4	5: Always 5	0 .. 252	8 bits unsigned NUMBER	
8	Spare			16 bits SPARE	
9	Mode	13: Color	0 .. 252	8 bits lookup GARMIN COLOR MODE	
10	Spare			8 bits SPARE	
11	Color		0 .. 252	8 bits lookup GARMIN COLOR	

0x1F000-0x1FEFF: Standardized mixed single/fast packet non-addressed

Standardized PGNs in PDU2 (non-addressed) mixed single/fast packet PGN range 0x1F000 to 0x1FEFF (126976 - 130815). When this is shown during analysis it means the PGN is not reverse engineered yet.

0x1F007: PGN 126983 - Alert

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 28 bytes long and contains 21 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Alert Type		0 .. 13	4 bits lookup ALERT TYPE	
2	Alert Category		0 .. 13	4 bits lookup ALERT CATEGORY	
3	Alert System		0 .. 252	8 bits unsigned NUMBER	
4	Alert Sub-System		0 .. 252	8 bits unsigned NUMBER	



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5	Alert ID		0 .. 65532	16 bits unsigned NUMBER	
6	Data Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
7	Data Source Instance		0 .. 252	8 bits unsigned NUMBER	true
8	Data Source Index-Source		0 .. 252	8 bits unsigned NUMBER	
9	Alert Occurrence Number		0 .. 252	8 bits unsigned NUMBER	
10	Temporary Silence Status		0 .. 1	1 bits lookup YES_NO	
11	Acknowledge Status		0 .. 1	1 bits lookup YES_NO	
12	Escalation Status		0 .. 1	1 bits lookup YES_NO	
13	Temporary Silence Support		0 .. 1	1 bits lookup YES_NO	
14	Acknowledge Support		0 .. 1	1 bits lookup YES_NO	
15	Escalation Support		0 .. 1	1 bits lookup YES_NO	
16	Reserved			2 bits RESERVED	
17	Acknowledge Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
18	Trigger Condition		0 .. 13	4 bits lookup ALERT_TRIGGER_CONDITION	
19	Threshold Status		0 .. 13	4 bits lookup ALERT_THRESHOLD_STATUS	
20	Alert Priority		0 .. 252	8 bits unsigned NUMBER	
21	Alert State		0 .. 252	8 bits lookup ALERT_STATE	

0x1F008: PGN 126984 - Alert Response

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 25 bytes long and contains 12 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Alert Type		0 .. 13	4 bits lookup ALERT_TYPE	
2	Alert Category		0 .. 13	4 bits lookup ALERT_CATEGORY	
3	Alert System		0 .. 252	8 bits unsigned NUMBER	
4	Alert Sub-System		0 .. 252	8 bits unsigned NUMBER	
5	Alert ID		0 .. 65532	16 bits unsigned NUMBER	
6	Data Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
7	Data Source Instance		0 .. 252	8 bits unsigned NUMBER	true
8	Data Source Index-Source		0 .. 252	8 bits unsigned NUMBER	
9	Alert Occurrence Number		0 .. 252	8 bits unsigned NUMBER	
10	Acknowledge Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
11	Response Command		0 .. 3	2 bits lookup ALERT_RESPONSE_COMMAND	
12	Reserved			6 bits RESERVED	

0x1F009: PGN 126985 - Alert Text

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known



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This fast-packet PGN is at least 17 bytes long and contains 12 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Alert Type		0 .. 13	4 bits lookup ALERT_TYPE	
2	Alert Category		0 .. 13	4 bits lookup ALERT_CATEGORY	
3	Alert System		0 .. 252	8 bits unsigned NUMBER	
4	Alert Sub-System		0 .. 252	8 bits unsigned NUMBER	
5	Alert ID		0 .. 65532	16 bits unsigned NUMBER	
6	Data Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
7	Data Source Instance		0 .. 252	8 bits unsigned NUMBER	true
8	Data Source Index-Source		0 .. 252	8 bits unsigned NUMBER	
9	Alert Occurrence Number		0 .. 252	8 bits unsigned NUMBER	
10	Language ID		0 .. 252	8 bits lookup ALERT_LANGUAGE_ID	
11	Alert Text Description			Variable length STRING_LAU	
12	Alert Location Text Description			Variable length STRING_LAU	

0x1F00A: PGN 126986 - Alert Configuration

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 20 bytes long and contains 15 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Alert Type		0 .. 13	4 bits lookup ALERT_TYPE	
2	Alert Category		0 .. 13	4 bits lookup ALERT_CATEGORY	
3	Alert System		0 .. 252	8 bits unsigned NUMBER	
4	Alert Sub-System		0 .. 252	8 bits unsigned NUMBER	
5	Alert ID		0 .. 65532	16 bits unsigned NUMBER	
6	Data Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
7	Data Source Instance		0 .. 252	8 bits unsigned NUMBER	true
8	Data Source Index-Source		0 .. 252	8 bits unsigned NUMBER	
9	Alert Occurrence Number		0 .. 252	8 bits unsigned NUMBER	
10	Alert Control		0 .. 2	2 bits unsigned NUMBER	
11	User Defined Alert Assignment		0 .. 2	2 bits unsigned NUMBER	
12	Reserved			4 bits RESERVED	
13	Reactivation Period		0 .. 252	8 bits unsigned NUMBER	
14	Temporary Silence Period		0 .. 252	8 bits unsigned NUMBER	
15	Escalation Period		0 .. 252	8 bits unsigned NUMBER	

0x1F00B: PGN 126987 - Alert Threshold

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known



This fast-packet PGN is at least 17 bytes long and contains 14 fields. The 4 fields starting at field 11 (with name "Parameter Number") form repeating set 1. The set is repeated n times, where n is determined by the value of field 10 (with name "Number of Parameters".)

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Alert Type		0 .. 13	4 bits lookup ALERT_TYPE	
2	Alert Category		0 .. 13	4 bits lookup ALERT_CATEGORY	
3	Alert System		0 .. 252	8 bits unsigned NUMBER	
4	Alert Sub-System		0 .. 252	8 bits unsigned NUMBER	
5	Alert ID		0 .. 65532	16 bits unsigned NUMBER	
6	Data Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
7	Data Source Instance		0 .. 252	8 bits unsigned NUMBER	true
8	Data Source Index-Source		0 .. 252	8 bits unsigned NUMBER	
9	Alert Occurrence Number		0 .. 252	8 bits unsigned NUMBER	
10	Number of Parameters	Total Number of Threshold Parameters	0 .. 252	8 bits unsigned NUMBER	
11 Set 1	Parameter Number		0 .. 252	8 bits unsigned NUMBER	
12 Set 1	Trigger Method		0 .. 252	8 bits unsigned NUMBER	
13 Set 1	Threshold Data Format		0 .. 252	8 bits unsigned NUMBER	
14 Set 1	Threshold Level		0 .. 18446744073709551615	64 bits unsigned NUMBER	

0x1F00C: PGN 126988 - Alert Value

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is at least 17 bytes long and contains 13 fields. The 3 fields starting at field 11 (with name "Value Parameter Number") form repeating set 1. The set is repeated n times, where n is determined by the value of field 10 (with name "Number of Parameters".)

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Alert Type		0 .. 13	4 bits lookup ALERT_TYPE	
2	Alert Category		0 .. 13	4 bits lookup ALERT_CATEGORY	
3	Alert System		0 .. 252	8 bits unsigned NUMBER	
4	Alert Sub-System		0 .. 252	8 bits unsigned NUMBER	
5	Alert ID		0 .. 65532	16 bits unsigned NUMBER	
6	Data Source Network ID NAME		0 .. 1.84467440737096e+19	64 bits ISO_NAME	
7	Data Source Instance		0 .. 252	8 bits unsigned NUMBER	true
8	Data Source Index-Source		0 .. 252	8 bits unsigned NUMBER	
9	Alert Occurrence Number		0 .. 252	8 bits unsigned NUMBER	



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10	Number of Parameters	Total Number of Value Parameters	0 .. 252	8 bits unsigned NUMBER	
<i>11 Set 1</i>	Value Parameter Number		0 .. 252	8 bits unsigned NUMBER	
<i>12 Set 1</i>	Value Data Format		0 .. 252	8 bits unsigned NUMBER	
<i>13 Set 1</i>	Value Data		0 .. 18446744073709551615	64 bits unsigned NUMBER	

0x1F010: PGN 126992 - System Time

The purpose of this PGN is twofold: To provide a regular transmission of UTC time and date. To provide synchronism for measurement data.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID	An upward counting number used to tie related information together between different PGNs. For example the SID would be used to tie together the COG, SOG and RAIM values to a given position.	0 .. 252	8 bits unsigned NUMBER	
2	Source		0 .. 13	4 bits lookup SYSTEM TIME	true
3	Reserved			4 bits RESERVED	
4	Date		d 0 .. 65532	16 bits unsigned DATE	
5	Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	

0x1F011: PGN 126993 - Heartbeat

Reception of this PGN confirms that a device is still present on the network. Reception of this PGN may also be used to maintain an address to NAME association table within the receiving device. The transmission interval may be used by the receiving unit to determine the time-out value for the connection supervision. The value contained in Field 1 of this PGN reflects the PGN's current Transmission Interval. Changes to this PGN's Transmission Interval shall be reflected in Field 1. The transmission interval can only be changed by using the Request Group Function PGN 126208 with no pairs of request parameters provided. Field 3 of the Request Group Function PGN 126208 may contain values between 1,000ms and 60,000ms. This PGN cannot be requested by the ISO Request PGN 059904 or Request Group Function PGN 126208. In Request Group Function PGN 126208, setting Field 3 to a value of 0xFFFF FFFF and Field 4 to a value of 0xFFFF: 'Transmit now without changing timing variables.' is prohibited. The Command Group Function PGN 126208 shall not be used with this PGN. Fields 3 and 4 of this PGN provide information which can be used to distinguish short duration disturbances from permanent failures. See ISO 11898 -1 Sections 6.12, 6.13, 6.14, 13.1.1, 13.1.4, 13.1.4.3 and Figure 16 (node status transition diagram) for additional context.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 7. It is normally transmitted every 60000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Data transmit offset	Offset in transmit time from time of request command: 0x0 = transmit immediately, 0xFFFF = Do not change offset.	0.01 s 0 .. 65532	16 bits unsigned DURATION	
2	Sequence Counter		0 .. 252	8 bits unsigned NUMBER	



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3	Controller 1 State		0..2	2 bits lookup CONTROLLER STATE	
4	Controller 2 State		0..2	2 bits lookup CONTROLLER STATE	
5	Equipment Status		0..2	2 bits lookup EQUIPMENT STATUS	
6	Reserved			34 bits RESERVED	

Source: <https://web.archive.org/web/20170609023206/http://www.nmea.org/Assets/20140102%20nmea-2000-126993%20heartbeat%20pgn%20corrigendum.pdf>

0x1F014: PGN 126996 - Product Information

Provides product information onto the network that could be important for determining quality of data coming from this product.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 134 bytes long and contains 8 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	NMEA 2000 Version	Binary number containing a decimal number of format AABBB, where AA is the major and BBB is the minor release. The decimal point position is assumed.	0.001 0..65.532	16 bits unsigned NUMBER	
2	Product Code		0..65532	16 bits unsigned NUMBER	
3	Model ID			256 bits STRING_FIX	
4	Software Version Code			256 bits STRING_FIX	
5	Model Version			256 bits STRING_FIX	
6	Model Serial Code			256 bits STRING_FIX	
7	Certification Level		0..252	8 bits lookup CERTIFICATION LEVEL	
8	Load Equivalency	Garantueed maximum power consumption, 50 mA per LEN	0..252	8 bits unsigned NUMBER	

0x1F016: PGN 126998 - Configuration Information

Free-form alphanumeric fields describing the installation (e.g., starboard engine room location) of the device and installation notes (e.g., calibration data).

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 0 bytes long and contains 3 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Installation Description #1			Variable length STRING_LAU	
2	Installation Description #2			Variable length STRING_LAU	
3	Manufacturer Information			Variable length STRING_LAU	

0x1F101: PGN 127233 - Man Overboard Notification

The MOB PGN is intended to provide notification from a MOB monitoring system. The included position information may be that of the vessel or the MOB device itself as identified in field 'X', position source. Additional information may include the current state of the MOB device, time of activation, and MOB device battery status. This PGN may be used to set a MOB waypoint, or to initiate an alert process. This PGN may be used to command or register a MOB device emitter Ids or other applicable fields in the message with an MOB System



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or other equipment. If the fields in this PGN are configured over the network, the Command Group Function (PGN 126208) shall be used. Queries for this PGN shall be requested using either the ISO Request (PGN 059904) or the NMEA Request Group Function (PGN 126208). A device receiving an ISO (PGN 059904) for this PGN (127233), shall respond by providing as many of these PGNs (127233) as necessary for every MOB Emitter ID that has associated data fields. If a Request Group Function (PGN 126208) requesting this PGN (127233) is received, the receiving device shall respond in the following manner: * If no requested fields have been included with the Request Group Function then the response is to return one or more PGNs, just like responding to the ISO Request (PGN 055904) described above. * If the Request Group Function (PGN 126208) includes the MOB Emitter ID field or MOB Status field, then the response shall be filtered by these fields contained within this request resulting in one or more PGN (127233) responses. If the MOB Emitter ID requested is not considered a valid MOB Emitter ID by the receiving device, then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of '0x3 = Access denied.' And the requested MOB Emitter ID field parameter error code (Field 6) of '0x3 = Requested or command parameter out-of-range;'. The Default update rate of this PGN is autonomous, as it is dependant upon notification rates of MOB devices.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 35 bytes long and contains 18 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	MOB Emitter ID	Identifier for each MOB emitter, unique to the vessel	0 .. 4294967292	32 bits unsigned NUMBER	
3	Man Overboard Status		0 .. 6	3 bits lookup MOB_STATUS	
4	Reserved			5 bits RESERVED	
5	Activation Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
6	Position Source		0 .. 6	3 bits lookup MOB_POSITION_SOURCE	
7	Reserved			5 bits RESERVED	
8	Position Date		d 0 .. 65532	16 bits unsigned DATE	
9	Position Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
10	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
11	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
12	COG Reference		0 .. 2	2 bits lookup DIRECTION_REFERENCE	
13	Reserved			6 bits RESERVED	
14	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
15	SOG		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
16	MMSI of vessel of origin		2000000 .. 999999999	32 bits unsigned MMSI	true
17	MOB Emitter Battery Low Status		0 .. 6	3 bits lookup LOW_BATTERY	
18	Reserved			5 bits RESERVED	

0x1F105: PGN 127237 - Heading/Track control

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 21 bytes long and contains 18 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 250 milliseconds.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Rudder Limit Exceeded		0..2	2 bits lookup YES NO	
2	Off-Heading Limit Exceeded		0..2	2 bits lookup YES NO	
3	Off-Track Limit Exceeded		0..2	2 bits lookup YES NO	
4	Override		0..2	2 bits lookup YES NO	
5	Steering Mode		0..6	3 bits lookup STEERING MODE	
6	Turn Mode		0..6	3 bits lookup TURN MODE	
7	Heading Reference		0..2	2 bits lookup DIRECTION REFERENCE	
8	Reserved			5 bits RESERVED	
9	Commanded Rudder Direction		0..6	3 bits lookup DIRECTION RUDDER	
10	Commanded Rudder Angle		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
11	Heading-To-Steer (Course)		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
12	Track		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
13	Rudder Limit		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
14	Off-Heading Limit		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
15	Radius of Turn Order		m -32767 .. 32764	16 bits signed NUMBER	
16	Rate of Turn Order		3.125e-05 rad/s -1.02396875 .. 1.023875	16 bits signed NUMBER	
17	Off-Track Limit		m -32767 .. 32764	16 bits signed NUMBER	
18	Vessel Heading		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	

0x1F10D: PGN 127245 - Rudder

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0..252	8 bits unsigned NUMBER	true
2	Direction Order		0..6	3 bits lookup DIRECTION RUDDER	
3	Reserved			5 bits RESERVED	
4	Angle Order		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
5	Position		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
6	Reserved			16 bits RESERVED	

0x1F112: PGN 127250 - Vessel Heading

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0..252	8 bits unsigned NUMBER	
2	Heading		0.0001 rad	16 bits unsigned NUMBER	



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			0 .. 6.2831852		
3	Deviation		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
4	Variation		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
5	Reference		0 .. 2	2 bits lookup DIRECTION REFERENCE	
6	Reserved			6 bits RESERVED	

0x1F113: PGN 127251 - Rate of Turn

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 3 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Rate		3.125e-08 rad/s -67.10886396875 .. 67.108863875	32 bits signed NUMBER	
3	Reserved			24 bits RESERVED	

0x1F114: PGN 127252 - Heave

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 3 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Heave		0.01 m -327.67 .. 327.64	16 bits signed NUMBER	
3	Reserved			40 bits RESERVED	

0x1F119: PGN 127257 - Attitude

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Yaw		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
3	Pitch		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
4	Roll		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
5	Reserved			8 bits RESERVED	

0x1F11A: PGN 127258 - Magnetic Variation

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 7. It is normally transmitted every 1000 milliseconds.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Source		0 .. 13	4 bits lookup MAGNETIC VARIATION	true
3	Reserved			4 bits RESERVED	
4	Age of service		d 0 .. 65532	16 bits unsigned DATE	
5	Variation		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
6	Reserved			16 bits RESERVED	

0x1F200: PGN 127488 - Engine Parameters, Rapid Update

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits lookup ENGINE INSTANCE	true
2	Speed		0.25 rpm 0 .. 16383	16 bits unsigned NUMBER	
3	Boost Pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
4	Tilt/Trim		% -127 .. 124	8 bits signed NUMBER	
5	Reserved			16 bits RESERVED	

0x1F201: PGN 127489 - Engine Parameters, Dynamic

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 26 bytes long and contains 14 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits lookup ENGINE INSTANCE	true
2	Oil pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
3	Oil temperature		0.1 K 0 .. 6553.2	16 bits unsigned NUMBER	
4	Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
5	Alternator Potential		0.01 V 327.67 .. 327.64	16 bits signed NUMBER	
6	Fuel Rate		0.1 L/h 3276.7 .. 3276.4	16 bits signed NUMBER	
7	Total Engine hours		s 0 .. 4294967292	32 bits unsigned DURATION	
8	Coolant Pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
9	Fuel Pressure		1000 Pa 0 .. 65532000	16 bits unsigned NUMBER	
10	Reserved			8 bits RESERVED	
11	Discrete Status 1		0 .. 65535	16 bits bitfield ENGINE STATUS 1	
12	Discrete Status 2		0 .. 65535	16 bits bitfield ENGINE STATUS 2	
13	Engine Load		% -127 .. 124	8 bits signed NUMBER	



14	Engine Torque		% -127 .. 124	8 bits signed NUMBER	
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0x1F202: PGN 127490 - Electric Drive Status, Dynamic

This PGN is used to report status of Electric Drive Status control and can be used with Command Group Function (PGN Electric propulsion motor status) to command equipment.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Inverter/Motor Identifier		0 .. 252	8 bits unsigned NUMBER	true
2	Operating Mode		0 .. 13	4 bits unsigned NUMBER	
3	Reserved			4 bits RESERVED	
4	Motor Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
5	Inverter Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Coolant Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
7	Gear Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
8	Shaft Torque		0 .. 65532	16 bits unsigned NUMBER	

0x1F203: PGN 127491 - Electric Energy Storage Status, Dynamic

This PGN is used to provide electric propulsion motor status and relevant data.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 15 bytes long and contains 10 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Energy Storage Identifier		0 .. 252	8 bits unsigned NUMBER	true
2	State of Charge		% 0 .. 252	8 bits unsigned NUMBER	
3	Time Remaining	Time remaining at current rate of discharge	60 s 0 .. 3931920	16 bits unsigned DURATION	
4	Highest Cell Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
5	Lowest Cell Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Average Cell Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
7	Max Discharge Current		0.1 A	16 bits signed NUMBER	



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8	Max Charge Current		0.1 A -3276.7 .. 3276.4	16 bits signed NUMBER	
9	Cooling System Status		0 .. 13	4 bits unsigned NUMBER	
10	Heating System Status		0 .. 13	4 bits unsigned NUMBER	

0x1F205: PGN 127493 - Transmission Parameters, Dynamic

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits lookup ENGINE_INSTANCE	true
2	Transmission Gear		0 .. 2	2 bits lookup GEAR_STATUS	
3	Reserved			6 bits RESERVED	
4	Oil pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
5	Oil temperature		0.1 K 0 .. 6553.2	16 bits unsigned NUMBER	
6	Discrete Status 1		0 .. 252	8 bits unsigned NUMBER	
7	Reserved			8 bits RESERVED	

0x1F206: PGN 127494 - Electric Drive Information

This PGN is used to provide information about electric motor specifications and ratings.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 28 bytes long and contains 13 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Inverter/Motor Identifier		0 .. 252	8 bits unsigned NUMBER	true
2	Motor Type		0 .. 13	4 bits unsigned NUMBER	
3	Reserved			4 bits RESERVED	
4	Motor Voltage Rating		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
5	Maximum Continuous Motor Power		W 0 .. 4294967292	32 bits unsigned NUMBER	
6	Maximum Boost Motor Power		W 0 .. 4294967292	32 bits unsigned NUMBER	
7	Maximum Motor Temperature Rating		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
8	Rated Motor Speed		0.25 rpm 0 .. 16383	16 bits unsigned NUMBER	
9	Maximum Controller Temperature Rating		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
10	Motor Shaft Torque Rating		0 .. 65532	16 bits unsigned NUMBER	
11	Motor DC-Voltage Derating Threshold		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	



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12	Motor DC-Voltage Cut Off Threshold		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
13	Drive/Motor Hours		S 0 .. 4294967292	32 bits unsigned DURATION	

0x1F207: PGN 127495 - Electric Energy Storage Information

This PGN is used to provide the status on power storage sources such as batteries. This PGN is new in v3.0 and has not been observed yet; field lengths and precisions are guesses.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 21 bytes long and contains 16 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Energy Storage Identifier		0 .. 252	8 bits unsigned NUMBER	true
2	Motor Type		0 .. 13	4 bits unsigned NUMBER	
3	Reserved			4 bits RESERVED	
4	Storage Chemistry/Conversion		0 .. 252	8 bits unsigned NUMBER	
5	Maximum Temperature Derating		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Maximum Temperature Shut Off		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
7	Minimum Temperature Derating		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
8	Minimum Temperature Shut Off		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
9	Usable Battery Energy		kWh 0 .. 4294967292	32 bits unsigned NUMBER	
10	State of Health		0 .. 252	8 bits unsigned NUMBER	
11	Battery Cycle Counter		0 .. 65532	16 bits unsigned NUMBER	
12	Battery Full Status		0 .. 2	2 bits unsigned NUMBER	
13	Battery Empty Status		0 .. 2	2 bits unsigned NUMBER	
14	Reserved			4 bits RESERVED	
15	Maximum Charge (SOC)		0 .. 252	8 bits unsigned NUMBER	
16	Minimum Charge (SOC)		0 .. 252	8 bits unsigned NUMBER	

0x1F208: PGN 127496 - Trip Parameters, Vessel

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 14 bytes long and contains 4 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Time to Empty		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	
2	Distance to Empty		0.01 m 0 .. 42949672.92	32 bits unsigned NUMBER	
3	Estimated Fuel Remaining		L 0 .. 65532	16 bits unsigned NUMBER	
4	Trip Run Time		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	



0x1F209: PGN 127497 - Trip Parameters, Engine

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 9 bytes long and contains 5 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits lookup ENGINE_INSTANCE	true
2	Trip Fuel Used		L 0 .. 65532	16 bits unsigned NUMBER	
3	Fuel Rate, Average		0.1 L/h -3276.7 .. 3276.4	16 bits signed NUMBER	
4	Fuel Rate, Economy		0.1 L/h -3276.7 .. 3276.4	16 bits signed NUMBER	
5	Instantaneous Fuel Economy		0.1 L/h -3276.7 .. 3276.4	16 bits signed NUMBER	

0x1F20A: PGN 127498 - Engine Parameters, Static

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 3 bytes long and contains 4 fields. The default priority (as observed in the field) is 5. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits lookup ENGINE_INSTANCE	true
2	Rated Engine Speed		0.25 rpm 0 .. 16383	16 bits unsigned NUMBER	
3	VIN			Variable length STRING_LAU	
4	Software ID			Variable length STRING_LAU	

0x1F20C: PGN 127500 - Load Controller Connection State/Control

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Sequence ID		0 .. 252	8 bits unsigned NUMBER	
2	Connection ID		0 .. 252	8 bits unsigned NUMBER	
3	State		0 .. 252	8 bits unsigned NUMBER	
4	Status		0 .. 252	8 bits unsigned NUMBER	
5	Operational Status & Control		0 .. 252	8 bits unsigned NUMBER	
6	PWM Duty Cycle		0 .. 252	8 bits unsigned NUMBER	
7	TimeON		0 .. 252	8 bits unsigned NUMBER	
8	TimeOFF		0 .. 252	8 bits unsigned NUMBER	

Source: <https://github.com/canboat/canboat/issues/366>

0x1F20D: PGN 127501 - Binary Switch Bank Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 29 fields. The default priority (as observed in the field) is 3.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0..252	8 bits unsigned NUMBER	true
2	Indicator1		0..2	2 bits lookup OFF ON	
3	Indicator2		0..2	2 bits lookup OFF ON	
4	Indicator3		0..2	2 bits lookup OFF ON	
5	Indicator4		0..2	2 bits lookup OFF ON	
6	Indicator5		0..2	2 bits lookup OFF ON	
7	Indicator6		0..2	2 bits lookup OFF ON	
8	Indicator7		0..2	2 bits lookup OFF ON	
9	Indicator8		0..2	2 bits lookup OFF ON	
10	Indicator9		0..2	2 bits lookup OFF ON	
11	Indicator10		0..2	2 bits lookup OFF ON	
12	Indicator11		0..2	2 bits lookup OFF ON	
13	Indicator12		0..2	2 bits lookup OFF ON	
14	Indicator13		0..2	2 bits lookup OFF ON	
15	Indicator14		0..2	2 bits lookup OFF ON	
16	Indicator15		0..2	2 bits lookup OFF ON	
17	Indicator16		0..2	2 bits lookup OFF ON	
18	Indicator17		0..2	2 bits lookup OFF ON	
19	Indicator18		0..2	2 bits lookup OFF ON	
20	Indicator19		0..2	2 bits lookup OFF ON	
21	Indicator20		0..2	2 bits lookup OFF ON	
22	Indicator21		0..2	2 bits lookup OFF ON	
23	Indicator22		0..2	2 bits lookup OFF ON	
24	Indicator23		0..2	2 bits lookup OFF ON	
25	Indicator24		0..2	2 bits lookup OFF ON	
26	Indicator25		0..2	2 bits lookup OFF ON	
27	Indicator26		0..2	2 bits lookup OFF ON	
28	Indicator27		0..2	2 bits lookup OFF ON	
29	Indicator28		0..2	2 bits lookup OFF ON	

0x1F20E: PGN 127502 - Switch Bank Control

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 29 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0..252	8 bits unsigned NUMBER	true
2	Switch1		0..3	2 bits lookup OFF ON CONTROL	
3	Switch2		0..3	2 bits lookup OFF ON CONTROL	
4	Switch3		0..3	2 bits lookup OFF ON CONTROL	
5	Switch4		0..3	2 bits lookup OFF ON CONTROL	
6	Switch5		0..3	2 bits lookup OFF ON CONTROL	
7	Switch6		0..3	2 bits lookup OFF ON CONTROL	



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9	Switch8		0..3	2 bits lookup OFF ON CONTROL	
10	Switch9		0..3	2 bits lookup OFF ON CONTROL	
11	Switch10		0..3	2 bits lookup OFF ON CONTROL	
12	Switch11		0..3	2 bits lookup OFF ON CONTROL	
13	Switch12		0..3	2 bits lookup OFF ON CONTROL	
14	Switch13		0..3	2 bits lookup OFF ON CONTROL	
15	Switch14		0..3	2 bits lookup OFF ON CONTROL	
16	Switch15		0..3	2 bits lookup OFF ON CONTROL	
17	Switch16		0..3	2 bits lookup OFF ON CONTROL	
18	Switch17		0..3	2 bits lookup OFF ON CONTROL	
19	Switch18		0..3	2 bits lookup OFF ON CONTROL	
20	Switch19		0..3	2 bits lookup OFF ON CONTROL	
21	Switch20		0..3	2 bits lookup OFF ON CONTROL	
22	Switch21		0..3	2 bits lookup OFF ON CONTROL	
23	Switch22		0..3	2 bits lookup OFF ON CONTROL	
24	Switch23		0..3	2 bits lookup OFF ON CONTROL	
25	Switch24		0..3	2 bits lookup OFF ON CONTROL	
26	Switch25		0..3	2 bits lookup OFF ON CONTROL	
27	Switch26		0..3	2 bits lookup OFF ON CONTROL	
28	Switch27		0..3	2 bits lookup OFF ON CONTROL	
29	Switch28		0..3	2 bits lookup OFF ON CONTROL	

0x1F20F: PGN 127503 - AC Input Status

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 2 bytes long and contains 12 fields. The 10 fields starting at field 3 (with name "Line") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "Number of Lines".) The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0..252	8 bits unsigned NUMBER	true
2	Number of Lines		0..252	8 bits unsigned NUMBER	
<i>3 Set 1</i>	Line		0..2	2 bits lookup AC LINE	
<i>4 Set 1</i>	Acceptability		0..3	2 bits lookup ACCEPTABILITY	
<i>5 Set 1</i>	Reserved			4 bits RESERVED	
<i>6 Set 1</i>	Voltage		0.01 V 0..655.32	16 bits unsigned NUMBER	
<i>7 Set 1</i>	Current		0.1 A 0..6553.2	16 bits unsigned NUMBER	
<i>8 Set 1</i>	Frequency		0.01 Hz 0..655.32	16 bits unsigned NUMBER	
<i>9 Set 1</i>	Breaker Size		0.1 A 0..6553.2	16 bits unsigned NUMBER	



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10 Set 1	Real Power		<u>W</u> 0 .. 4294967292	32 bits unsigned NUMBER	
11 Set 1	Reactive Power		<u>VAR</u> 0 .. 4294967292	32 bits unsigned NUMBER	
12 Set 1	Power factor		0.01 Cos Phi -1 .. 1	8 bits signed NUMBER	

Source: <https://web.archive.org/web/20161130030724/http://www.nmea.org/Assets/nmea-2000-corrigendum-1-2010-1.pdf>

0x1F210: PGN 127504 - AC Output Status

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 2 bytes long and contains 12 fields. The 10 fields starting at field 3 (with name "Line") form repeating set 1. The set is repeated *n* times, where *n* is determined by the value of field 2 (with name "Number of Lines".) The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Number of Lines		0 .. 252	8 bits unsigned NUMBER	
3 Set 1	Line		0 .. 2	2 bits lookup LINE	
4 Set 1	Waveform		0 .. 6	3 bits lookup WAVEFORM	
5 Set 1	Reserved			3 bits RESERVED	
6 Set 1	Voltage		0.01 <u>V</u> 0 .. 655.32	16 bits unsigned NUMBER	
7 Set 1	Current		0.1 <u>A</u> 0 .. 6553.2	16 bits unsigned NUMBER	
8 Set 1	Frequency		0.01 <u>Hz</u> 0 .. 655.32	16 bits unsigned NUMBER	
9 Set 1	Breaker Size		0.1 <u>A</u> 0 .. 6553.2	16 bits unsigned NUMBER	
10 Set 1	Real Power		<u>W</u> 0 .. 4294967292	32 bits unsigned NUMBER	
11 Set 1	Reactive Power		<u>VAR</u> 0 .. 4294967292	32 bits unsigned NUMBER	
12 Set 1	Power factor		0.01 Cos Phi -1 .. 1	8 bits signed NUMBER	

Source: <https://web.archive.org/web/20161130030724/http://www.nmea.org/Assets/nmea-2000-corrigendum-1-2010-1.pdf>

0x1F211: PGN 127505 - Fluid Level

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 2500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 13	4 bits unsigned NUMBER	true
2	Type		0 .. 13	4 bits lookup TANK TYPE	
3	Level		0.004 % -131.068 .. 131.056	16 bits signed NUMBER	
4	Capacity		0.1 <u>L</u> 0 .. 429496729.2	32 bits unsigned NUMBER	



5	Reserved		8 bits RESERVED	
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Source: https://web.archive.org/web/20060511065306/http://www.maretron.com/support/manuals/TLA100UM_1.2.pdf

0x1F212: PGN 127506 - DC Detailed Status

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 11 bytes long and contains 8 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Instance		0 .. 252	8 bits unsigned NUMBER	true
3	DC Type		0 .. 252	8 bits lookup DC SOURCE	true
4	State of Charge		% 0 .. 252	8 bits unsigned NUMBER	
5	State of Health		% 0 .. 252	8 bits unsigned NUMBER	
6	Time Remaining	Time remaining at current rate of discharge	60 S 0 .. 3931920	16 bits unsigned DURATION	
7	Ripple Voltage		0.001 V 0 .. 65.532	16 bits unsigned NUMBER	
8	Remaining capacity		Ah 0 .. 65532	16 bits unsigned NUMBER	

0x1F213: PGN 127507 - Charger Status

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 6 bytes long and contains 8 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Battery Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Operating State		0 .. 13	4 bits lookup CHARGER STATE	
4	Charge Mode		0 .. 13	4 bits lookup CHARGER MODE	
5	Enabled		0 .. 2	2 bits lookup OFF ON	
6	Equalization Pending		0 .. 2	2 bits lookup OFF ON	
7	Reserved			4 bits RESERVED	
8	Equalization Time Remaining		60 S 0 .. 3931920	16 bits unsigned DURATION	

0x1F214: PGN 127508 - Battery Status

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Voltage		0.01 V -327.67 .. 327.64	16 bits signed NUMBER	
3	Current		0.1 A -3276.7 .. 3276.4	16 bits signed NUMBER	
4	Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	

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0x1F215: PGN 127509 - Inverter Status

The NMEA wrote in the link in the URL that this PGN is obsolete and superceded by PGN 127751, but that PGN reference is obviously incorrect. They probably meant PGN 127511. The other interesting thing is that this PGN is only four bytes long but still referenced as a Fast PGN, which matches various sources; see github issue #428.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 4 bytes long and contains 6 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	AC Instance		0 .. 252	8 bits unsigned NUMBER	true
3	DC Instance		0 .. 252	8 bits unsigned NUMBER	true
4	Operating State		0 .. 13	4 bits lookup INVERTER STATE	
5	Inverter Enable		0 .. 2	2 bits lookup OFF ON	
6	Reserved			2 bits RESERVED	

Source: <https://web.archive.org/web/20140913025729/https://www.nmea.org/Assets/20140102%20nmea-2000-127509%20pgn%20corrigendum.pdf>

0x1F216: PGN 127510 - Charger Configuration Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 8 bytes long and contains 11 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Battery Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Charger Enable/Disable		0 .. 2	2 bits lookup OFF ON	
4	Reserved			6 bits RESERVED	
5	Charge Current Limit		% 0 .. 252	8 bits unsigned NUMBER	
6	Charging Algorithm		0 .. 13	4 bits lookup CHARGING ALGORITHM	
7	Charger Mode		0 .. 13	4 bits lookup CHARGER MODE	
8	Estimated Temperature	If there is no battery temperature sensor the charger will use this field to steer the charging algorithm	0 .. 13	4 bits lookup DEVICE TEMP STATE	
9	Equalize One Time Enable/Disable		0 .. 2	2 bits lookup OFF ON	
10	Over Charge Enable/Disable		0 .. 2	2 bits lookup OFF ON	
11	Equalize Time		60 s 0 .. 3931920	16 bits unsigned DURATION	

0x1F217: PGN 127511 - Inverter Configuration Status



This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	AC Instance		0 .. 252	8 bits unsigned NUMBER	true
3	DC Instance		0 .. 252	8 bits unsigned NUMBER	true
4	Inverter Enable/Disable		0 .. 2	2 bits lookup OFF_ON	
5	Inverter Mode		0 .. 13	4 bits lookup INVERTER_MODE	
6	Load Sense Enable/Disable		0 .. 2	2 bits lookup OFF_ON	
7	Load Sense Power Threshold		W 0 .. 65532	16 bits unsigned NUMBER	
8	Load Sense Interval		0.01 s 0 .. 655.32	16 bits unsigned DURATION	

0x1F218: PGN 127512 - AGS Configuration Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 3 bytes long and contains 4 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Generator Instance		0 .. 252	8 bits unsigned NUMBER	true
3	AGS Mode		0 .. 13	4 bits lookup AGS_MODE	
4	Reserved			4 bits RESERVED	

0x1F219: PGN 127513 - Battery Configuration Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 8 bytes long and contains 10 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Battery Type		0 .. 13	4 bits lookup BATTERY_TYPE	
3	Supports Equalization		0 .. 2	2 bits lookup YES_NO	
4	Reserved			2 bits RESERVED	
5	Nominal Voltage		0 .. 13	4 bits lookup BATTERY_VOLTAGE	
6	Chemistry		0 .. 13	4 bits lookup BATTERY_CHEMISTRY	
7	Capacity		Ah 0 .. 65532	16 bits unsigned NUMBER	
8	Temperature Coefficient		% -127 .. 124	8 bits signed NUMBER	
9	Peukert Exponent		0.002 1 .. 1.5	8 bits unsigned NUMBER stored with offset 1	
10	Charge Efficiency Factor		%	8 bits signed NUMBER	



0x1F21A: PGN 127514 - AGS Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 5 bytes long and contains 6 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Instance		0 .. 252	8 bits unsigned NUMBER	true
2	Generator Instance		0 .. 252	8 bits unsigned NUMBER	true
3	AGS Operating State		0 .. 13	4 bits lookup AGS_OPERATING_STATE	
4	Generator State		0 .. 13	4 bits lookup AGS_GENERATING_STATE	
5	Generator On Reason		0 .. 252	8 bits lookup AGS_ON_REASON	
6	Generator Off Reason		0 .. 252	8 bits lookup AGS_OFF_REASON	

0x1F300: PGN 127744 - AC Power / Current - Phase A

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	AC RMS Current		0.1 A 0 .. 6553.2	16 bits unsigned NUMBER	
4	Power		W -2147483647 .. 2147483644	32 bits signed NUMBER	

0x1F301: PGN 127745 - AC Power / Current - Phase B

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	AC RMS Current		0.1 A 0 .. 6553.2	16 bits unsigned NUMBER	
4	Power		W -2147483647 .. 2147483644	32 bits signed NUMBER	

0x1F302: PGN 127746 - AC Power / Current - Phase C

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	

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2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	AC RMS Current		0.1 A 0 .. 6553.2	16 bits unsigned NUMBER	
4	Power		W -2147483647 .. 2147483644	32 bits signed NUMBER	

0x1F303: PGN 127747 - AC Voltage / Frequency - Phase A

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	AC Voltage Line to Neutral		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
4	AC Voltage Line to Line		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
5	Frequency		0.1 Hz 0 .. 6553.2	16 bits unsigned NUMBER	

0x1F304: PGN 127748 - AC Voltage / Frequency - Phase B

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	AC Voltage Line to Neutral		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
4	AC Voltage Line to Line		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
5	Frequency		0.1 Hz 0 .. 6553.2	16 bits unsigned NUMBER	

0x1F305: PGN 127749 - AC Voltage / Frequency - Phase C

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	AC Voltage Line to Neutral		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
4	AC Voltage Line to Line		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
5	Frequency		0.1 Hz 0 .. 6553.2	16 bits unsigned NUMBER	

0x1F306: PGN 127750 - Converter Status



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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID			8 bits BINARY	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	Operating State		0 .. 252	8 bits lookup CONVERTER STATE	
4	Temperature State		0 .. 2	2 bits lookup GOOD WARNING ERROR	
5	Overload State		0 .. 2	2 bits lookup GOOD WARNING ERROR	
6	Low DC Voltage State		0 .. 2	2 bits lookup GOOD WARNING ERROR	
7	Ripple State		0 .. 2	2 bits lookup GOOD WARNING ERROR	
8	Reserved			32 bits RESERVED	

0x1F307: PGN 127751 - DC Voltage/Current

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID			8 bits BINARY	
2	Connection Number		0 .. 252	8 bits unsigned NUMBER	
3	DC Voltage		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
4	DC Current		0.01 A -83886.07 .. 83886.04	24 bits signed NUMBER	
5	Reserved			8 bits RESERVED	

0x1F400: PGN 128000 - Leeway Angle

This PGN provides the Nautical Leeway Angle. Nautical leeway angle is defined as the angle between the direction a vessel is heading (pointing) and the direction it is actually travelling (tracking thru the water). It is commonly provided by dual-axis speed sensors.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Leeway Angle		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
3	Reserved			40 bits RESERVED	

Source:

<https://web.archive.org/web/20170609074636/https://www.nmea.org/Assets/20170204%20nmea%202000%20leeway%20pgn%20final.pdf>

0x1F401: PGN 128001 - Vessel Acceleration

The Vessel Acceleration PGN transmits the acceleration of the vessel in all three axes, ahead/astern, port/starboard, and up/down.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect



- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

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This single-frame PGN is 8 bytes long and contains 5 fields.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Longitudinal Acceleration		-32767 .. 32764	16 bits signed NUMBER	
3	Transverse Acceleration		-32767 .. 32764	16 bits signed NUMBER	
4	Vertical Acceleration		-32767 .. 32764	16 bits signed NUMBER	
5	Reserved			8 bits RESERVED	

0x1F402: PGN 128002 - Electric Drive Status, Rapid Update

This PGN is used to provide the Electric Propulsion Drive System Status.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Inverter/Motor Controller		0 .. 252	8 bits unsigned NUMBER	
2	Active Motor Mode		0 .. 2	2 bits unsigned NUMBER	
3	Brake Mode		0 .. 2	2 bits unsigned NUMBER	
4	Reserved			4 bits RESERVED	
5	Rotational Shaft Speed		0.25 rpm 0 .. 16383	16 bits unsigned NUMBER	
6	Motor DC Voltage		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
7	Motor DC Current		0.1 A -3276.7 .. 3276.4	16 bits signed NUMBER	

0x1F403: PGN 128003 - Electric Energy Storage Status, Rapid Update

Electric Energy Storage Status message provides important energy storage information global at a rapid update rate.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Energy Storage Identifier		0 .. 252	8 bits unsigned NUMBER	
2	Battery Status		0 .. 2	2 bits unsigned NUMBER	
3	Isolation Status		0 .. 2	2 bits unsigned NUMBER	
4	Battery Error		0 .. 13	4 bits unsigned NUMBER	



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5	Battery Voltage		0.1 V 0 .. 6553.2	16 bits unsigned NUMBER	
6	Battery Current		0.1 A -3276.7 .. 3276.4	16 bits signed NUMBER	
7	Reserved			16 bits RESERVED	

0x1F406: PGN 128006 - Thruster Control Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Identifier		0 .. 252	8 bits unsigned NUMBER	
3	Direction Control		0 .. 13	4 bits lookup THRUSTER DIRECTION CONTROL	
4	Power Enabled		0 .. 2	2 bits lookup OFF ON	
5	Retract Control		0 .. 2	2 bits lookup THRUSTER RETRACT CONTROL	
6	Speed Control		% 0 .. 252	8 bits unsigned NUMBER	
7	Control Events		0 .. 255	8 bits bitfield THRUSTER CONTROL EVENTS	
8	Command Timeout		0.005 s 0 .. 1.26	8 bits unsigned DURATION	
9	Azimuth Control		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	

0x1F407: PGN 128007 - Thruster Information

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Identifier		0 .. 252	8 bits unsigned NUMBER	
2	Motor Type		0 .. 13	4 bits lookup THRUSTER MOTOR TYPE	
3	Reserved			4 bits RESERVED	
4	Power Rating		W 0 .. 65532	16 bits unsigned NUMBER	
5	Maximum Temperature Rating		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Maximum Rotational Speed		0.25 rpm 0 .. 16383	16 bits unsigned NUMBER	

0x1F408: PGN 128008 - Thruster Motor Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Identifier		0 .. 252	8 bits unsigned NUMBER	
3	Motor Events		0 .. 255	8 bits bitfield THRUSTER MOTOR EVENTS	

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4	Current		A 0 .. 252	8 bits unsigned NUMBER	
5	Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Operating Time		60 s 0 .. 3931920	16 bits unsigned DURATION	

0x1F503: PGN 128259 - Speed

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Speed Water Referenced		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
3	Speed Ground Referenced		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
4	Speed Water Referenced Type		0 .. 252	8 bits lookup WATER REFERENCE	
5	Speed Direction		0 .. 13	4 bits unsigned NUMBER	
6	Reserved			12 bits RESERVED	

0x1F50B: PGN 128267 - Water Depth

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Depth	Depth below transducer	0.01 m 0 .. 42949672.92	32 bits unsigned NUMBER	
3	Offset	Distance between transducer and surface (positive) or keel (negative)	0.001 m -32.767 .. 32.764	16 bits signed NUMBER	
4	Range	Max measurement range	10 m 0 .. 2520	8 bits unsigned NUMBER	

0x1F513: PGN 128275 - Distance Log

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 14 bytes long and contains 4 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Date		d 0 .. 65532	16 bits unsigned DATE	
2	Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
3	Log	Total cumulative distance	m 0 .. 4294967292	32 bits unsigned NUMBER	
4	Trip Log	Distance since last reset	m 0 .. 4294967292	32 bits unsigned NUMBER	

0x1F608: PGN 128520 - Tracked Target Data



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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 27 bytes long and contains 17 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Target ID #	Number of route, waypoint, event, mark, etc.	0 .. 65532	16 bits unsigned NUMBER	
3	Track Status		0 .. 3	2 bits lookup TRACKING	
4	Reported Target		0 .. 1	1 bits lookup YES NO	
5	Target Acquisition		0 .. 1	1 bits lookup TARGET ACQUISITION	
6	Bearing Reference		0 .. 2	2 bits lookup DIRECTION REFERENCE	
7	Reserved			2 bits RESERVED	
8	Bearing		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
9	Distance		0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
10	Course		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
11	Speed		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
12	CPA		0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
13	TCPA	negative = time elapsed since event, positive = time to go	0.001 s -2147483.647 .. 2147483.644	32 bits signed DURATION	
14	UTC of Fix	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
15	Name			Variable length STRING LAU	
16	Reference Target		0 .. 2	2 bits lookup YES NO	
17	Reserved			6 bits RESERVED	

0x1F61A: PGN 128538 - Elevator Car Status

This PGN provides the status information of an elevator car. This includes the elevator car id and type, sensors for load and weight limits, smoke detection, door status, motor status, and brake status. Also provided are weight and speed measurements, current and destination deck location, proximity switch status, inertial measurement unit status and Emergency button and buzzer status.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 22 bytes long and contains 32 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	



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2	Elevator Car ID		0..252	8 bits unsigned NUMBER	
3	Elevator Car Usage		0..252	8 bits unsigned NUMBER	
4	Smoke Sensor Status		0..2	2 bits unsigned NUMBER	
5	Limit Switch Sensor Status		0..2	2 bits unsigned NUMBER	
6	Proximity Switch Sensor Status		0..2	2 bits unsigned NUMBER	
7	Inertial Measurement Unit (IMU) Sensor Status		0..2	2 bits unsigned NUMBER	
8	Elevator Load Limit Status		0..2	2 bits unsigned NUMBER	
9	Elevator Load Balance Status		0..2	2 bits unsigned NUMBER	
10	Elevator Load Sensor 1 Status		0..2	2 bits unsigned NUMBER	
11	Elevator Load Sensor 2 Status		0..2	2 bits unsigned NUMBER	
12	Elevator Load Sensor 3 Status		0..2	2 bits unsigned NUMBER	
13	Elevator Load Sensor 4 Status		0..2	2 bits unsigned NUMBER	
14	Reserved			4 bits RESERVED	
15	Elevator Car Motion Status		0..2	2 bits unsigned NUMBER	
16	Elevator Car Door Status		0..2	2 bits unsigned NUMBER	
17	Elevator Car Emergency Button Status		0..2	2 bits unsigned NUMBER	
18	Elevator Car Buzzer Status		0..2	2 bits unsigned NUMBER	
19	Open Door Button Status		0..2	2 bits unsigned NUMBER	
20	Close Door Button Status		0..2	2 bits unsigned NUMBER	
21	Reserved			4 bits RESERVED	
22	Current Deck		0..252	8 bits unsigned NUMBER	
23	Destination Deck		0..252	8 bits unsigned NUMBER	
24	Total Number of Decks		0..252	8 bits unsigned NUMBER	
25	Weight of Load Cell 1		0..65532	16 bits unsigned NUMBER	
26	Weight of Load Cell 2		0..65532	16 bits unsigned NUMBER	
27	Weight of Load Cell 3		0..65532	16 bits unsigned NUMBER	
28	Weight of Load Cell 4		0..65532	16 bits unsigned NUMBER	
29	Speed of Elevator Car		0.01 m/s -327.67 .. 327.64	16 bits signed NUMBER	
30	Elevator Brake Status		0..2	2 bits unsigned NUMBER	
31	Elevator Motor rotation control Status		0..2	2 bits unsigned NUMBER	
32	Reserved			4 bits RESERVED	

0x1F700: PGN 128768 - Elevator Motor Control

This PGN provides the status of an elevator motor controller. Settings of the elevator motor controller may be changed using the NMEA Command Group Function.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0..252	8 bits unsigned NUMBER	



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2	Elevator Car ID		0..252	8 bits unsigned NUMBER	
3	Elevator Car Usage		0..252	8 bits unsigned NUMBER	
4	Motor Acceleration/Deceleration profile selection		0..13	4 bits unsigned NUMBER	
5	Motor Rotational Control Status		0..2	2 bits unsigned NUMBER	
6	Reserved			34 bits RESERVED	

0x1F701: PGN 128769 - Elevator Deck Push Button

Transmit data of Deck controller to Elevator Main controller.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0..252	8 bits unsigned NUMBER	
2	Elevator Call Button ID		0..252	8 bits unsigned NUMBER	
3	Deck Button ID		0..252	8 bits unsigned NUMBER	
4	Elevator Car Usage		0..252	8 bits unsigned NUMBER	
5	Elevator Car Button Selection		0..252	8 bits unsigned NUMBER	
6	Reserved			24 bits RESERVED	

0x1F708: PGN 128776 - Windlass Control Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 14 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0..252	8 bits unsigned NUMBER	
2	Windlass ID		0..252	8 bits unsigned NUMBER	
3	Windlass Direction Control		0..2	2 bits lookup WINDLASS_DIRECTION	
4	Anchor Docking Control		0..2	2 bits lookup OFF_ON	
5	Speed Control Type		0..2	2 bits lookup SPEED_TYPE	
6	Reserved			2 bits RESERVED	
7	Speed Control	0=Off,Single speed:1-100=On,Dual Speed:1-49=Slow/50-100=Fast,Proportional:10-100		8 bits BINARY	
8	Power Enable		0..2	2 bits lookup OFF_ON	
9	Mechanical Lock		0..2	2 bits lookup OFF_ON	
10	Deck and Anchor Wash		0..2	2 bits lookup OFF_ON	
11	Anchor Light		0..2	2 bits lookup OFF_ON	



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12	Command Timeout	If timeout elapses the thruster stops operating and reverts to static mode	0.005 $\frac{s}{0 \dots 1.26}$	8 bits unsigned DURATION	
13	Windlass Control Events		0..15	4 bits bitfield WINDLASS_CONTROL	
14	Reserved			12 bits RESERVED	

Source: <https://www.nmea.org/Assets/20190613%20windlass%20amendment,%20128776,%20128777,%20128778.pdf>

0x1F709: PGN 128777 - Anchor Windlass Operating Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 10 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0..252	8 bits unsigned NUMBER	
2	Windlass ID		0..252	8 bits unsigned NUMBER	
3	Windlass Direction Control		0..2	2 bits lookup WINDLASS_DIRECTION	
4	Windlass Motion Status		0..2	2 bits lookup WINDLASS_MOTION	
5	Rode Type Status		0..2	2 bits lookup RODE_TYPE	
6	Reserved			2 bits RESERVED	
7	Rode Counter Value		0.1 $\frac{m}{0 \dots 6553.2}$	16 bits unsigned NUMBER	
8	Windlass Line Speed		0.01 $\frac{m}{s} \frac{0 \dots 655.32}$	16 bits unsigned NUMBER	
9	Anchor Docking Status		0..2	2 bits lookup DOCKING_STATUS	
10	Windlass Operating Events		0..63	6 bits bitfield WINDLASS_OPERATION	

Source: <https://www.nmea.org/Assets/20190613%20windlass%20amendment,%20128776,%20128777,%20128778.pdf>

0x1F70A: PGN 128778 - Anchor Windlass Monitoring Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0..252	8 bits unsigned NUMBER	
2	Windlass ID		0..252	8 bits unsigned NUMBER	
3	Windlass Monitoring Events		0..255	8 bits bitfield WINDLASS_MONITORING	
4	Controller voltage		0.2 $\frac{V}{0 \dots 50.4}$	8 bits unsigned NUMBER	
5	Motor current		$\frac{A}{0 \dots 252}$	8 bits unsigned NUMBER	
6	Total Motor Time		60 $\frac{s}{0 \dots 3931920}$	16 bits unsigned DURATION	
7	Reserved			8 bits RESERVED	

Source: <https://www.nmea.org/Assets/20190613%20windlass%20amendment,%20128776,%20128777,%20128778.pdf>

0x1F70C: PGN 128780 - Linear Actuator Control/Status

Actuator is a broad description of any device that embodies moving an object between two fixed limits, such as raising or lowering an outboard engine assembly. In the context of this PGN, the word "Device" refers to the object being moved. In the case of multiple



Actuators per controller, the Actuator Identifier field specifies which Actuator the PGN message is intended for, and all following data fields refer only to that Actuator. This PGN supports manufacturer calibrated systems and retrofit systems where it is impractical for the installer to enter the Maximum Travel distance of the device.

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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Actuator Identifier		0 .. 252	8 bits unsigned NUMBER	
2	Commanded Device Position		0 .. 252	8 bits unsigned NUMBER	
3	Device Position		0 .. 252	8 bits unsigned NUMBER	
4	Maximum Device Travel		0 .. 65532	16 bits unsigned NUMBER	
5	Direction of Travel		0 .. 252	8 bits unsigned NUMBER	
6	Reserved			16 bits RESERVED	

0x1F801: PGN 129025 - Position, Rapid Update

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 2 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
2	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	

0x1F802: PGN 129026 - COG & SOG, Rapid Update

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 250 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	COG Reference		0 .. 2	2 bits lookup DIRECTION_REFERENCE	
3	Reserved			6 bits RESERVED	
4	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
5	SOG		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
6	Reserved			16 bits RESERVED	

Source: http://www.maretron.com/support/manuals/GPS100UM_1.2.pdf

0x1F803: PGN 129027 - Position Delta, Rapid Update

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet



This single-frame PGN is 8 bytes long and contains 4 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Time Delta		0.005 s 0 .. 1.26	8 bits unsigned DURATION	
3	Latitude Delta		2.77778e-09 deg -0.0233016861111111 .. 0.0233016777777778	24 bits signed NUMBER	
4	Longitude Delta		2.77778e-09 deg -0.0233016861111111 .. 0.0233016777777778	24 bits signed NUMBER	

0x1F804: PGN 129028 - Altitude Delta, Rapid Update

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Time Delta		0.005 s 0 .. 1.26	8 bits unsigned DURATION	
3	GNSS Quality		0 .. 13	4 bits lookup GNS METHOD	
4	Direction		0 .. 2	2 bits lookup DIRECTION REFERENCE	
5	Reserved			2 bits RESERVED	
6	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
7	Altitude Delta		0.001 m -8388.607 .. 8388.604	24 bits signed NUMBER	

0x1F805: PGN 129029 - GNSS Position Data

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 43 bytes long and contains 18 fields. The 3 fields starting at field 16 (with name "Reference Station Type") form repeating set 1. The set is repeated n times, where n is determined by the value of field 15 (with name "Reference Stations".) The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Date		d 0 .. 65532	16 bits unsigned DATE	
3	Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
4	Latitude		1e-16 deg -90 .. 90	64 bits signed NUMBER	
5	Longitude		1e-16 deg -180 .. 180	64 bits signed NUMBER	
6	Altitude	Altitude referenced to WGS-84	1e-06 m -9223372036854.78 .. 9223372036854.78	64 bits signed NUMBER	
7	GNSS type		0 .. 13	4 bits lookup GNS	
8	Method		0 .. 13	4 bits lookup GNS METHOD	
9	Integrity		0 .. 3	2 bits lookup GNS INTEGRITY	



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10	Reserved			6 bits RESERVED	
11	Number of SVs	Number of satellites used in solution	0 .. 252	8 bits unsigned NUMBER	
12	HDOP	Horizontal dilution of precision	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
13	PDOP	Positional dilution of precision	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
14	Geoidal Separation	Geoidal Separation	0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
15	Reference Stations	Number of reference stations	0 .. 252	8 bits unsigned NUMBER	
16 Set 1	Reference Station Type		0 .. 13	4 bits lookup GNS	
17 Set 1	Reference Station ID		0 .. 4092	12 bits unsigned NUMBER	
18 Set 1	Age of DGNSS Corrections		0.01 s 0 .. 655.32	16 bits unsigned DURATION	

0x1F809: PGN 129033 - Time & Date

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 3 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Date		d 0 .. 65532	16 bits unsigned DATE	
2	Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
3	Local Offset		60 s -1966020 .. 1965840	16 bits signed DURATION	

0x1F80E: PGN 129038 - AIS Class A Position Report

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 28 bytes long and contains 20 fields. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
5	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Position Accuracy		0 .. 1	1 bits lookup POSITION_ACCURACY	
7	RAIM		0 .. 1	1 bits lookup RAIM_FLAG	
8	Time Stamp	0-59 = UTC second when the report was generated	0 .. 63	6 bits lookup TIME_STAMP	
9	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	



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10	SOG		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
11	Communication State	Information used by the TDMA slot allocation algorithm and synchronization information		19 bits BINARY	
12	AIS Transceiver information		0 .. 29	5 bits lookup AIS_TRANSCIVER	
13	Heading	True heading	0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
14	Rate of Turn		3.125e-05 rad/s -1.02396875 .. 1.023875	16 bits signed NUMBER	
15	Nav Status		0 .. 14	4 bits lookup NAV_STATUS	
16	Special Maneuver Indicator		0 .. 3	2 bits lookup AIS_SPECIAL_MANEUVER	
17	Reserved			2 bits RESERVED	
18	Spare			3 bits SPARE	
19	Reserved			5 bits RESERVED	
20	Sequence ID		0 .. 252	8 bits unsigned NUMBER	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1F80F: PGN 129039 - AIS Class B Position Report

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 27 bytes long and contains 23 fields. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
5	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Position Accuracy		0 .. 1	1 bits lookup POSITION_ACCURACY	
7	RAIM		0 .. 1	1 bits lookup RAIM_FLAG	
8	Time Stamp		0 .. 63	6 bits lookup TIME_STAMP	
9	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
10	SOG		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
11	Communication State	Information used by the TDMA slot allocation algorithm and synchronization information		19 bits BINARY	
12	AIS Transceiver information		0 .. 29	5 bits lookup AIS_TRANSCIVER	
13	Heading	True heading	0.0001 rad	16 bits unsigned NUMBER	



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			0 .. 6.2831852		
14	Regional Application			8 bits SPARE	
15	Regional Application B			2 bits SPARE	
16	Unit type		0 .. 1	1 bits lookup AIS_TYPE	
17	Integrated Display	Whether the unit can show messages 12 and 14	0 .. 1	1 bits lookup YES_NO	
18	DSC		0 .. 1	1 bits lookup YES_NO	
19	Band		0 .. 1	1 bits lookup AIS_BAND	
20	Can handle Msg 22		0 .. 1	1 bits lookup YES_NO	
21	AIS mode		0 .. 1	1 bits lookup AIS_MODE	
22	AIS communication state		0 .. 1	1 bits lookup AIS_COMMUNICATION_STATE	
23	Reserved			15 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1F810: PGN 129040 - AIS Class B Extended Position Report

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 54 bytes long and contains 27 fields. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
5	Latitude		1e-07 deg 90 .. 90	32 bits signed NUMBER	
6	Position Accuracy		0 .. 1	1 bits lookup POSITION_ACCURACY	
7	RAIM		0 .. 1	1 bits lookup RAIM_FLAG	
8	Time Stamp		0 .. 63	6 bits lookup TIME_STAMP	
9	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
10	SOG		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
11	Regional Application			8 bits SPARE	
12	Regional Application B			4 bits SPARE	
13	Reserved			4 bits RESERVED	
14	Type of ship		0 .. 252	8 bits lookup SHIP_TYPE	
15	True Heading		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
16	Reserved			4 bits RESERVED	
17	GNSS type		0 .. 13	4 bits lookup POSITION_FIX_DEVICE	
18	Length		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
19	Beam		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	



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20	Position reference from Starboard		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
21	Position reference from Bow		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
22	Name			160 bits STRING FIX	
23	DTE		0 .. 1	1 bits lookup AVAILABLE	
24	AIS mode		0 .. 1	1 bits lookup AIS MODE	
25	Spare			4 bits SPARE	
26	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
27	Reserved			5 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1F811: PGN 129041 - AIS Aids to Navigation (AtoN) Report

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 26 bytes long and contains 23 fields. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	User ID		20000000 .. 999999999	32 bits unsigned MMSI	true
4	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
5	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Position Accuracy		0 .. 1	1 bits lookup POSITION ACCURACY	
7	RAIM		0 .. 1	1 bits lookup RAIM FLAG	
8	Time Stamp		0 .. 63	6 bits lookup TIME STAMP	
9	Length/Diameter		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
10	Beam/Diameter		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
11	Position Reference from Starboard Edge		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
12	Position Reference from True North Facing Edge		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
13	AtoN Type		0 .. 31	5 bits lookup ATON TYPE	
14	Off Position Indicator		0 .. 1	1 bits lookup YES NO	
15	Virtual AtoN Flag		0 .. 1	1 bits lookup YES NO	
16	Assigned Mode Flag		0 .. 1	1 bits lookup AIS ASSIGNED MODE	
17	Spare			1 bits SPARE	
18	Position Fixing Device Type		0 .. 13	4 bits lookup POSITION FIX DEVICE	
19	Reserved			3 bits RESERVED	
20	AtoN Status	00000000 = default		8 bits BINARY	



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21	AIS Transceiver information		0 .. 29	5 bits lookup AIS_TRANSCIVER	
22	Reserved			3 bits RESERVED	
23	AtoN Name			Variable length STRING_LAU	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1F814: PGN 129044 - Datum

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 20 bytes long and contains 5 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 10000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Local Datum	defined in IHO Publication S-60, Appendices B and C. First three chars are datum ID as per IHO tables. Fourth char is local datum subdivision code.		32 bits STRING_FIX	
2	Delta Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
3	Delta Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
4	Delta Altitude		0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
5	Reference Datum	defined in IHO Publication S-60, Appendices B and C. First three chars are datum ID as per IHO tables. Fourth char is local datum subdivision code.		32 bits STRING_FIX	

0x1F815: PGN 129045 - User Datum

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 40 bytes long and contains 10 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Delta X	Delta shift in X axis from WGS 84	0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
2	Delta Y	Delta shift in Y axis from WGS 84	0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
3	Delta Z	Delta shift in Z axis from WGS 84	0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
4	Rotation in X	Rotational shift in X axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.	rad -3.1415926 .. 3.1415926	32 bits signed FLOAT	
5	Rotation in Y	Rotational shift in Y axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.	rad -3.1415926 .. 3.1415926	32 bits signed FLOAT	
6	Rotation in Z	Rotational shift in Z axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.	rad -3.1415926 .. 3.1415926	32 bits signed FLOAT	
7	Scale		ppm -3.40282346638529e+38 .. 3.40282346638529e+38	32 bits signed FLOAT	

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8	Ellipsoid Semi-major Axis	Semi-major axis (a) of the User Datum ellipsoid	0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
9	Ellipsoid Flattening Inverse	Flattening (1/f) of the User Datum ellipsoid	-3.40282346638529e+38 .. 3.40282346638529e+38	32 bits signed FLOAT	
10	Datum Name	4 character code from IHO Publication S-60,Appendices B and C. First three chars are datum ID as per IHO tables. Fourth char is local datum subdivision code.		32 bits STRING_FIX	

0x1F903: PGN 129283 - Cross Track Error

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	XTE mode		0 .. 13	4 bits lookup RESIDUAL_MODE	
3	Reserved			2 bits RESERVED	
4	Navigation Terminated		0 .. 2	2 bits lookup YES_NO	
5	XTE		0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
6	Reserved			16 bits RESERVED	

0x1F904: PGN 129284 - Navigation Data

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 34 bytes long and contains 15 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Distance to Waypoint		0.01 m 0 .. 42949672.92	32 bits unsigned NUMBER	
3	Course/Bearing reference		0 .. 2	2 bits lookup DIRECTION_REFERENCE	
4	Perpendicular Crossed		0 .. 2	2 bits lookup YES_NO	
5	Arrival Circle Entered		0 .. 2	2 bits lookup YES_NO	
6	Calculation Type		0 .. 2	2 bits lookup BEARING_MODE	
7	ETA Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
8	ETA Date		d 0 .. 65532	16 bits unsigned DATE	
9	Bearing, Origin to Destination Waypoint		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
10	Bearing, Position to Destination Waypoint		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
11	Origin Waypoint Number		0 .. 4294967292	32 bits unsigned NUMBER	
12	Destination Waypoint Number		0 .. 4294967292	32 bits unsigned NUMBER	
13	Destination Latitude		1e-07 deg	32 bits signed NUMBER	



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			-90 .. 90		
14	Destination Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
15	Waypoint Closing Velocity		0.01 m/s -327.67 .. 327.64	16 bits signed NUMBER	

0x1F905: PGN 129285 - Navigation - Route/WP Information

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 10 bytes long and contains 13 fields. The 4 fields starting at field 10 (with name "WP ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start RPS#		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Database ID		0 .. 65532	16 bits unsigned NUMBER	
4	Route ID		0 .. 65532	16 bits unsigned NUMBER	
5	Navigation direction in route		0 .. 6	3 bits lookup DIRECTION	
6	Supplementary Route/WP data available		0 .. 2	2 bits lookup OFF_ON	
7	Reserved			3 bits RESERVED	
8	Route Name			Variable length STRING_LAU	
9	Reserved			8 bits RESERVED	
10 Set 1	WP ID		0 .. 65532	16 bits unsigned NUMBER	
11 Set 1	WP Name			Variable length STRING_LAU	
12 Set 1	WP Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
13 Set 1	WP Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	

0x1F90B: PGN 129291 - Set & Drift, Rapid Update

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Set Reference		0 .. 2	2 bits lookup DIRECTION_REFERENCE	
3	Reserved			6 bits RESERVED	
4	Set		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
5	Drift		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
6	Reserved			16 bits RESERVED	

0x1F915: PGN 129301 - Navigation - Route / Time to+from Mark

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:



- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 10 bytes long and contains 5 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Time to mark	negative = elapsed since event, positive = time to go	0.001 s -2147483.647 .. 2147483.644	32 bits signed DURATION	
3	Mark Type		0 .. 13	4 bits lookup MARK_TYPE	
4	Reserved			4 bits RESERVED	
5	Mark ID		0 .. 4294967292	32 bits unsigned NUMBER	

0x1F916: PGN 129302 - Bearing and Distance between two Marks

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 17 bytes long and contains 10 fields. The default priority (as observed in the field) is 6.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Bearing Reference		0 .. 2	2 bits lookup DIRECTION_REFERENCE	
3	Calculation Type		0 .. 2	2 bits lookup BEARING_MODE	
4	Reserved			4 bits RESERVED	
5	Bearing, Origin to Destination		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	Distance		0.01 m 0 .. 42949672.92	32 bits unsigned NUMBER	
7	Origin Mark Type		0 .. 13	4 bits lookup MARK_TYPE	
8	Destination Mark Type		0 .. 13	4 bits lookup MARK_TYPE	
9	Origin Mark ID		0 .. 4294967292	32 bits unsigned NUMBER	
10	Destination Mark ID		0 .. 4294967292	32 bits unsigned NUMBER	

0x1FA02: PGN 129538 - GNSS Control Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 16 bytes long and contains 11 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SV Elevation Mask	Will not use SV below this elevation	0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
2	PDOP Mask	Will not report position above this PDOP	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
3	PDOP Switch	Will report 2D position above this PDOP	0.01 -327.67 .. 327.64	16 bits signed NUMBER	

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4	SNR Mask	Will not use SV below this SNR	0.01 dB -327.67 .. 327.64	16 bits signed NUMBER	
5	GNSS Mode (desired)		0 .. 6	3 bits lookup GNSS MODE	
6	DGNSS Mode (desired)		0 .. 6	3 bits lookup DGNSS MODE	
7	Position/Velocity Filter		0 .. 2	2 bits lookup YES_NO	
8	Max Correction Age		0.01 s 0 .. 655.32	16 bits unsigned DURATION	
9	Antenna Altitude for 2D Mode		0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
10	Use Antenna Altitude for 2D Mode		0 .. 2	2 bits lookup YES_NO	
11	Reserved			6 bits RESERVED	

0x1FA03: PGN 129539 - GNSS DOPs

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Desired Mode		0 .. 6	3 bits lookup GNSS MODE	
3	Actual Mode		0 .. 6	3 bits lookup GNSS MODE	
4	Reserved			2 bits RESERVED	
5	HDOP	Horizontal dilution of precision	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
6	VDOP	Vertical dilution of precision	0.01 -327.67 .. 327.64	16 bits signed NUMBER	
7	TDOP	Time dilution of precision	0.01 -327.67 .. 327.64	16 bits signed NUMBER	

0x1FA04: PGN 129540 - GNSS Sats in View

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 3 bytes long and contains 11 fields. The 7 fields starting at field 5 (with name "PRN") form repeating set 1. The set is repeated n times, where n is determined by the value of field 4 (with name "Sats in View".) The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Range Residual Mode		0 .. 2	2 bits lookup RANGE_RESIDUAL_MODE	
3	Reserved			6 bits RESERVED	
4	Sats in View		0 .. 252	8 bits unsigned NUMBER	
5 Set 1	PRN		0 .. 252	8 bits unsigned NUMBER	
6 Set 1	Elevation		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
7 Set 1	Azimuth		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
8 Set 1	SNR		0.01 dB -327.67 .. 327.64	16 bits signed NUMBER	



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9 Set 1	Range residuals		1e-05 m -21474.83647 .. 21474.83644	32 bits signed NUMBER	
10 Set 1	Status		0 .. 13	4 bits lookup SATELLITE STATUS	
11 Set 1	Reserved			4 bits RESERVED	

0x1FA05: PGN 129541 - GPS Almanac Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 26 bytes long and contains 14 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	PRN		0 .. 252	8 bits unsigned NUMBER	
2	GPS Week number		0 .. 65532	16 bits unsigned NUMBER	
3	SV Health Bits			8 bits BINARY	
4	Eccentricity	'e' in table 20-VI in ICD-GPS-200	4.76837e-07 m/m 0 .. 0.0312480926513672	16 bits unsigned NUMBER	
5	Almanac Reference Time	't oa' in table 20-VI in ICD-GPS-200	4096 s 0 .. 1032192	8 bits unsigned NUMBER	
6	Inclination Angle	'delta i' in table 20-VI in ICD-GPS-200	1.90735e-06 semi-circle -0.0624980926513672 .. 0.0624923706054688	16 bits signed NUMBER	
7	Rate of Right Ascension	'OMEGADOT' in table 20-VI in ICD-GPS-200	3.63798e-12 semi-circle/s -1.19205651571974e-07 .. 1.19194737635553e-07	16 bits signed NUMBER	
8	Root of Semi-major Axis	'(A)^0.5' in table 20-VI in ICD-GPS-200	0.000488281 sqrt(m) 0 .. 8191.998046875	24 bits unsigned NUMBER	
9	Argument of Perigee	'(OMEGA)0' in table 20-VI in ICD-GPS-200	1.19209e-07 semi-circle -0.99999988079071 .. 0.999999523162842	24 bits signed NUMBER	
10	Longitude of Ascension Node	'small-omega' in table 20-VI in ICD-GPS-200	1.19209e-07 semi-circle -0.99999988079071 .. 0.999999523162842	24 bits signed NUMBER	
11	Mean Anomaly	'M 0' in table 20-VI in ICD-GPS-200	1.19209e-07 semi-circle -0.99999988079071 .. 0.999999523162842	24 bits signed NUMBER	
12	Clock Parameter 1	'a f0' in table 20-VI in ICD-GPS-200	9.53674e-07 s -0.000975608825683594 .. 0.000972747802734375	11 bits signed NUMBER	
13	Clock Parameter 2	'a f1' in table 20-VI in ICD-GPS-200	3.63798e-12 s/s -3.72165231965482e-09 .. 3.71073838323355e-09	11 bits signed NUMBER	
14	Reserved			2 bits RESERVED	

Source: <https://www.gps.gov/technical/icwg/ICD-GPS-200C.pdf>

0x1FA06: PGN 129542 - GNSS Pseudorange Noise Statistics

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 15 bytes long and contains 8 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	RMS of Position Uncertainty		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
3	STD of Major axis		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
4	STD of Minor axis		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
5	Orientation of Major axis		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	STD of Lat Error		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
7	STD of Lon Error		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
8	STD of Alt Error		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	

0x1FA09: PGN 129545 - GNSS RAIM Output

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 15 bytes long and contains 10 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Integrity flag		0 .. 3	2 bits lookup GNS_INTEGRITY	
3	Reserved			6 bits RESERVED	
4	Latitude expected error		0.01 m -327.67 .. 327.64	16 bits signed NUMBER	
5	Longitude expected error		0.01 m -327.67 .. 327.64	16 bits signed NUMBER	
6	Altitude expected error		0.01 m -327.67 .. 327.64	16 bits signed NUMBER	
7	SV ID of most likely failed sat		0 .. 252	8 bits unsigned NUMBER	
8	Probability of missed detection		0.01 m -327.67 .. 327.64	16 bits signed NUMBER	
9	Estimate of pseudorange bias		0.01 m 327.67 .. 327.64	16 bits signed NUMBER	
10	Std Deviation of bias		0.01 m -327.67 .. 327.64	16 bits signed NUMBER	

0x1FA0A: PGN 129546 - GNSS RAIM Settings

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Radial Position Error Maximum Threshold		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
2	Probability of False Alarm		% -127 .. 124	8 bits signed NUMBER	
3	Probability of Missed Detection		% -127 .. 124	8 bits signed NUMBER	
4	Pseudorange Residual Filtering Time Constant		S 0 .. 65532	16 bits unsigned DURATION	



5	Reserved			16 bits RESERVED	
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0x1FA0B: PGN 129547 - GNSS Pseudorange Error Statistics

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 15 bytes long and contains 8 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	RMS Std Dev of Range Inputs		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
3	Std Dev of Major error ellipse		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
4	Std Dev of Minor error ellipse		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
5	Orientation of error ellipse		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	Std Dev Lat Error		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
7	Std Dev Lon Error		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
8	Std Dev Alt Error		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	

0x1FA0D: PGN 129549 - DGNSS Corrections

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 16 bytes long and contains 11 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Reference Station ID		0 .. 4092	12 bits unsigned NUMBER	
3	Reference Station Type		0 .. 13	4 bits lookup GNS	
4	Time of corrections		0.1 s 0 .. 6553.2	16 bits unsigned DURATION	
5	Station Health		0 .. 13	4 bits lookup STATION HEALTH	
6	Reserved			4 bits RESERVED	
7	Satellite ID		0 .. 252	8 bits unsigned NUMBER	
8	PRC		0.0001 m -214748.3647 .. 214748.3644	32 bits signed NUMBER	
9	RRC		0.0001 m/s -3.2767 .. 3.2764	16 bits signed NUMBER	
10	UDRE		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
11	IOD		0 .. 252	8 bits unsigned NUMBER	

0x1FA0E: PGN 129550 - GNSS Differential Correction Receiver Interface

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 7 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Channel		0 .. 252	8 bits unsigned NUMBER	



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2	Frequency		10 Hz 0 .. 42949672920	32 bits unsigned NUMBER	
3	Serial Interface Bit Rate		0 .. 29	5 bits lookup SERIAL_BIT_RATE	
4	Serial Interface Detection Mode		0 .. 6	3 bits lookup SERIAL_DETECTION_MODE	
5	Differential Source		0 .. 13	4 bits lookup DIFFERENTIAL_SOURCE	
6	Differential Operation Mode		0 .. 13	4 bits lookup DIFFERENTIAL_MODE	
7	Reserved			8 bits RESERVED	

0x1FA0F: PGN 129551 - GNSS Differential Correction Receiver Signal

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 20 bytes long and contains 14 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Channel		0 .. 252	8 bits unsigned NUMBER	
3	Signal Strength	Signal strength in dB relative to 1 uV/m	0.01 dB -21474836.47 .. 21474836.44	32 bits signed NUMBER	
4	Signal SNR		0.01 dB 327.67 .. 327.64	16 bits signed NUMBER	
5	Frequency		10 Hz 0 .. 42949672920	32 bits unsigned NUMBER	
6	Station Type		0 .. 13	4 bits lookup GNS	
7	Reference Station ID		0 .. 4092	12 bits unsigned NUMBER	
8	Differential Signal Bit Rate		0 .. 29	5 bits lookup SERIAL_BIT_RATE	
9	Differential Signal Detection Mode		0 .. 6	3 bits lookup SERIAL_DETECTION_MODE	
10	Used as Correction Source		0 .. 2	2 bits lookup YES_NO	
11	Reserved			2 bits RESERVED	
12	Differential Source		0 .. 13	4 bits lookup DIFFERENTIAL_SOURCE	
13	Time since Last Sat Differential Sync	Age of differential corrections	0.01 s 0 .. 65532	16 bits unsigned DURATION	
14	Satellite Service ID No.		0 .. 65532	16 bits unsigned NUMBER	

0x1FA14: PGN 129556 - GLONASS Almanac Data

Almanac data for GLONASS products. The almanac contains satellite vehicle course orbital parameters. These parameters are described in the GLONASS ICS Section 4.5 Table 4.3. See URL.

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 26 bytes long and contains 14 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	PRN	Satellite ID number	0 .. 252	8 bits unsigned NUMBER	
2	NA	Calendar day count within the four year period beginning with the previous leap year	0 .. 65532	16 bits unsigned NUMBER	

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3	Reserved			2 bits RESERVED	
4	CnA	Generalized health of the satellite	0 .. 1	1 bits unsigned NUMBER	
5	HnA	Carrier frequency number	0 .. 29	5 bits unsigned NUMBER	
6	(epsilon)nA	Eccentricity	0 .. 65532	16 bits unsigned NUMBER	
7	(deltaTnA)DOT	Rate of change of the draconitic circling time	0 .. 252	8 bits unsigned NUMBER	
8	(omega)nA	Rate of change of the draconitic circling time	0 .. 65532	16 bits unsigned NUMBER	
9	(delta)TnA	Correction to the average value of the draconitic circling time	0 .. 16777212	24 bits unsigned NUMBER	
10	tnA	Time of the ascension node	0 .. 16777212	24 bits unsigned NUMBER	
11	(lambda)nA	Greenwich longitude of the ascension node	0 .. 16777212	24 bits unsigned NUMBER	
12	(delta)inA	Correction to the average value of the inclination angle	0 .. 16777212	24 bits unsigned NUMBER	
13	(tau)cA	System time scale correction	0 .. 268435452	28 bits unsigned NUMBER	
14	(tau)nA	Course value of the time scale shift	0 .. 4092	12 bits unsigned NUMBER	

Source: https://www.unavco.org/help/glossary/docs/ICD_GLONASS_5.1_%282008%29_en.pdf

0x1FB00: PGN 129792 - AIS DGNSS Broadcast Binary Message

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 17 bytes long and contains 12 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 2	2 bits unsigned NUMBER	
3	Source ID		2000000 .. 99999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
8	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
9	Reserved			3 bits RESERVED	
10	Spare			5 bits SPARE	
11	Number of Bits in Binary Data Field		0 .. 65532	16 bits unsigned NUMBER	
12	Binary Data			Field 11 defines the Variable length BINARY	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>



0x1FB01: PGN 129793 - AIS UTC and Date Report

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 24 bytes long and contains 14 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
5	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Position Accuracy		0 .. 1	1 bits lookup POSITION_ACCURACY	
7	RAIM		0 .. 1	1 bits lookup RAIM_FLAG	
8	Reserved			6 bits RESERVED	
9	Position Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
10	Communication State	Information used by the TDMA slot allocation algorithm and synchronization information		19 bits BINARY	
11	AIS Transceiver information		0 .. 29	5 bits lookup AIS_TRANSCIVER	
12	Position Date		d 0 .. 65532	16 bits unsigned DATE	
13	Reserved			4 bits RESERVED	
14	GNSS type		0 .. 13	4 bits lookup POSITION_FIX_DEVICE	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB02: PGN 129794 - AIS Class A Static and Voyage Related Data

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 75 bytes long and contains 21 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	IMO number		0 .. 4294967292	32 bits unsigned NUMBER	
5	Callsign			56 bits STRING_FIX	
6	Name			160 bits STRING_FIX	

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7	Type of ship		0..252	8 bits lookup SHIP_TYPE	
8	Length		0.1 m 0..6553.2	16 bits unsigned NUMBER	
9	Beam		0.1 m 0..6553.2	16 bits unsigned NUMBER	
10	Position reference from Starboard		0.1 m 0..6553.2	16 bits unsigned NUMBER	
11	Position reference from Bow		0.1 m 0..6553.2	16 bits unsigned NUMBER	
12	ETA Date		d 0..65532	16 bits unsigned DATE	
13	ETA Time	Seconds since midnight	0.0001 s 0..86401	32 bits TIME	
14	Draft		0.01 m 0..655.32	16 bits unsigned NUMBER	
15	Destination			160 bits STRING_FIX	
16	AIS version indicator		0..3	2 bits lookup AIS_VERSION	
17	GNSS type		0..13	4 bits lookup POSITION_FIX_DEVICE	
18	DTE		0..1	1 bits lookup AVAILABLE	
19	Reserved			1 bits RESERVED	
20	AIS Transceiver information		0..29	5 bits lookup AIS_TRANSCIVER	
21	Reserved			3 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB03: PGN 129795 - AIS Addressed Binary Message

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 13 bytes long and contains 12 fields. The default priority (as observed in the field) is 5. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0..61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0..3	2 bits lookup REPEAT_INDICATOR	
3	Source ID		2000000.. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0..29	5 bits lookup AIS_TRANSCIVER	
6	Sequence Number		0..3	2 bits unsigned NUMBER	
7	Destination ID		2000000.. 999999999	32 bits unsigned MMSI	true
8	Reserved			6 bits RESERVED	
9	Retransmit flag		0..1	1 bits lookup YES_NO	
10	Reserved			1 bits RESERVED	
11	Number of Bits in Binary Data Field		0..65532	16 bits unsigned NUMBER	
12	Binary Data			Field 11 defines the Variable length BINARY	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB04: PGN 129796 - AIS Acknowledge



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This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 6 bytes long and contains 9 fields. The 3 fields starting at field 7 (with name "Destination ID") form repeating set 1. The set is repeated until there is no more data in the PGN. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7 Set 1	Destination ID		2000000 .. 999999999	32 bits unsigned MMSI	true
8 Set 1	Sequence Number		0 .. 3	2 bits unsigned NUMBER	
9 Set 1	Reserved			6 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB05: PGN 129797 - AIS Binary Broadcast Message

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 5. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Number of Bits in Binary Data Field		0 .. 65532	16 bits unsigned NUMBER	
8	Binary Data			Field 7 defines the Variable length BINARY	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB06: PGN 129798 - AIS SAR Aircraft Position Report

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 27 bytes long and contains 17 fields. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	



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2	Repeat Indicator		0..3	2 bits lookup REPEAT_INDICATOR	
3	User ID		2000000.. 999999999	32 bits unsigned MMSI	true
4	Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
5	Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Position Accuracy		0..1	1 bits lookup POSITION_ACCURACY	
7	RAIM		0..1	1 bits lookup RAIM_FLAG	
8	Time Stamp		0..63	6 bits lookup TIME_STAMP	
9	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
10	SOG	Originally this used the same resolution as SOG for boats (0.01 m/s), but this did not allow for speeds over 180 km/h so in some addendum this has been amended to have a resolution of 0.1 m/s	0.1 m/s 0 .. 6553.2	16 bits unsigned NUMBER	
11	Communication State	Information used by the TDMA slot allocation algorithm and synchronization information		19 bits BINARY	
12	AIS Transceiver information		0..29	5 bits lookup AIS_TRANSCIVER	
13	Altitude		0.01 m -21474836.47 .. 21474836.44	32 bits signed NUMBER	
14	Reserved for Regional Applications			8 bits BINARY	
15	DTE		0..1	1 bits lookup AVAILABLE	
16	Spare			5 bits SPARE	
17	Reserved			2 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB07: PGN 129799 - Radio Frequency/Mode/Power

The Radio Channel is NOT a numeric field, it has been observed to contain values such as 9000L1-L3 and 9000F1-F3 (indicating private channels as allowed in some countries.)

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 19 bytes long and contains 6 fields. The default priority (as observed in the field) is 3. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Rx Frequency		10 Hz 0.. 42949672920	32 bits unsigned NUMBER	
2	Tx Frequency		10 Hz 0.. 42949672920	32 bits unsigned NUMBER	
3	Radio Channel	MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF		48 bits STRING_FIX	

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		channels to have the first digit 9 followed by zero. The next digit is 1 indicating the ship stations transmit frequency is being used as a simplex channel frequency, or 2 indicating the coast stations transmit frequency is being used as a simplex channel frequency, 0 otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.			
4	Tx Power		W 0 .. 65532	16 bits unsigned NUMBER	
5	Mode		0 .. 252	8 bits lookup TELEPHONE_MODE	
6	Channel Bandwidth		Hz 0 .. 65532	16 bits unsigned NUMBER	

0x1FB08: PGN 129800 - AIS UTC/Date Inquiry

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 10 bytes long and contains 7 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS_TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Destination ID		2000000 .. 999999999	32 bits unsigned MMSI	true

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB09: PGN 129801 - AIS Addressed Safety Related Message

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 11 bytes long and contains 11 fields. The default priority (as observed in the field) is 5. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS_MESSAGE_ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS_TRANSCEIVER	
6	Sequence Number		0 .. 3	2 bits unsigned NUMBER	
7	Destination ID		2000000 .. 999999999	32 bits unsigned MMSI	true
8	Reserved			6 bits RESERVED	
9	Retransmit flag		0 .. 1	1 bits lookup YES_NO	
10	Spare			1 bits SPARE	
11	Safety Related Text			Variable length STRING_LAU	

Source: <https://navcen.uscg.gov/ais-addressed-safety-related-message12>



0x1FB0A: PGN 129802 - AIS Safety Related Broadcast Message

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 6 bytes long and contains 7 fields. The default priority (as observed in the field) is 5. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Safety Related Text			Variable length STRING LAU	

Source: <https://www.navcen.uscg.gov/ais-safety-related-broadcast-message14>

0x1FB0B: PGN 129803 - AIS Interrogation

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 26 bytes long and contains 22 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Destination ID 1		2000000 .. 999999999	32 bits unsigned MMSI	true
8	Reserved			2 bits RESERVED	
9	Message ID 1.1		0 .. 61	6 bits lookup AIS MESSAGE ID	
10	Slot Offset 1.1		0 .. 65532	16 bits unsigned NUMBER	
11	Spare			2 bits SPARE	
12	Message ID 1.2		0 .. 61	6 bits lookup AIS MESSAGE ID	
13	Slot Offset 1.2		0 .. 65532	16 bits unsigned NUMBER	
14	Reserved			6 bits RESERVED	
15	Reserved			2 bits RESERVED	
16	Destination ID 2		2000000 .. 999999999	32 bits unsigned MMSI	true
17	Reserved			2 bits RESERVED	
18	Message ID 2.1		0 .. 61	6 bits lookup AIS MESSAGE ID	
19	Slot Offset 2.1		0 .. 65532	16 bits unsigned NUMBER	
20	Spare			2 bits SPARE	

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21	Reserved			6 bits RESERVED	
22	SID		0 .. 252	8 bits unsigned NUMBER	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB0C: PGN 129804 - AIS Assignment Mode Command

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 23 bytes long and contains 14 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Destination ID A		2000000 .. 999999999	32 bits unsigned MMSI	true
8	Offset A	Offset from current slot to first assigned slot	0 .. 65532	16 bits unsigned NUMBER	
9	Increment A	Increment to next assigned slot	0 .. 65532	16 bits unsigned NUMBER	
10	Destination ID B		2000000 .. 999999999	32 bits unsigned MMSI	true
11	Offset B	Offset from current slot to first assigned slot	0 .. 65532	16 bits unsigned NUMBER	
12	Increment B	Increment to next assigned slot	0 .. 65532	16 bits unsigned NUMBER	
13	Spare			4 bits SPARE	
14	Reserved			4 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB0D: PGN 129805 - AIS Data Link Management Message

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 6 bytes long and contains 10 fields. The 4 fields starting at field 7 (with name "Offset") form repeating set 1. The set is repeated until there is no more data in the PGN. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	

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7 Set 1	Offset	Reserved offset number	0 .. 65532	16 bits unsigned NUMBER	
8 Set 1	Number of Slots	Reserved consecutive slots	0 .. 252	8 bits unsigned NUMBER	
9 Set 1	Timeout		60 s 0 .. 15120	8 bits unsigned DURATION	
10 Set 1	Increment		0 .. 65532	16 bits unsigned NUMBER	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB0E: PGN 129806 - AIS Channel Management

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 31 bytes long and contains 23 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Reserved			1 bits RESERVED	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Spare			2 bits SPARE	
7	Channel A		0 .. 65532	16 bits unsigned NUMBER	
8	Channel B		0 .. 65532	16 bits unsigned NUMBER	
9	Reserved			3 bits RESERVED	
10	Power		0 .. 1	1 bits lookup POWER MODE	
11	Tx/Rx Mode		0 .. 13	4 bits lookup TX RX MODE	
12	North East Longitude Corner 1		1e-07 deg -180 .. 180	32 bits signed NUMBER	
13	North East Latitude Corner 1		1e-07 deg -90 .. 90	32 bits signed NUMBER	
14	South West Longitude Corner 2		1e-07 deg -180 .. 180	32 bits signed NUMBER	
15	South West Latitude Corner 2		1e-07 deg -90 .. 90	32 bits signed NUMBER	
16	Reserved			1 bits RESERVED	
17	Addressed or Broadcast Message Indicator		0 .. 1	1 bits lookup BROADCAST INDICATOR	
18	Channel A Bandwidth		0 .. 1	1 bits lookup BANDWIDTH	
19	Channel B Bandwidth		0 .. 1	1 bits lookup BANDWIDTH	
20	Reserved			1 bits RESERVED	
21	Transitional Zone Size		0 .. 6	3 bits lookup ZONE SIZE	
22	Spare			23 bits SPARE	
23	Reserved			1 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB0F: PGN 129807 - AIS Class B Group Assignment



This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

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This fast-packet PGN is 29 bytes long and contains 19 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	Source ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Spare			2 bits SPARE	
5	Tx/Rx Mode		0 .. 13	4 bits lookup TX_RX MODE	
6	Reserved			2 bits RESERVED	
7	North East Longitude Corner 1		1e-07 deg -180 .. 180	32 bits signed NUMBER	
8	North East Latitude Corner 1		1e-07 deg -90 .. 90	32 bits signed NUMBER	
9	South West Longitude Corner 2		1e-07 deg -180 .. 180	32 bits signed NUMBER	
10	South West Latitude Corner 2		1e-07 deg -90 .. 90	32 bits signed NUMBER	
11	Station Type		0 .. 13	4 bits lookup STATION TYPE	
12	Reserved			4 bits RESERVED	
13	Ship and Cargo Filter		0 .. 252	8 bits lookup SHIP TYPE	
14	Spare			22 bits SPARE	
15	Reserved			2 bits RESERVED	
16	Reporting Interval		0 .. 13	4 bits lookup REPORTING INTERVAL	
17	Quiet Time	Commanded quiet time	60 s 0 .. 780	4 bits unsigned DURATION	
18	Spare			6 bits SPARE	
19	Reserved			2 bits RESERVED	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB10: PGN 129808 - DSC Distress Call Information

This PGN description applies when the following field(s) match:

DSC Category	112
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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 60 bytes long and contains 22 fields. The 2 fields starting at field 21 (with name "DSC Expansion Field Symbol") form repeating set 1. The set is repeated until there is no more data in the PGN. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	DSC Format		0 .. 252	8 bits lookup DSC FORMAT	



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2	DSC Category	112: Distress	0 .. 252	8 bits lookup DSC_CATEGORY	
3	DSC Message Address	MMSI, Geographic Area or blank	0 .. 9999999999	40 bits unsigned DECIMAL	
4	Nature of Distress		0 .. 252	8 bits lookup DSC_NATURE	
5	Subsequent Communication Mode or 2nd Telecommand		0 .. 252	8 bits lookup DSC_SECOND_TELECOMMAND	
6	Proposed Rx Frequency/Channel			48 bits STRING_FIX	
7	Proposed Tx Frequency/Channel			48 bits STRING_FIX	
8	Telephone Number			Variable length STRING_LAU	
9	Latitude of Vessel Reported		1e-07 deg -90 .. 90	32 bits signed NUMBER	
10	Longitude of Vessel Reported		1e-07 deg -180 .. 180	32 bits signed NUMBER	
11	Time of Position	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
12	MMSI of Ship In Distress		0 .. 9999999999	40 bits unsigned DECIMAL	
13	DSC EOS Symbol		0 .. 252	8 bits unsigned NUMBER	
14	Expansion Enabled		0 .. 2	2 bits lookup YES_NO	
15	Reserved			6 bits RESERVED	
16	Calling Rx Frequency/Channel			48 bits STRING_FIX	
17	Calling Tx Frequency/Channel			48 bits STRING_FIX	
18	Time of Receipt	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
19	Date of Receipt		d 0 .. 65532	16 bits unsigned DATE	
20	DSC Equipment Assigned Message ID		0 .. 65532	16 bits unsigned NUMBER	
21 Set 1	DSC Expansion Field Symbol		0 .. 252	8 bits lookup DSC_EXPANSION_DATA	
22 Set 1	DSC Expansion Field Data			Variable length STRING_LAU	

Source:

<https://web.archive.org/web/20160408121445/http://www.nmea.org/Assets/20150328%20dsc%20technical%20corrigendum%20v2.data>

0x1FB10: PGN 129808 - DSC Call Information

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 60 bytes long and contains 22 fields. The 2 fields starting at field 21 (with name "DSC Expansion Field Symbol") form repeating set 1. The set is repeated until there is no more data in the PGN. The default priority (as observed in the field) is 4. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	DSC Format Symbol		0 .. 252	8 bits lookup DSC_FORMAT	
2	DSC Category Symbol		0 .. 252	8 bits lookup DSC_CATEGORY	



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3	DSC Message Address	MMSI, Geographic Area or blank	0 .. 9999999999	40 bits unsigned DECIMAL	
4	1st Telecommand		0 .. 252	8 bits lookup DSC FIRST TELECOMMAND	
5	Subsequent Communication Mode or 2nd Telecommand		0 .. 252	8 bits lookup DSC SECOND TELECOMMAND	
6	Proposed Rx Frequency/Channel			48 bits STRING FIX	
7	Proposed Tx Frequency/Channel			48 bits STRING FIX	
8	Telephone Number			Variable length STRING LAU	
9	Latitude of Vessel Reported		1e-07 deg -90 .. 90	32 bits signed NUMBER	
10	Longitude of Vessel Reported		1e-07 deg -180 .. 180	32 bits signed NUMBER	
11	Time of Position	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
12	MMSI of Ship In Distress		0 .. 9999999999	40 bits unsigned DECIMAL	
13	DSC EOS Symbol		0 .. 252	8 bits unsigned NUMBER	
14	Expansion Enabled		0 .. 2	2 bits lookup YES NO	
15	Reserved			6 bits RESERVED	
16	Calling Rx Frequency/Channel			48 bits STRING FIX	
17	Calling Tx Frequency/Channel			48 bits STRING FIX	
18	Time of Receipt	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
19	Date of Receipt		d 0 .. 65532	16 bits unsigned DATE	
20	DSC Equipment Assigned Message ID		0 .. 65532	16 bits unsigned NUMBER	
21 Set 1	DSC Expansion Field Symbol		0 .. 252	8 bits lookup DSC EXPANSION DATA	
22 Set 1	DSC Expansion Field Data			Variable length STRING LAU	

Source:

<https://web.archive.org/web/20160408121445/http://www.nmea.org/Assets/20150328%20dsc%20technical%20corrigendum%20v2.data>

0x1FB11: PGN 129809 - AIS Class B static data (msg 24 Part A)

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 27 bytes long and contains 7 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Name			160 bits STRING FIX	
5	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCEIVER	
6	Reserved			3 bits RESERVED	
7	Sequence ID		0 .. 252	8 bits unsigned NUMBER	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FB12: PGN 129810 - AIS Class B static data (msg 24 Part B)

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 35 bytes long and contains 17 fields. The default priority (as observed in the field) is 6. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Message ID		0 .. 61	6 bits lookup AIS MESSAGE ID	
2	Repeat Indicator		0 .. 3	2 bits lookup REPEAT INDICATOR	
3	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
4	Type of ship		0 .. 252	8 bits lookup SHIP TYPE	
5	Vendor ID			56 bits STRING FIX	
6	Callsign			56 bits STRING FIX	
7	Length		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
8	Beam		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
9	Position reference from Starboard		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
10	Position reference from Bow		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
11	Mothership User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
12	Reserved			2 bits RESERVED	
13	Spare			2 bits SPARE	
14	GNSS type		0 .. 13	4 bits lookup POSITION FIX DEVICE	
15	AIS Transceiver information		0 .. 29	5 bits lookup AIS TRANSCIVER	
16	Reserved			3 bits RESERVED	
17	Sequence ID		0 .. 252	8 bits unsigned NUMBER	

Source: <https://www.itu.int/rec/R-REC-M.1371-5-201402-I/en>

0x1FC04: PGN 130052 - Loran-C TD Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 32 bytes long and contains 15 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Repetition Interval (GRI)		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
2	Master Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
3	V Secondary TD		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
4	W Secondary TD		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
5	X Secondary TD		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
6	Y Secondary TD		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
7	Z Secondary TD		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	

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8	Station status: Master		0..15	4 bits bitfield STATION STATUS	
9	Station status: V		0..15	4 bits bitfield STATION STATUS	
10	Station status: W		0..15	4 bits bitfield STATION STATUS	
11	Station status: X		0..15	4 bits bitfield STATION STATUS	
12	Station status: Y		0..15	4 bits bitfield STATION STATUS	
13	Station status: Z		0..15	4 bits bitfield STATION STATUS	
14	Mode		0..13	4 bits lookup RESIDUAL MODE	
15	Reserved			4 bits RESERVED	

0x1FC05: PGN 130053 - Loran-C Range Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 32 bytes long and contains 15 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Repetition Interval (GRI)		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
2	Master Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
3	V Secondary Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
4	W Secondary Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
5	X Secondary Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
6	Y Secondary Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
7	Z Secondary Range		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
8	Station status: Master		0..15	4 bits bitfield STATION STATUS	
9	Station status: V		0..15	4 bits bitfield STATION STATUS	
10	Station status: W		0..15	4 bits bitfield STATION STATUS	
11	Station status: X		0..15	4 bits bitfield STATION STATUS	
12	Station status: Y		0..15	4 bits bitfield STATION STATUS	
13	Station status: Z		0..15	4 bits bitfield STATION STATUS	
14	Mode		0..13	4 bits lookup RESIDUAL MODE	
15	Reserved			4 bits RESERVED	

0x1FC06: PGN 130054 - Loran-C Signal Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 15 bytes long and contains 5 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Group Repetition Interval (GRI)		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
2	Station identifier			8 bits STRING FIX	
3	Station SNR		0.01 dB	16 bits signed NUMBER	



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4	Station ECD		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	
5	Station ASF		1e-09 s -2.147483647 .. 2.147483644	32 bits signed DURATION	

0x1FC0C: PGN 130060 - Label

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 9 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Hardware Channel ID		0 .. 252	8 bits unsigned NUMBER	
2	PGN		0 .. 262143	24 bits PGN	
3	Data Source Instance Field Number		0 .. 252	8 bits unsigned NUMBER	
4	Data Source Instance Value		0 .. 252	8 bits unsigned NUMBER	
5	Secondary Enumeration Field Number		0 .. 252	8 bits unsigned NUMBER	
6	Secondary Enumeration Field Value		0 .. 252	8 bits unsigned NUMBER	
7	Parameter Field Number		0 .. 252	8 bits unsigned NUMBER	
8	Label			Variable length STRING LAU	

0x1FC0D: PGN 130061 - Channel Source Configuration

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 19 bytes long and contains 11 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Data Source Channel ID		0 .. 252	8 bits unsigned NUMBER	
2	Source Selection Status		0 .. 2	2 bits unsigned NUMBER	
3	Reserved			2 bits RESERVED	
4	NAME Selection Criteria Mask			12 bits BINARY	
5	Source NAME		0 .. 1.84467440737096e+19	64 bits ISO NAME	
6	PGN		0 .. 262143	24 bits PGN	
7	Data Source Instance Field Number		0 .. 252	8 bits unsigned NUMBER	
8	Data Source Instance Value		0 .. 252	8 bits unsigned NUMBER	
9	Secondary Enumeration Field Number		0 .. 252	8 bits unsigned NUMBER	
10	Secondary Enumeration Field Value		0 .. 252	8 bits unsigned NUMBER	
11	Parameter Field Number		0 .. 252	8 bits unsigned NUMBER	

0x1FC10: PGN 130064 - Route and WP Service - Database List

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 6 bytes long and contains 12 fields. The 9 fields starting at field 4 (with name "Database ID") form repeating set 1. The set is repeated *n* times, where *n* is determined by the value of field 2 (with name "nItems"). The default priority (as

observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start Database ID		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of Databases Available		0 .. 65532	16 bits unsigned NUMBER	
4 Set 1	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5 Set 1	Database Name			Variable length STRING LAU	
6 Set 1	Database Timestamp	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
7 Set 1	Database Datestamp		d 0 .. 65532	16 bits unsigned DATE	
8 Set 1	WP Position Resolution		0 .. 13	4 bits lookup WP_POSITION_RESOLUTION	
9 Set 1	Reserved			4 bits RESERVED	
10 Set 1	Number of Routes in Database		0 .. 65532	16 bits unsigned NUMBER	
11 Set 1	Number of WPs in Database		0 .. 4294967292	32 bits unsigned NUMBER	
12 Set 1	Number of Bytes in Database		0 .. 4294967292	32 bits unsigned NUMBER	

0x1FC11: PGN 130065 - Route and WP Service - Route List

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 8 bytes long and contains 9 fields. The 5 fields starting at field 5 (with name "Route ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems".) The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start Route ID		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of Routes in Database		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5 Set 1	Route ID		0 .. 65532	16 bits unsigned NUMBER	
6 Set 1	Route Name			Variable length STRING LAU	
7 Set 1	Reserved			2 bits RESERVED	
8 Set 1	WP Identification Method		0 .. 2	2 bits lookup WP_IDENTIFICATION_METHOD	
9 Set 1	Route Status		0 .. 13	4 bits lookup WP_ROUTE_STATUS	

0x1FC12: PGN 130066 - Route and WP Service - Route/WP-List Attributes

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.



This fast-packet PGN is at least 17 bytes long and contains 12 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Database ID		0 .. 65532	16 bits unsigned NUMBER	
2	Route ID		0 .. 65532	16 bits unsigned NUMBER	
3	Route/WP-List Name			Variable length STRING LAU	
4	Route/WP-List Timestamp	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
5	Route/WP-List Datestamp		d 0 .. 65532	16 bits unsigned DATE	
6	Change at Last Timestamp		0 .. 255	8 bits bitfield WP CHANGE	
7	Number of WPs in the Route/WP-List		0 .. 65532	16 bits unsigned NUMBER	
8	Critical supplementary parameters		0 .. 255	8 bits bitfield WP CRITICAL PARAMETERS	
9	Navigation Method		0 .. 2	2 bits lookup WP NAVIGATION METHOD	
10	WP Identification Method		0 .. 2	2 bits lookup WP IDENTIFICATION METHOD	
11	Route Status		0 .. 13	4 bits lookup WP ROUTE STATUS	
12	XTE Limit for the Route		m -32767 .. 32764	16 bits signed NUMBER	

0x1FC13: PGN 130067 - Route and WP Service - Route - WP Name & Position

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 10 bytes long and contains 9 fields. The 4 fields starting at field 6 (with name "WP ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems".) The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start RPS#		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of WPs in the Route/WP-List		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5	Route ID		0 .. 65532	16 bits unsigned NUMBER	
6 Set 1	WP ID		0 .. 65532	16 bits unsigned NUMBER	
7 Set 1	WP Name			Variable length STRING LAU	
8 Set 1	WP Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
9 Set 1	WP Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	

0x1FC14: PGN 130068 - Route and WP Service - Route - WP Name

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 10 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "WP ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems".) The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start RPS#		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of WPs in the Route/WP-List		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5	Route ID		0 .. 65532	16 bits unsigned NUMBER	
6 Set 1	WP ID		0 .. 65532	16 bits unsigned NUMBER	
7 Set 1	WP Name			Variable length STRING LAU	

0x1FC15: PGN 130069 - Route and WP Service - XTE Limit & Navigation Method

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 6 bytes long and contains 9 fields. The 6 fields starting at field 4 (with name "Database ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start RPS#		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of WPs with a specific XTE Limit or Nav. Method		0 .. 65532	16 bits unsigned NUMBER	
4 Set 1	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5 Set 1	Route ID		0 .. 65532	16 bits unsigned NUMBER	
6 Set 1	RPS#		0 .. 65532	16 bits unsigned NUMBER	
7 Set 1	XTE Limit in the leg after WP		m 32767 .. 32764	16 bits signed NUMBER	
8 Set 1	Nav. Method in the leg after WP		0 .. 2	2 bits lookup WP NAVIGATION METHOD	
9 Set 1	Reserved			6 bits RESERVED	

0x1FC16: PGN 130070 - Route and WP Service - WP Comment

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 10 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "WP ID / RPS#") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start ID		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of WPs with Comments		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5	Route ID		0 .. 65532	16 bits unsigned NUMBER	



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6 Set 1	WP ID / RPS#		0 .. 65532	16 bits unsigned NUMBER	
7 Set 1	Comment			Variable length STRING LAU	

0x1FC17: PGN 130071 - Route and WP Service - Route Comment

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 8 bytes long and contains 6 fields. The 2 fields starting at field 5 (with name "Route ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start Route ID		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of Routes with Comments		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5 Set 1	Route ID		0 .. 65532	16 bits unsigned NUMBER	
6 Set 1	Comment			Variable length STRING LAU	

0x1FC18: PGN 130072 - Route and WP Service - Database Comment

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 6 bytes long and contains 5 fields. The 2 fields starting at field 4 (with name "Database ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start Database ID		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of Databases with Comments		0 .. 65532	16 bits unsigned NUMBER	
4 Set 1	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5 Set 1	Comment			Variable length STRING LAU	

0x1FC19: PGN 130073 - Route and WP Service - Radius of Turn

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 10 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "RPS#") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start RPS#		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of WPs with a specific Radius of Turn		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5	Route ID		0 .. 65532	16 bits unsigned NUMBER	
6 Set 1	RPS#		0 .. 65532	16 bits unsigned NUMBER	



7 Set 1	Radius of Turn		m -32767 .. 32764	16 bits signed NUMBER	
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0x1FC1A: PGN 130074 - Route and WP Service - WP List - WP Name & Position

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 10 bytes long and contains 9 fields. The 4 fields starting at field 6 (with name "WP ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "nItems"). The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Start WP ID		0 .. 65532	16 bits unsigned NUMBER	
2	nItems		0 .. 65532	16 bits unsigned NUMBER	
3	Number of valid WPs in the WP-List		0 .. 65532	16 bits unsigned NUMBER	
4	Database ID		0 .. 65532	16 bits unsigned NUMBER	
5	Reserved			16 bits RESERVED	
6 Set 1	WP ID		0 .. 65532	16 bits unsigned NUMBER	
7 Set 1	WP Name			Variable length STRING LAU	
8 Set 1	WP Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
9 Set 1	WP Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	

0x1FD02: PGN 130306 - Wind Data

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 100 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Wind Speed		0.01 m/s 0 .. 65532	16 bits unsigned NUMBER	
3	Wind Angle		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
4	Reference		0 .. 6	3 bits lookup WIND REFERENCE	true
5	Reserved			21 bits RESERVED	

Source: http://askjackrabbit.typepad.com/ask_jack_rabbit/page/7/

0x1FD06: PGN 130310 - Environmental Parameters (obsolete)

This PGN was succeeded by PGN 130310, but it should no longer be generated and separate PGNs in range 130312..130315 should be used

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Water Temperature		0.01 K 0 .. 65532	16 bits unsigned NUMBER	



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3	Outside Ambient Air Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
4	Atmospheric Pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	

0x1FD07: PGN 130311 - Environmental Parameters

This PGN was introduced as a better version of PGN 130310, but it should no longer be generated and separate PGNs in range 130312..130315 should be used

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Temperature Source		0 .. 61	6 bits lookup TEMPERATURE_SOURCE	true
3	Humidity Source		0 .. 2	2 bits lookup HUMIDITY_SOURCE	
4	Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
5	Humidity		0.004 % -131.068 .. 131.056	16 bits signed NUMBER	
6	Atmospheric Pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	

0x1FD08: PGN 130312 - Temperature

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 2000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Source		0 .. 252	8 bits lookup TEMPERATURE_SOURCE	true
4	Actual Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
5	Set Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Reserved			8 bits RESERVED	

0x1FD09: PGN 130313 - Humidity

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 2000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Source		0 .. 252	8 bits lookup HUMIDITY_SOURCE	true
4	Actual Humidity		0.004 % -131.068 .. 131.056	16 bits signed NUMBER	
5	Set Humidity		0.004 % -131.068 .. 131.056	16 bits signed NUMBER	



6	Reserved			8 bits RESERVED	
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0x1FD0A: PGN 130314 - Actual Pressure

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 2000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Source		0 .. 252	8 bits lookup PRESSURE_SOURCE	true
4	Pressure		0.1 Pa -214748364.7 .. 214748364.4	32 bits signed NUMBER	
5	Reserved			8 bits RESERVED	

0x1FD0B: PGN 130315 - Set Pressure

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 5. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Source		0 .. 252	8 bits lookup PRESSURE_SOURCE	true
4	Pressure		0.1 Pa -214748364.7 .. 214748364.4	32 bits signed NUMBER	
5	Reserved			8 bits RESERVED	

0x1FD0C: PGN 130316 - Temperature Extended Range

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This single-frame PGN is 8 bytes long and contains 5 fields. The default priority (as observed in the field) is 5. It is normally transmitted every 2000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Instance		0 .. 252	8 bits unsigned NUMBER	true
3	Source		0 .. 252	8 bits lookup TEMPERATURE_SOURCE	true
4	Temperature		0.001 K 0 .. 16777.212	24 bits unsigned NUMBER	
5	Set Temperature		0.1 K 0 .. 6553.2	16 bits unsigned NUMBER	

Source:

<https://web.archive.org/web/20220516160001/https://www.nmea.org/Assets/nmea%202000%20pgn%20130316%20corrigenda%20nmd>

0x1FD10: PGN 130320 - Tide Station Data

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is at least 19 bytes long and contains 11 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
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1	Mode		0 .. 13	4 bits lookup RESIDUAL_MODE	
2	Tide Tendency		0 .. 2	2 bits lookup TIDE	
3	Reserved			2 bits RESERVED	
4	Measurement Date		d 0 .. 65532	16 bits unsigned DATE	
5	Measurement Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
6	Station Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
7	Station Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
8	Tide Level	Relative to MLLW	0.001 m -32.767 .. 32.764	16 bits signed NUMBER	
9	Tide Level standard deviation		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
10	Station ID			Variable length STRING_LAU	true
11	Station Name			Variable length STRING_LAU	

0x1FD11: PGN 130321 - Salinity Station Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 21 bytes long and contains 10 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Mode		0 .. 13	4 bits lookup RESIDUAL_MODE	
2	Reserved			4 bits RESERVED	
3	Measurement Date		d 0 .. 65532	16 bits unsigned DATE	
4	Measurement Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
5	Station Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Station Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
7	Salinity		ppt -3.40282346638529e+38 .. 3.40282346638529e+38	32 bits signed FLOAT	
8	Water Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
9	Station ID			Variable length STRING_LAU	
10	Station Name			Variable length STRING_LAU	

0x1FD12: PGN 130322 - Current Station Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 25 bytes long and contains 13 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Mode		0 .. 13	4 bits lookup RESIDUAL_MODE	

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2	State		0 .. 6	3 bits lookup FLOOD_STATE	
3	Reserved			1 bits RESERVED	
4	Measurement Date		d 0 .. 65532	16 bits unsigned DATE	
5	Measurement Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
6	Station Latitude		1e-07 deg 90 .. 90	32 bits signed NUMBER	
7	Station Longitude		1e-07 deg 180 .. 180	32 bits signed NUMBER	
8	Measurement Depth	Depth below transducer	0.01 m 0 .. 42949672.92	32 bits unsigned NUMBER	
9	Current speed		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
10	Current flow direction		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
11	Water Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
12	Station ID			Variable length STRING LAU	true
13	Station Name			Variable length STRING LAU	

0x1FD13: PGN 130323 - Meteorological Station Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 26 bytes long and contains 15 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Mode		0 .. 13	4 bits lookup RESIDUAL MODE	
2	Reserved			4 bits RESERVED	
3	Measurement Date		d 0 .. 65532	16 bits unsigned DATE	
4	Measurement Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
5	Station Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Station Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
7	Wind Speed		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
8	Wind Direction		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
9	Wind Reference		0 .. 6	3 bits lookup WIND_REFERENCE	
10	Reserved			5 bits RESERVED	
11	Wind Gusts		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
12	Atmospheric Pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
13	Ambient Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
14	Station ID			Variable length STRING LAU	true
15	Station Name			Variable length STRING LAU	

0x1FD14: PGN 130324 - Moored Buoy Station Data



This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

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This fast-packet PGN is at least 34 bytes long and contains 18 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Mode		0 .. 13	4 bits lookup RESIDUAL_MODE	
2	Reserved			4 bits RESERVED	
3	Measurement Date		d 0 .. 65532	16 bits unsigned DATE	
4	Measurement Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
5	Station Latitude		1e-07 deg -90 .. 90	32 bits signed NUMBER	
6	Station Longitude		1e-07 deg -180 .. 180	32 bits signed NUMBER	
7	Wind Speed		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
8	Wind Direction		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
9	Wind Reference		0 .. 6	3 bits lookup WIND_REFERENCE	
10	Reserved			5 bits RESERVED	
11	Wind Gusts		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
12	Wave Height		0.01 m 0 .. 655.32	16 bits unsigned NUMBER	
13	Dominant Wave Period		0.01 s 0 .. 655.32	16 bits unsigned DURATION	
14	Atmospheric Pressure		100 Pa 0 .. 6553200	16 bits unsigned NUMBER	
15	Pressure Tendency Rate		10 Pa/hr -327670 .. 327640	16 bits signed NUMBER	
16	Air Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
17	Water Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
18	Station ID			Variable length STRING_LAU	true

0x1FD1A: PGN 130330 - Lighting System Settings

This PGN provides a lighting controller settings and number of supported capabilities.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 12 bytes long and contains 12 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Global Enable		0 .. 2	2 bits unsigned NUMBER	
2	Default Settings/Command		0 .. 6	3 bits lookup LIGHTING_COMMAND	
3	Reserved			3 bits RESERVED	
4	Name of the lighting controller			Variable length STRING_LAU	

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5	Max Scenes		0 .. 252	8 bits unsigned NUMBER	
6	Max Scene Configuration Count		0 .. 252	8 bits unsigned NUMBER	
7	Max Zones		0 .. 252	8 bits unsigned NUMBER	
8	Max Color Sequences		0 .. 252	8 bits unsigned NUMBER	
9	Max Color Sequence Color Count		0 .. 252	8 bits unsigned NUMBER	
10	Number of Programs		0 .. 252	8 bits unsigned NUMBER	
11	Controller Capabilities		0 .. 252	8 bits unsigned NUMBER	
12	Identify Device		0 .. 4294967292	32 bits unsigned NUMBER	

0x1FE00: PGN 130560 - Payload Mass

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	SID		0 .. 252	8 bits unsigned NUMBER	
2	Measurement Status		0 .. 6	3 bits unsigned NUMBER	
3	Reserved			5 bits RESERVED	
4	Measurement ID		0 .. 252	8 bits unsigned NUMBER	
5	Payload Mass		0 .. 4294967292	32 bits unsigned NUMBER	
6	Reserved			8 bits RESERVED	

0x1FE01: PGN 130561 - Lighting Zone

This PGN is used to report or configure a name for a given zone. A zone is a grouping of devices that are controlled by a Scene. This PGN is only sent upon request.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 13 bytes long and contains 14 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Zone Index		0 .. 252	8 bits unsigned NUMBER	
2	Zone Name			Variable length STRING LAU	
3	Red Component		0 .. 252	8 bits unsigned NUMBER	
4	Green Component		0 .. 252	8 bits unsigned NUMBER	
5	Blue Component		0 .. 252	8 bits unsigned NUMBER	
6	Color Temperature		0 .. 65532	16 bits unsigned NUMBER	
7	Intensity		0 .. 252	8 bits unsigned NUMBER	
8	Program ID		0 .. 252	8 bits unsigned NUMBER	
9	Program Color Sequence Index		0 .. 252	8 bits unsigned NUMBER	
10	Program Intensity		0 .. 252	8 bits unsigned NUMBER	
11	Program Rate		0 .. 252	8 bits unsigned NUMBER	



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12	Program Color Sequence		0 .. 252	8 bits unsigned NUMBER	
13	Zone Enabled		0 .. 2	2 bits lookup OFF ON	
14	Reserved			6 bits RESERVED	

0x1FE02: PGN 130562 - Lighting Scene

A Lighting Scene is a sequence of zone program configurations.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 3 bytes long and contains 12 fields. The 8 fields starting at field 5 (with name "Configuration Index") form repeating set 1. The set is repeated n times, where n is determined by the value of field 4 (with name "Configuration Count".)

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Scene Index		0 .. 252	8 bits unsigned NUMBER	
2	Zone Name			Variable length STRING LAU	
3	Control		0 .. 252	8 bits unsigned NUMBER	
4	Configuration Count		0 .. 252	8 bits unsigned NUMBER	
<i>5 Set 1</i>	Configuration Index		0 .. 252	8 bits unsigned NUMBER	
<i>6 Set 1</i>	Zone Index		0 .. 252	8 bits unsigned NUMBER	
<i>7 Set 1</i>	Devices ID		0 .. 4294967292	32 bits unsigned NUMBER	
<i>8 Set 1</i>	Program Index		0 .. 252	8 bits unsigned NUMBER	
<i>9 Set 1</i>	Program Color Sequence Index		0 .. 252	8 bits unsigned NUMBER	
<i>10 Set 1</i>	Program Intensity		0 .. 252	8 bits unsigned NUMBER	
<i>11 Set 1</i>	Program Rate		0 .. 252	8 bits unsigned NUMBER	
<i>12 Set 1</i>	Program Color Sequence Rate		0 .. 252	8 bits unsigned NUMBER	

0x1FE03: PGN 130563 - Lighting Device

This PGN is used to provide status and capabilities of a lighting device. A lighting device may be a virtual device connected to a lighting controller or physical device on the network.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 20 bytes long and contains 18 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Device ID		0 .. 4294967292	32 bits unsigned NUMBER	



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2	Device Capabilities		0 .. 252	8 bits unsigned NUMBER	
3	Color Capabilities		0 .. 252	8 bits unsigned NUMBER	
4	Zone Index		0 .. 252	8 bits unsigned NUMBER	
5	Name of Lighting Device			Variable length STRING LAU	
6	Status		0 .. 252	8 bits unsigned NUMBER	
7	Red Component		0 .. 252	8 bits unsigned NUMBER	
8	Green Component		0 .. 252	8 bits unsigned NUMBER	
9	Blue Component		0 .. 252	8 bits unsigned NUMBER	
10	Color Temperature		0 .. 65532	16 bits unsigned NUMBER	
11	Intensity		0 .. 252	8 bits unsigned NUMBER	
12	Program ID		0 .. 252	8 bits unsigned NUMBER	
13	Program Color Sequence Index		0 .. 252	8 bits unsigned NUMBER	
14	Program Intensity		0 .. 252	8 bits unsigned NUMBER	
15	Program Rate		0 .. 252	8 bits unsigned NUMBER	
16	Program Color Sequence Rate		0 .. 252	8 bits unsigned NUMBER	
17	Enabled		0 .. 2	2 bits lookup OFF ON	
18	Reserved			6 bits RESERVED	

0x1FE04: PGN 130564 - Lighting Device Enumeration

This PGN allows for enumeration of the lighting devices on the controller.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 6 bytes long and contains 5 fields. The 2 fields starting at field 4 (with name "Device ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 3 (with name "Number of Devices").

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Index of First Device		0 .. 65532	16 bits unsigned NUMBER	
2	Total Number of Devices		0 .. 65532	16 bits unsigned NUMBER	
3	Number of Devices		0 .. 65532	16 bits unsigned NUMBER	
4 Set 1	Device ID		0 .. 4294967292	32 bits unsigned NUMBER	
5 Set 1	Status		0 .. 252	8 bits unsigned NUMBER	

0x1FE05: PGN 130565 - Lighting Color Sequence

Sequences could be 1 to (PGN Lighting - System Configuration) Max Color Sequence Color Count colors.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 2 bytes long and contains 8 fields. The 6 fields starting at field 3 (with name "Color Index") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "Color Count").

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Sequence Index		0..252	8 bits unsigned NUMBER	
2	Color Count		0..252	8 bits unsigned NUMBER	
3 Set 1	Color Index		0..252	8 bits unsigned NUMBER	
4 Set 1	Red Component		0..252	8 bits unsigned NUMBER	
5 Set 1	Green Component		0..252	8 bits unsigned NUMBER	
6 Set 1	Blue Component		0..252	8 bits unsigned NUMBER	
7 Set 1	Color Temperature		0..65532	16 bits unsigned NUMBER	
8 Set 1	Intensity		0..252	8 bits unsigned NUMBER	

0x1FE06: PGN 130566 - Lighting Program

This PGN describes an available program on the controller. Can be a built in required NMEA one or a custom vendor program.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- One or more of the lookup fields contain missing or incorrect values
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 2 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Program ID		0..252	8 bits unsigned NUMBER	
2	Name of Program			Variable length STRING LAU	
3	Description			Variable length STRING LAU	
4	Program Capabilities		0..13	4 bits unsigned NUMBER	
5	Reserved			4 bits RESERVED	

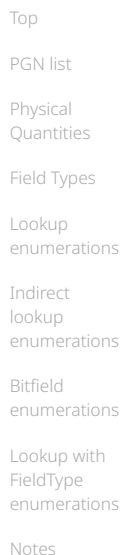
0x1FE07: PGN 130567 - Watermaker Input Setting and Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 24 bytes long and contains 23 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Watermaker Operating State		0..61	6 bits lookup WATERMAKER STATE	
2	Production Start/Stop		0..2	2 bits lookup YES NO	
3	Rinse Start/Stop		0..2	2 bits lookup YES NO	
4	Low Pressure Pump Status		0..2	2 bits lookup YES NO	
5	High Pressure Pump Status		0..2	2 bits lookup YES NO	
6	Emergency Stop		0..2	2 bits lookup YES NO	
7	Product Solenoid Valve Status		0..2	2 bits lookup OK WARNING	
8	Flush Mode Status		0..2	2 bits lookup YES NO	
9	Salinity Status		0..2	2 bits lookup OK WARNING	



Source:
<https://web.archive.org/web/20170609065450/https://www.nmea.org/Assets/20130905%20amendment%20at%202000%20201309051%20201309051.pdf>

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- This fast-packet PGN is 26 bytes long and contains 17 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Zone		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
2	Source		0 .. 252	8 bits lookup ENTERTAINMENT_SOURCE	
3	Number	Source number per type	0 .. 252	8 bits unsigned NUMBER	
4	ID	Unique file ID	0 .. 4294967292	32 bits unsigned NUMBER	
5	Play status		0 .. 252	8 bits lookup ENTERTAINMENT_PLAY_STATUS	
6	Elapsed Track Time		S 0 .. 65532	16 bits unsigned DURATION	
7	Track Time		S 0 .. 65532	16 bits unsigned DURATION	
8	Repeat Status		0 .. 13	4 bits lookup ENTERTAINMENT_REPEAT_STATUS	
9	Shuffle Status		0 .. 13	4 bits lookup ENTERTAINMENT_SHUFFLE_STATUS	
10	Save Favorite Number	Used to command AV to save current station as favorite	0 .. 252	8 bits unsigned NUMBER	



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11	Play Favorite Number	Used to command AV to play indicated favorite station	0 .. 65532	16 bits unsigned NUMBER	
12	Thumbs Up/Down		0 .. 252	8 bits lookup ENTERTAINMENT LIKE STATUS	
13	Signal Strength		% 0 .. 252	8 bits unsigned NUMBER	
14	Radio Frequency		10 Hz 0 .. 42949672920	32 bits unsigned NUMBER	
15	HD Frequency Multicast	Digital sub channel	0 .. 252	8 bits unsigned NUMBER	
16	Delete Favorite Number	Used to command AV to delete current station as favorite	0 .. 252	8 bits unsigned NUMBER	
17	Total Number of Tracks		0 .. 65532	16 bits unsigned NUMBER	

Source:

<https://web.archive.org/web/20170609024809/https://www.nmea.org/Assets/20160725%20corrigenda%20pgn%20130569%20published>

0x1FE0A: PGN 130570 - Library Data File

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 19 bytes long and contains 17 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Source		0 .. 252	8 bits lookup ENTERTAINMENT_SOURCE	
2	Number	Source number per type	0 .. 252	8 bits unsigned NUMBER	
3	ID	Unique file ID	0 .. 4294967292	32 bits unsigned NUMBER	
4	Type		0 .. 252	8 bits lookup ENTERTAINMENT_TYPE	
5	Name			Variable length STRING_LAU	
6	Track		0 .. 65532	16 bits unsigned NUMBER	
7	Station		0 .. 65532	16 bits unsigned NUMBER	
8	Favorite		0 .. 252	8 bits unsigned NUMBER	
9	Radio Frequency		10 Hz 0 .. 42949672920	32 bits unsigned NUMBER	
10	HD Frequency		0 .. 252	8 bits unsigned NUMBER	
11	Zone		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
12	In play queue		0 .. 2	2 bits lookup YES_NO	
13	Locked		0 .. 2	2 bits lookup YES_NO	
14	Reserved			4 bits RESERVED	
15	Artist Name			Variable length STRING_LAU	
16	Album Name			Variable length STRING_LAU	
17	Station Name			Variable length STRING_LAU	

Source:

<https://web.archive.org/web/20170609110901/https://www.nmea.org/Assets/20160715%20corrigenda%20entertainment%20pgns%20p>

0x1FE0B: PGN 130571 - Library Data Group



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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 15 bytes long and contains 12 fields. The 3 fields starting at field 9 (with name "ID type") form repeating set 1. The set is repeated n times, where n is determined by the value of field 7 (with name "ID count").

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Source		0 .. 252	8 bits lookup ENTERTAINMENT_SOURCE	
2	Number	Source number per type	0 .. 252	8 bits unsigned NUMBER	
3	Type		0 .. 252	8 bits lookup ENTERTAINMENT_TYPE	
4	Zone		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
5	Group ID	Unique group ID	0 .. 4294967292	32 bits unsigned NUMBER	
6	ID offset	First ID in this PGN	0 .. 65532	16 bits unsigned NUMBER	
7	ID count	Number of IDs in this PGN	0 .. 65532	16 bits unsigned NUMBER	
8	Total ID count	Total IDs in group	0 .. 65532	16 bits unsigned NUMBER	
9 Set 1	ID type		0 .. 252	8 bits lookup ENTERTAINMENT_ID_TYPE	
10 Set 1	ID		0 .. 4294967292	32 bits unsigned NUMBER	
11 Set 1	Name			Variable length STRING_LAU	
12	Artist			Variable length STRING_LAU	

Source:

<https://web.archive.org/web/20170609110901/https://www.nmea.org/Assets/20160715%20corrigenda%20entertainment%20pgns%20.p>

0x1FE0C: PGN 130572 - Library Data Search

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 9 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Source		0 .. 252	8 bits lookup ENTERTAINMENT_SOURCE	
2	Number	Source number per type	0 .. 252	8 bits unsigned NUMBER	
3	Group ID	Unique group ID	0 .. 4294967292	32 bits unsigned NUMBER	
4	Group type 1		0 .. 252	8 bits lookup ENTERTAINMENT_GROUP	
5	Group name 1			Variable length STRING_LAU	
6	Group type 2		0 .. 252	8 bits lookup ENTERTAINMENT_GROUP	
7	Group name 2			Variable length STRING_LAU	
8	Group type 3		0 .. 252	8 bits lookup ENTERTAINMENT_GROUP	
9	Group name 3			Variable length STRING_LAU	

Source:

<https://web.archive.org/web/20170609110901/https://www.nmea.org/Assets/20160715%20corrigenda%20entertainment%20pgns%20.p>



0x1FE0D: PGN 130573 - Supported Source Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 6 bytes long and contains 13 fields. The 10 fields starting at field 4 (with name "ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "ID count".)

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	ID offset	First ID in this PGN	0 .. 65532	16 bits unsigned NUMBER	
2	ID count	Number of IDs in this PGN	0 .. 65532	16 bits unsigned NUMBER	
3	Total ID count	Total IDs in group	0 .. 65532	16 bits unsigned NUMBER	
4 Set 1	ID	Source ID	0 .. 252	8 bits unsigned NUMBER	
5 Set 1	Source		0 .. 252	8 bits lookup ENTERTAINMENT_SOURCE	
6 Set 1	Number	Source number per type	0 .. 252	8 bits unsigned NUMBER	
7 Set 1	Name			Variable length STRING_LAU	
8 Set 1	Play support		0 .. 4294967295	32 bits bitfield ENTERTAINMENT_PLAY_STATUS_BITFIELD	
9 Set 1	Browse support		0 .. 65535	16 bits bitfield ENTERTAINMENT_GROUP_BITFIELD	
10 Set 1	Thumbs support		0 .. 2	2 bits lookup YES_NO	
11 Set 1	Connected		0 .. 2	2 bits lookup YES_NO	
12 Set 1	Repeat support		0 .. 3	2 bits bitfield ENTERTAINMENT_REPEAT_BITFIELD	
13 Set 1	Shuffle support		0 .. 3	2 bits bitfield ENTERTAINMENT_SHUFFLE_BITFIELD	

Source:

<https://web.archive.org/web/20170609110901/https://www.nmea.org/Assets/20160715%20corrigenda%20entertainment%20pgns%20.p>

0x1FE0E: PGN 130574 - Supported Zone Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is at least 3 bytes long and contains 5 fields. The 2 fields starting at field 4 (with name "Zone ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 2 (with name "Zone count".) The default priority (as observed in the field) is 6. It is normally transmitted every 500 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	First zone ID	First Zone in this PGN	0 .. 252	8 bits unsigned NUMBER	
2	Zone count	Number of Zones in this PGN	0 .. 252	8 bits unsigned NUMBER	

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3	Total zone count	Total Zones supported by this device	0 .. 252	8 bits unsigned NUMBER	
4 Set 1	Zone ID		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
5 Set 1	Name			Variable length STRING_LAU	

Source:

<https://web.archive.org/web/20170609110901/https://www.nmea.org/Assets/20160715%20corrigenda%20entertainment%20pgns%20.p>

0x1FE10: PGN 130576 - Small Craft Status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This single-frame PGN is 8 bytes long and contains 3 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 200 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Port trim tab		% -127 .. 124	8 bits signed NUMBER	
2	Starboard trim tab		% -127 .. 124	8 bits signed NUMBER	
3	Reserved			48 bits RESERVED	

0x1FE11: PGN 130577 - Direction Data

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 14 bytes long and contains 10 fields. The default priority (as observed in the field) is 3. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Data Mode		0 .. 13	4 bits lookup RESIDUAL_MODE	
2	COG Reference		0 .. 2	2 bits lookup DIRECTION_REFERENCE	
3	Reserved			2 bits RESERVED	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	COG		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
6	SOG		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
7	Heading		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
8	Speed through Water		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	
9	Set		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
10	Drift		0.01 m/s 0 .. 655.32	16 bits unsigned NUMBER	

0x1FE12: PGN 130578 - Vessel Speed Components

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 12 bytes long and contains 6 fields. The default priority (as observed in the field) is 2. It is normally transmitted every 250 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Longitudinal Speed, Water-referenced		0.001 m/s	16 bits signed NUMBER	



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			-32.767 .. 32.764		
2	Transverse Speed, Water-referenced		0.001 m/s -32.767 .. 32.764	16 bits signed NUMBER	
3	Longitudinal Speed, Ground-referenced		0.001 m/s -32.767 .. 32.764	16 bits signed NUMBER	
4	Transverse Speed, Ground-referenced		0.001 m/s -32.767 .. 32.764	16 bits signed NUMBER	
5	Stern Speed, Water-referenced		0.001 m/s -32.767 .. 32.764	16 bits signed NUMBER	
6	Stern Speed, Ground-referenced		0.001 m/s -32.767 .. 32.764	16 bits signed NUMBER	

0x1FE13: PGN 130579 - System Configuration

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 6 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Power		0 .. 2	2 bits lookup YES_NO	
2	Default Settings		0 .. 2	2 bits lookup ENTERTAINMENT_DEFAULT_SETTINGS	
3	Tuner regions		0 .. 13	4 bits lookup ENTERTAINMENT_REGIONS	
4	Max favorites		0 .. 252	8 bits unsigned NUMBER	
5	Video protocols		0 .. 13	4 bits lookup VIDEO_PROTOCOLS	
6	Reserved			44 bits RESERVED	

0x1FE14: PGN 130580 - System Configuration (deprecated)

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Power		0 .. 2	2 bits lookup YES_NO	
2	Default Settings		0 .. 2	2 bits lookup ENTERTAINMENT_DEFAULT_SETTINGS	
3	Tuner regions		0 .. 13	4 bits lookup ENTERTAINMENT_REGIONS	
4	Max favorites		0 .. 252	8 bits unsigned NUMBER	

0x1FE15: PGN 130581 - Zone Configuration (deprecated)

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 3 bytes long and contains 5 fields. The 2 fields starting at field 4 (with name "Zone ID") form repeating set 1. The set is repeated *n* times, where *n* is determined by the value of field 2 (with name "Zone count".)

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	First zone ID	First Zone in this PGN	0 .. 252	8 bits unsigned NUMBER	
2	Zone count	Number of Zones in this PGN	0 .. 252	8 bits unsigned NUMBER	

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3	Total zone count	Total Zones supported by this device	0 .. 252	8 bits unsigned NUMBER	
4 <i>Set 1</i>	Zone ID		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
5 <i>Set 1</i>	Zone name			Variable length STRING_LAU	

0x1FE16: PGN 130582 - Zone Volume

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Zone ID		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
2	Volume		% 0 .. 252	8 bits unsigned NUMBER	
3	Volume change	Write only	0 .. 2	2 bits lookup ENTERTAINMENT_VOLUME_CONTROL	
4	Mute		0 .. 2	2 bits lookup YES_NO	
5	Reserved			4 bits RESERVED	
6	Channel		0 .. 252	8 bits lookup ENTERTAINMENT_CHANNEL	
7	Reserved			32 bits RESERVED	

0x1FE17: PGN 130583 - Available Audio EQ presets

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 3 bytes long and contains 5 fields. The 2 fields starting at field 4 (with name "Preset type") form repeating set 1. The set is repeated *n* times, where *n* is determined by the value of field 2 (with name "Preset count".)

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	First preset	First preset in this PGN	0 .. 252	8 bits unsigned NUMBER	
2	Preset count		0 .. 252	8 bits unsigned NUMBER	
3	Total preset count		0 .. 252	8 bits unsigned NUMBER	
4 <i>Set 1</i>	Preset type		0 .. 252	8 bits lookup ENTERTAINMENT_EQ	
5 <i>Set 1</i>	Preset name			Variable length STRING_LAU	

0x1FE18: PGN 130584 - Available Bluetooth addresses

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is at least 3 bytes long and contains 7 fields. The 4 fields starting at field 4 (with name "Bluetooth address") form repeating set 1. The set is repeated *n* times, where *n* is determined by the value of field 2 (with name "Address count".)

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	First address	First address in this PGN	0 .. 252	8 bits unsigned NUMBER	

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2	Address count		0 .. 252	8 bits unsigned NUMBER	
3	Total address count		0 .. 252	8 bits unsigned NUMBER	
4 <i>Set 1</i>	Bluetooth address			48 bits BINARY	
5 <i>Set 1</i>	Status		0 .. 252	8 bits lookup BLUETOOTH STATUS	
6 <i>Set 1</i>	Device name			Variable length STRING_LAU	
7 <i>Set 1</i>	Signal strength		% 0 .. 252	8 bits unsigned NUMBER	

0x1FE19: PGN 130585 - Bluetooth source status

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This single-frame PGN is 8 bytes long and contains 5 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Source number		0 .. 252	8 bits unsigned NUMBER	
2	Status		0 .. 13	4 bits lookup BLUETOOTH_SOURCE_STATUS	
3	Forget device		0 .. 2	2 bits lookup YES_NO	
4	Discovering		0 .. 2	2 bits lookup YES_NO	
5	Bluetooth address			48 bits BINARY	

0x1FE1A: PGN 130586 - Zone Configuration

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 15 bytes long and contains 13 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Zone ID		0 .. 252	8 bits lookup ENTERTAINMENT_ZONE	
2	Volume limit		% 0 .. 252	8 bits unsigned NUMBER	
3	Fade		% -127 .. 124	8 bits signed NUMBER	
4	Balance		% -127 .. 124	8 bits signed NUMBER	
5	Sub volume		% 0 .. 252	8 bits unsigned NUMBER	
6	EQ - Treble		% -127 .. 124	8 bits signed NUMBER	
7	EQ - Mid range		% -127 .. 124	8 bits signed NUMBER	
8	EQ - Bass		% -127 .. 124	8 bits signed NUMBER	
9	Preset type		0 .. 252	8 bits lookup ENTERTAINMENT_EQ	
10	Audio filter		0 .. 252	8 bits lookup ENTERTAINMENT_FILTER	
11	High pass filter frequency		Hz 0 .. 65532	16 bits unsigned NUMBER	



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12	Low pass filter frequency		Hz 0 .. 65532	16 bits unsigned NUMBER	
13	Channel		0 .. 252	8 bits lookup ENTERTAINMENT_CHANNEL	

0x1FF00-0x1FFFF: Manufacturer Specific fast-packet non-addressed

This definition is used for Manufacturer Specific PGNs in PDU2 (non-addressed) fast-packet PGN range 0x1FF00 to 0x1FFFF (130816 - 131071). When this is shown during analysis it means the PGN is not reverse engineered yet.

0x1FF00: PGN 130816 - SonicHub: Init #2

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	1

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	1: Init #2	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	A		0 .. 65532	16 bits unsigned NUMBER	
8	B		0 .. 65532	16 bits unsigned NUMBER	

0x1FF00: PGN 130816 - SonicHub: AM Radio

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 43 bytes long and contains 12 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	



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3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	4: AM Radio	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Item		0 .. 252	8 bits lookup SONICHUB_TUNING	
8	Frequency		Hz 0 .. 4294967292	32 bits unsigned NUMBER	
9	Noise level		0 .. 2	2 bits unsigned NUMBER	
10	Signal level		0 .. 13	4 bits unsigned NUMBER	
11	Reserved			2 bits RESERVED	
12	Text			256 bits STRING_LZ	

0x1FF00: PGN 130816 - SonicHub: Zone info

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	5

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	5: Zone Info	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Zone		0 .. 252	8 bits unsigned NUMBER	

0x1FF00: PGN 130816 - SonicHub: Source

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	6

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
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1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	6: Source	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Source		0 .. 252	8 bits lookup SONICHUB_SOURCE	

0x1FF00: PGN 130816 - SonicHub: Source List

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	8

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 39 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	8: Source List	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Source ID		0 .. 252	8 bits unsigned NUMBER	
8	A		0 .. 252	8 bits unsigned NUMBER	
9	Text			256 bits STRING_LZ	

0x1FF00: PGN 130816 - SonicHub: Control

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	9

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	



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2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	9: Control	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Item		0 .. 252	8 bits lookup FUSION_MUTE_COMMAND	

0x1FF00: PGN 130816 - SonicHub: FM Radio

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	12

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 43 bytes long and contains 12 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	12: FM Radio	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Item		0 .. 252	8 bits lookup SONICHUB_TUNING	
8	Frequency		Hz 0 .. 4294967292	32 bits unsigned NUMBER	
9	Noise level		0 .. 2	2 bits unsigned NUMBER	
10	Signal level		0 .. 13	4 bits unsigned NUMBER	
11	Reserved			2 bits RESERVED	
12	Text			256 bits STRING_LZ	

0x1FF00: PGN 130816 - SonicHub: Playlist

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	13

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 23 bytes long and contains 12 fields. The default priority (as observed in the field) is 7.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	13: Playlist	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Item		0 .. 252	8 bits lookup SONICHUB_PLAYLIST	
8	A		0 .. 252	8 bits unsigned NUMBER	
9	Current Track		0 .. 4294967292	32 bits unsigned NUMBER	
10	Tracks		0 .. 4294967292	32 bits unsigned NUMBER	
11	Length		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	
12	Position in track		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	

0x1FF00: PGN 130816 - SonicHub: Track

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	14

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 41 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	14: Track	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Item		0 .. 4294967292	32 bits unsigned NUMBER	
8	Text			256 bits STRING_LZ	

0x1FF00: PGN 130816 - SonicHub: Artist

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	15

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect



- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This fast-packet PGN is 41 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	15: Artist	0 .. 252	8 bits lookup SONICHUB COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB CONTROL	
7	Item		0 .. 4294967292	32 bits unsigned NUMBER	
8	Text			256 bits STRING LZ	

0x1FF00: PGN 130816 - SonicHub: Album

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	16

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 41 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	16: Album	0 .. 252	8 bits lookup SONICHUB COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB CONTROL	
7	Item		0 .. 4294967292	32 bits unsigned NUMBER	
8	Text			256 bits STRING LZ	

0x1FF00: PGN 130816 - SonicHub: Menu Item

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	19

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

- The default transmission interval is not known

This fast-packet PGN is 44 bytes long and contains 11 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	19: Menu Item	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Item		0 .. 4294967292	32 bits unsigned NUMBER	
8	C		0 .. 252	8 bits unsigned NUMBER	
9	D		0 .. 252	8 bits unsigned NUMBER	
10	E		0 .. 252	8 bits unsigned NUMBER	
11	Text			256 bits STRING_LZ	

0x1FF00: PGN 130816 - SonicHub: Zones

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	20

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	20: Zones	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Zones		0 .. 252	8 bits unsigned NUMBER	

0x1FF00: PGN 130816 - SonicHub: Max Volume

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	23

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect



- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This fast-packet PGN is 7 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	23: Max Volume	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Zone		0 .. 252	8 bits unsigned NUMBER	
8	Level		0 .. 252	8 bits unsigned NUMBER	

0x1FF00: PGN 130816 - SonicHub: Volume

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	24

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 7 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	24: Volume	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Zone		0 .. 252	8 bits unsigned NUMBER	
8	Level		0 .. 252	8 bits unsigned NUMBER	

0x1FF00: PGN 130816 - SonicHub: Init #1

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	25

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	25: Init #1	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	

0x1FF00: PGN 130816 - SonicHub: Position

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	48

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	48: Position	0 .. 252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0 .. 252	8 bits lookup SONICHUB_CONTROL	
7	Position		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	

0x1FF00: PGN 130816 - SonicHub: Init #3

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4
Proprietary ID	50

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 7 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	



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3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	50: Init #3	0..252	8 bits lookup SONICHUB_COMMAND	true
6	Control		0..252	8 bits lookup SONICHUB_CONTROL	
7	A		0..252	8 bits unsigned NUMBER	
8	B		0..252	8 bits unsigned NUMBER	

0x1FF00: PGN 130816 - Furuno: Status and Version Report

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4
Proprietary ID	0

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is at least 4 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Proprietary ID	0: Version Report	0..252	8 bits unsigned NUMBER	true
5	A		0..252	8 bits unsigned NUMBER	
6	Status			Variable length STRING_LAU	

0x1FF00: PGN 130816 - Simrad: Text Message

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	50

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 41 bytes long and contains 11 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Reserved			8 bits RESERVED	
5	Proprietary ID	50: Text	0..252	8 bits lookup SIMNET_COMMAND	true

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6	A		0 .. 252	8 bits unsigned NUMBER	
7	B		0 .. 252	8 bits unsigned NUMBER	
8	C		0 .. 252	8 bits unsigned NUMBER	
9	SID		0 .. 252	8 bits unsigned NUMBER	
10	Prio		0 .. 252	8 bits unsigned NUMBER	
11	Text			256 bits STRING FIX	

0x1FF01: PGN 130817 - Navico: Unknown

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 7 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 252	8 bits unsigned NUMBER	
7	D		0 .. 252	8 bits unsigned NUMBER	
8	E		0 .. 252	8 bits unsigned NUMBER	

0x1FF01: PGN 130817 - Lowrance: Product Information

This PGN description applies when the following field(s) match:

Manufacturer Code	140
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 113 bytes long and contains 11 fields. The default priority (as observed in the field) is 7. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	140: Lowrance	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Product Code		0 .. 65532	16 bits unsigned NUMBER	
5	Model			256 bits STRING FIX	

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6	A		0 .. 252	8 bits unsigned NUMBER	
7	B		0 .. 252	8 bits unsigned NUMBER	
8	C		0 .. 252	8 bits unsigned NUMBER	
9	Firmware version			80 bits STRING_FIX	
10	Firmware date			256 bits STRING_FIX	
11	Firmware time			256 bits STRING_FIX	

0x1FF01: PGN 130817 - Furuno: SV control

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 24 bytes long and contains 12 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	F4	Unknown		8 bits BINARY	
5	F5	Unknown		8 bits BINARY	
6	SBAS mode		0 .. 252	8 bits lookup AUTOMATIC_MANUAL	
7	SBAS satellite		0 .. 252	8 bits lookup SBAS_SV	
8	F8	Unknown		16 bits BINARY	
9	GPS disable		0 .. 4294967295	32 bits bitfield DISABLED_SATELLITES	
10	GLONASS disable		0 .. 4294967295	32 bits bitfield DISABLED_SATELLITES	
11	Galileo disable		0 .. 1099511627775	40 bits bitfield DISABLED_SATELLITES	
12	QZSS disable		0 .. 16777215	24 bits bitfield DISABLED_SATELLITES	

0x1FF02: PGN 130818 - Simnet: Reprogram Data

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 223 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	



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2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Version		0 .. 65532	16 bits unsigned NUMBER	
5	Sequence		0 .. 65532	16 bits unsigned NUMBER	
6	Data			1736 bits BINARY	

0x1FF02: PGN 130818 - Furuno: Sensor setup

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 28 bytes long and contains 15 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Rotation smoothing		0.1 s 0 .. 6553.2	16 bits unsigned DURATION	
5	Heading offset		0.1 deg -3276.7 .. 3276.4	16 bits signed NUMBER	
6	Pitch offset		0.1 deg -3276.7 .. 3276.4	16 bits signed NUMBER	
7	Roll offset		0.1 deg -3276.7 .. 3276.4	16 bits signed NUMBER	
8	F8		0 .. 252	8 bits unsigned NUMBER	
9	F9		0 .. 252	8 bits unsigned NUMBER	
10	SOG and COG smoothing		0.001 s -2147483.647 .. 2147483.644	32 bits signed DURATION	
11	3 Axis speed smoothing		0.001 s -2147483.647 .. 2147483.644	32 bits signed DURATION	
12	FC		0 .. 65532	16 bits unsigned NUMBER	
13	3 axis offset		0.00032 % -12.5 .. 12.5	16 bits signed NUMBER	
14	Air pressure offset		10 Pa -99.9 .. 99.9	16 bits signed NUMBER	
15	Air temperature offset		0.1 K -99.9 .. 99.9	16 bits signed NUMBER	

0x1FF03: PGN 130819 - Simnet: Request Reprogram

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect



- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

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This fast-packet PGN is 2 bytes long and contains 3 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

0x1FF04: PGN 130820 - Simnet: Reprogram Status

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The length of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 7 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Reserved			8 bits RESERVED	
5	Status		0 .. 252	8 bits unsigned NUMBER	
6	Reserved			24 bits RESERVED	

0x1FF04: PGN 130820 - Furuno: Unknown 130820

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 7 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 252	8 bits unsigned NUMBER	
7	D		0 .. 252	8 bits unsigned NUMBER	



8	E		0 .. 252	8 bits unsigned NUMBER	
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0x1FF04: PGN 130820 - Fusion: Versions

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32769

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32769: API Version	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	
5	HW Version Major		0 .. 65532	16 bits unsigned NUMBER	
6	HW Version Minor		0 .. 65532	16 bits unsigned NUMBER	
7	SW Version Major		0 .. 252	8 bits unsigned NUMBER	
8	SW Version Minor		0 .. 252	8 bits unsigned NUMBER	
9	Build Number		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Source

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32770

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 8 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32770: Source	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	
5	Source ID		0 .. 252	8 bits unsigned NUMBER	
6	Current Source ID		0 .. 252	8 bits unsigned NUMBER	
7	Source Type		0 .. 252	8 bits lookup FUSION_SOURCE_TYPE	
8	Flags		0 .. 252	8 bits unsigned NUMBER	
9	Source			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: Source Count

This PGN description applies when the following field(s) match:



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Manufacturer Code	419
Industry Code	4
Message ID	32771

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 5 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32771: Source Count	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	
5	Source Count		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Media

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32772

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 23 bytes long and contains 10 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32772: Track Info	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Flags		0 .. 65532	16 bits lookup FUSION_PLAY_STATUS	
7	Track #		0 .. 4294967292	32 bits unsigned NUMBER	
8	Track Count		0 .. 4294967292	32 bits unsigned NUMBER	
9	Length		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	
10	Position in track		0.001 s 0 .. 4294967.292	32 bits unsigned DURATION	

0x1FF04: PGN 130820 - Fusion: Track Name

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32773

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known



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This fast-packet PGN is at least 9 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32773: Track Title	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Index		0 .. 4294967292	32 bits unsigned NUMBER	
7	Track			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: Artist Name

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32774

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 9 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32774: Track Artist	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Index		0 .. 4294967292	32 bits unsigned NUMBER	
7	Artist			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: Album Name

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32775

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 9 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32775: Track Album	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true



6	Index		0 .. 4294967292	32 bits unsigned NUMBER	
7	Album			Variable length STRING LZ	

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0x1FF04: PGN 130820 - Fusion: Device Name

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32801

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 4 bytes long and contains 5 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32801: Unit Name	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Name			Variable length STRING LZ	

0x1FF04: PGN 130820 - Fusion: Zone Name

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32813

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 5 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32813: Zone Name	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Number		0 .. 252	8 bits unsigned NUMBER	true
6	Name			Variable length STRING LZ	

0x1FF04: PGN 130820 - Fusion: Track Position

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32777

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known



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This fast-packet PGN is 8 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32777: Track Progress	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Progress		0.001 s 0 .. 16777.212	24 bits unsigned DURATION	

0x1FF04: PGN 130820 - Fusion: Tuner

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32779

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 11 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32779: Tuner	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits lookup FUSION_RADIO_SOURCE	true
6	Scanning		0 .. 252	8 bits unsigned NUMBER	
7	Frequency		Hz 0 .. 4294967292	32 bits unsigned NUMBER	
8	Signal Strength		0 .. 252	8 bits unsigned NUMBER	
9	Track			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: MARINE_TUNER

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32780

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 7 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	



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4	Message ID	32780: Marine Tuner	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Channel		0..252	8 bits unsigned NUMBER	
7	Signal Strength		0..252	8 bits unsigned NUMBER	
8	Name			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: Marine Squelch

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32781

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32781: Marine Squelch	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Squelch		0..252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Marine Scan Mode

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32782

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32782: Marine Scan Mode	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Scan		0..252	8 bits lookup YES_NO	

0x1FF04: PGN 130820 - Fusion: Menu Item

This PGN description applies when the following field(s) match:

Manufacturer Code	419
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Industry Code	4
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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 11 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32785: Menu Item	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Item Index		0 .. 4294967292	32 bits unsigned NUMBER	
7	Flags		0 .. 252	8 bits unsigned NUMBER	
8	Lock ID		0 .. 252	8 bits unsigned NUMBER	
9	Text			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: Aux Gain

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32787

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32787: Aux Gain	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Gain		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: USB Repeat Status

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32788
ID	9

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32788: Setting	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	ID	9: USB repeat	0 .. 4294967292	32 bits lookup FUSION_SETTING	
6	Status		0 .. 4294967292	32 bits lookup FUSION_REPEAT_STATUS	

0x1FF04: PGN 130820 - Fusion: Setting

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32788

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32788: Setting	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	ID		0 .. 4294967292	32 bits lookup FUSION_SETTING	
6	Value		0 .. 4294967292	32 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Settings

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32789

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 8 bytes long and contains 7 fields. The 2 fields starting at field 6 (with name "ID") form repeating set 1. The set is repeated n times, where n is determined by the value of field 5 (with name "Count".) The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32789: Settings	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Count		0 .. 4294967292	32 bits unsigned NUMBER	
6 Set 1	ID		0 .. 4294967292	32 bits lookup FUSION_SETTING	



7 Set 1	Value		0 .. 4294967292	32 bits unsigned NUMBER	
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0x1FF04: PGN 130820 - Fusion: Mute

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32791

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 5 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32791: Mute	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Mute		0 .. 252	8 bits lookup FUSION MUTE COMMAND	

0x1FF04: PGN 130820 - Fusion: Balance

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32792

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32792: Balance	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone		0 .. 252	8 bits unsigned NUMBER	true
6	VALUE		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Low Pass Filter

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32793

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known



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This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32793: Low Pass Filter	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone		0 .. 252	8 bits unsigned NUMBER	true
6	Filter		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Sublevels

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32794

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32794: Sublevels	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone 1		0 .. 252	8 bits unsigned NUMBER	
6	Zone 2		0 .. 252	8 bits unsigned NUMBER	
7	Zone 3		0 .. 252	8 bits unsigned NUMBER	
8	Zone 4		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: EQ

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32795

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32795: Tone	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone		0 .. 252	8 bits unsigned NUMBER	true



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6	Bass		-127 .. 124	8 bits signed NUMBER	
7	Mid		-127 .. 124	8 bits signed NUMBER	
8	Treble		-127 .. 124	8 bits signed NUMBER	

0x1FF04: PGN 130820 - Fusion: Volume Limits

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32796

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32796: Volume Limits	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone 1 Volume Limit		0 .. 252	8 bits unsigned NUMBER	
6	Zone 2 Volume Limit		0 .. 252	8 bits unsigned NUMBER	
7	Zone 3 Volume Limit		0 .. 252	8 bits unsigned NUMBER	
8	Zone 4 Volume Limit		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Volumes

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32797

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 8 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32797: Volume	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone 1		0 .. 252	8 bits unsigned NUMBER	
6	Zone 2		0 .. 252	8 bits unsigned NUMBER	
7	Zone 3		0 .. 252	8 bits unsigned NUMBER	
8	Zone 4		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Capabilities

This PGN description applies when the following field(s) match:



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Manufacturer Code	419
Industry Code	4
Message ID	32798

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 14 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32798: Capabilities	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone 1		0 .. 65532	16 bits unsigned NUMBER	
6	Zone 2		0 .. 65532	16 bits unsigned NUMBER	
7	Zone 3		0 .. 65532	16 bits unsigned NUMBER	
8	Zone 4		0 .. 65532	16 bits unsigned NUMBER	
9	Global		0 .. 65532	16 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Line Level Control

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32799

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Message ID	32799: Line Level Control	0 .. 65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Zone		0 .. 252	8 bits unsigned NUMBER	true
6	Control		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: Power State

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32800

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 5 bytes long and contains 5 fields. The default priority (as observed in the field) is 7.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Message ID	32800: Power	0..65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	State		0..252	8 bits lookup FUSION POWER STATE	

0x1FF04: PGN 130820 - Fusion: SiriusXM

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32802

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 10 bytes long and contains 9 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Message ID	32802: Sirius	0..65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Com State		0..255	8 bits lookup FUSION SIRIUS COM STATE	
7	Alert		0..255	8 bits lookup FUSION SIRIUS COM STATE	
8	Advisory Channel		0..65532	16 bits unsigned NUMBER	
9	Tuning Mode		0..252	8 bits lookup FUSION SIRIUS TUNING MODE	

0x1FF04: PGN 130820 - Fusion: SiriusXM Channel

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32804

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 7 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY CODE	
4	Message ID	32804: SiriusXM Channel	0..65532	16 bits lookup FUSION STATUS MESSAGE ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Channel Number		0..65532	16 bits unsigned NUMBER	



7	Channel			Variable length STRING LZ	
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0x1FF04: PGN 130820 - Fusion: SiriusXM Title

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32805

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 7 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32805: SiriusXM Title	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Channel		0..65532	16 bits unsigned NUMBER	
7	Title			Variable length STRING LZ	

0x1FF04: PGN 130820 - Fusion: SiriusXM Artist

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32806

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 7 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32806: SiriusXM Artist	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Channel		0..65532	16 bits unsigned NUMBER	
7	Artist			Variable length STRING LZ	

0x1FF04: PGN 130820 - Fusion: SiriusXM Content Info

This PGN description applies when the following field(s) match:

Manufacturer Code	419
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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 7 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32807: SiriusXM Genre	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Channel		0..65532	16 bits unsigned NUMBER	
7	Genre			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: SiriusXM Category

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32808

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 7 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32808: SiriusXM Category	0..65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0..252	8 bits unsigned NUMBER	true
6	Channel		0..65532	16 bits unsigned NUMBER	
7	Name			Variable length STRING_LZ	

0x1FF04: PGN 130820 - Fusion: SiriusXM Signal

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32809

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 6 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0..2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0..6	3 bits lookup INDUSTRY_CODE	



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4	Message ID	32809: SiriusXm Signal	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Signal		0 .. 252	8 bits unsigned NUMBER	

0x1FF04: PGN 130820 - Fusion: SiriusXM Presets

This PGN description applies when the following field(s) match:

Manufacturer Code	419
Industry Code	4
Message ID	32812

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is at least 6 bytes long and contains 7 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	419: Fusion Electronics	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	32812: SiriusXM Presets	0 .. 65532	16 bits lookup FUSION_STATUS_MESSAGE_ID	true
5	Source ID		0 .. 252	8 bits unsigned NUMBER	true
6	Count		0 .. 252	8 bits unsigned NUMBER	
7	Values	Preset Values. There will be Count / 4 presets stored		Field 6 defines the Variable length BINARY	

0x1FF05: PGN 130821 - Navico: ASCII Data

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 233 bytes long and contains 5 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	Message			1840 bits STRING_FIX	

0x1FF05: PGN 130821 - Furuno: Unknown 130821

This PGN description applies when the following field(s) match:



Manufacturer Code	1855
Industry Code	4

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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 13 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	A		0 .. 252	8 bits unsigned NUMBER	
6	B		0 .. 252	8 bits unsigned NUMBER	
7	C		0 .. 252	8 bits unsigned NUMBER	
8	D		0 .. 252	8 bits unsigned NUMBER	
9	E		0 .. 252	8 bits unsigned NUMBER	
10	F		0 .. 252	8 bits unsigned NUMBER	
11	G		0 .. 252	8 bits unsigned NUMBER	
12	H		0 .. 252	8 bits unsigned NUMBER	
13	I		0 .. 252	8 bits unsigned NUMBER	

0x1FF06: PGN 130822 - Navico: Unknown 1

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 233 bytes long and contains 4 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Data			1848 bits BINARY	

0x1FF07: PGN 130823 - Maretron: Proprietary Temperature High Range

This PGN description applies when the following field(s) match:

Manufacturer Code	137
Industry Code	4



This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 8 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	Instance		0 .. 252	8 bits unsigned NUMBER	true
6	Source		0 .. 252	8 bits lookup TEMPERATURE_SOURCE	true
7	Actual Temperature		0.1 K 0 .. 6553.2	16 bits unsigned NUMBER	
8	Set Temperature		0.1 K 0 .. 6553.2	16 bits unsigned NUMBER	

0x1FF08: PGN 130824 - B&G: key-value data

This PGN description applies when the following field(s) match:

Manufacturer Code	381
Industry Code	4

Contains any number of key/value pairs, sent by various B&G devices such as MFDs and Sailing Processors.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- One or more of the lookup fields contain missing or incorrect values

This fast-packet PGN is at least 2 bytes long and contains 6 fields. The 3 fields starting at field 4 (with name "Key") form repeating set 1. The set is repeated until there is no more data in the PGN. The default priority (as observed in the field) is 2. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	381: B & G	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4 Set 1	Key		0 .. 4092	12 bits lookup BANDG_KEY_VALUE that defines the next variable field's type.	true
5 Set 1	Length	Length of field 6	0 .. 13	4 bits unsigned DYNAMIC_FIELD_LENGTH	
6 Set 1	Value	Data value		Variable length DYNAMIC_FIELD_VALUE	

0x1FF08: PGN 130824 - Maretron: Annunciator

This PGN description applies when the following field(s) match:

Manufacturer Code	137
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect

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- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This fast-packet PGN is 9 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Field 4		0 .. 252	8 bits unsigned NUMBER	
5	Field 5		0 .. 252	8 bits unsigned NUMBER	
6	Field 6		0 .. 65532	16 bits unsigned NUMBER	
7	Field 7		0 .. 252	8 bits unsigned NUMBER	
8	Field 8		0 .. 65532	16 bits unsigned NUMBER	

0x1FF09: PGN 130825 - Navico: Unknown 2

This PGN description applies when the following field(s) match:

Manufacturer Code	275
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 4 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	275: Navico	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Data			80 bits BINARY	

0x1FF0B: PGN 130827 - Lowrance: unknown

This PGN description applies when the following field(s) match:

Manufacturer Code	140
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 10 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	140: Lowrance	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	

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3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 252	8 bits unsigned NUMBER	
7	D		0 .. 252	8 bits unsigned NUMBER	
8	E		0 .. 65532	16 bits unsigned NUMBER	
9	F		0 .. 65532	16 bits unsigned NUMBER	

0x1FF0C: PGN 130828 - Simnet: Set Serial Number

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF0F: PGN 130831 - Suzuki: Engine and Storage Device Config

This PGN description applies when the following field(s) match:

Manufacturer Code	586
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	586: Suzuki Motor Corporation	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF10: PGN 130832 - Simnet: Fuel Used - High Resolution

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF11: PGN 130833 - B&G: User and Remote rename

This PGN description applies when the following field(s) match:

Manufacturer Code	381
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 30 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	381: B & G	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Data Type		0 .. 4092	12 bits lookup BANDG_KEY_VALUE that defines the next variable field's type.	true
5	Length	Length of field 8	0 .. 13	4 bits unsigned NUMBER	
6	Reserved			8 bits RESERVED	
7	Decimals		0 .. 254	8 bits lookup BANDG_DECIMALS	
8	Short name			64 bits STRING_FIX	
9	Long name			128 bits STRING_FIX	

0x1FF12: PGN 130834 - Simnet: Engine and Tank Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet



- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

0x1FF13: PGN 130835 - Simnet: Set Engine and Tank Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

0x1FF14: PGN 130836 - Simnet: Fluid Level Sensor Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 14 bytes long and contains 12 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	C		0 .. 252	8 bits unsigned NUMBER	
5	Device		0 .. 252	8 bits unsigned NUMBER	true
6	Instance		0 .. 252	8 bits unsigned NUMBER	true
7	F		0 .. 13	4 bits unsigned NUMBER	
8	Tank type		0 .. 13	4 bits lookup TANK TYPE	
9	Capacity		0.1 L 0 .. 429496729.2	32 bits unsigned NUMBER	

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10	G		0 .. 252	8 bits unsigned NUMBER	
11	H		32767 .. 32764	16 bits signed NUMBER	
12	I		127 .. 124	8 bits signed NUMBER	

0x1FF14: PGN 130836 - Maretron: Switch Status Counter

This PGN description applies when the following field(s) match:

Manufacturer Code	137
Industry Code	4

This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 23 bytes long and contains 12 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 15000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Instance		0 .. 252	8 bits unsigned NUMBER	true
5	Indicator Number		0 .. 252	8 bits unsigned NUMBER	
6	Start Date		d 0 .. 65532	16 bits unsigned DATE	
7	Start Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
8	OFF Counter		0 .. 4294967292	32 bits unsigned NUMBER	
9	ON Counter		0 .. 4294967292	32 bits unsigned NUMBER	
10	ERROR Counter		0 .. 4294967292	32 bits unsigned NUMBER	
11	Switch Status		0 .. 2	2 bits lookup OFF ON	
12	Reserved			6 bits RESERVED	

0x1FF15: PGN 130837 - Simnet: Fuel Flow Turbine Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

0x1FF15: PGN 130837 - Maretron: Switch Status Timer

This PGN description applies when the following field(s) match:



Manufacturer Code	137
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This PGN is believed to be (nearly) fully reverse engineered and seems to work in most practical purposes.

This fast-packet PGN is 23 bytes long and contains 12 fields. The default priority (as observed in the field) is 6. It is normally transmitted every 15000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	137: Maretron	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Instance		0 .. 252	8 bits unsigned NUMBER	true
5	Indicator Number		0 .. 252	8 bits unsigned NUMBER	
6	Start Date		d 0 .. 65532	16 bits unsigned DATE	
7	Start Time	Seconds since midnight	0.0001 s 0 .. 86401	32 bits TIME	
8	Accumulated OFF Period		s 0 .. 4294967292	32 bits unsigned DURATION	
9	Accumulated ON Period		s 0 .. 4294967292	32 bits unsigned DURATION	
10	Accumulated ERROR Period		s 0 .. 4294967292	32 bits unsigned DURATION	
11	Switch Status		0 .. 2	2 bits lookup OFF_ON	
12	Reserved			6 bits RESERVED	

0x1FF16: PGN 130838 - Simnet: Fluid Level Warning

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF17: PGN 130839 - Simnet: Pressure Sensor Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect



- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

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This fast-packet PGN is 2 bytes long and contains 3 fields.

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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF18: PGN 130840 - Simnet: Data User Group Configuration

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields. The default priority (as observed in the field) is 3.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF1A: PGN 130842 - Simnet: AIS Class B static data (msg 24 Part A)

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Message ID	0

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 29 bytes long and contains 9 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	0: Msg 24 Part A	0 .. 61	6 bits unsigned NUMBER	true
5	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	



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6	D		0 .. 252	8 bits unsigned NUMBER	
7	E		0 .. 252	8 bits unsigned NUMBER	
8	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
9	Name			160 bits STRING FIX	

0x1FF1A: PGN 130842 - Furuno: Six Degrees Of Freedom Movement

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 29 bytes long and contains 12 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		-2147483647 .. 2147483644	32 bits signed NUMBER	
5	B		-2147483647 .. 2147483644	32 bits signed NUMBER	
6	C		-2147483647 .. 2147483644	32 bits signed NUMBER	
7	D		127 .. 124	8 bits signed NUMBER	
8	E		-2147483647 .. 2147483644	32 bits signed NUMBER	
9	F		-2147483647 .. 2147483644	32 bits signed NUMBER	
10	G		32767 .. 32764	16 bits signed NUMBER	
11	H		32767 .. 32764	16 bits signed NUMBER	
12	I		32767 .. 32764	16 bits signed NUMBER	

0x1FF1A: PGN 130842 - Simnet: AIS Class B static data (msg 24 Part B)

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Message ID	1

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 37 bytes long and contains 18 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	



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2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Message ID	1: Msg 24 Part B	0 .. 61	6 bits unsigned NUMBER	true
5	Repeat Indicator		0 .. 3	2 bits lookup REPEAT_INDICATOR	
6	D		0 .. 252	8 bits unsigned NUMBER	
7	E		0 .. 252	8 bits unsigned NUMBER	
8	User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
9	Type of ship		0 .. 252	8 bits lookup SHIP_TYPE	
10	Vendor ID			56 bits STRING_FIX	
11	Callsign			56 bits STRING_FIX	
12	Length		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
13	Beam		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
14	Position reference from Starboard		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
15	Position reference from Bow		0.1 m 0 .. 6553.2	16 bits unsigned NUMBER	
16	Mothership User ID		2000000 .. 999999999	32 bits unsigned MMSI	true
17	Spare			6 bits SPARE	
18	Reserved			2 bits RESERVED	

0x1FF1B: PGN 130843 - Furuno: Heel Angle, Roll Information

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 10 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	Yaw		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
7	Pitch		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	
8	Roll		0.0001 rad -3.1415926 .. 3.1415926	16 bits signed NUMBER	

0x1FF1B: PGN 130843 - Simnet: Sonar Status, Frequency and DSP Voltage



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This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

0x1FF1D: PGN 130845 - Furuno: Multi Sats In View Extended

This PGN description applies when the following field(s) match:

Manufacturer Code	1855
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

0x1FF1D: PGN 130845 - Simnet: Key Value

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is at least 10 bytes long and contains 11 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	

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2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Repeat Indicator		0 .. 252	8 bits lookup REPEAT_INDICATOR	
6	Display Group		0 .. 252	8 bits lookup SIMNET_DISPLAY_GROUP	
7	Reserved			8 bits RESERVED	
8	Key		0 .. 65532	16 bits lookup SIMNET_KEY_VALUE that defines the next variable field's type.	true
9	Spare			8 bits SPARE	
10	MinLength	Possibly the length of data field; probably something else	0 .. 252	8 bits unsigned NUMBER	
11	Value	Data value		Variable length DYNAMIC_FIELD_VALUE	

0x1FF1E: PGN 130846 - Simnet: Parameter Set

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is at least 11 bytes long and contains 11 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	B	00, 01 or FF observed	0 .. 252	8 bits unsigned NUMBER	
6	Display Group		0 .. 252	8 bits lookup SIMNET_DISPLAY_GROUP	
7	D	Various values observed	0 .. 65532	16 bits unsigned NUMBER	
8	Key		0 .. 65532	16 bits lookup SIMNET_KEY_VALUE that defines the next variable field's type.	true
9	Spare			8 bits SPARE	
10	Length	Possibly the length of data field; probably something else	0 .. 252	8 bits unsigned NUMBER	
11	Value	Data value		Variable length DYNAMIC_FIELD_VALUE	

0x1FF1E: PGN 130846 - Furuno: Motion Sensor Status Extended

This PGN description applies when the following field(s) match:

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This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 2 bytes long and contains 3 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1855: Furuno	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	

0x1FF1F: PGN 130847 - SeaTalk: Node Statistics

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 10 bytes long and contains 8 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Product Code		0 .. 65532	16 bits unsigned NUMBER	
5	Year		0 .. 252	8 bits unsigned NUMBER	
6	Month		0 .. 252	8 bits unsigned NUMBER	
7	Device Number		0 .. 65532	16 bits unsigned NUMBER	
8	Node Voltage		0.01 V 0 .. 655.32	16 bits unsigned NUMBER	

0x1FF20: PGN 130848 - SeaTalk: Waypoint Information

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 31 bytes long and contains 9 fields. The default priority (as observed in the field) is 7. It is normally transmitted every 1000 milliseconds.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	SID		0 .. 252	8 bits unsigned NUMBER	
5	Waypoint Name			128 bits STRING FIX	
6	Waypoint Sequence			32 bits STRING FIX	
7	Bearing to Waypoint, True		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
8	Bearing to Waypoint, Magnetic		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
9	Distance to Waypoint		0.01 m 0 .. 42949672.92	32 bits unsigned NUMBER	

0x1FF22: PGN 130850 - Simnet: Command AP Standby

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	10
Event	6

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 12 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	10: AP Command	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	6: Standby	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP NoDrift



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This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	10
Event	12

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 12 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	10: AP Command	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	12: No Drift mode	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP Wind

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	10
Event	15

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 12 fields. The default priority (as observed in the field) is 2.



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Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	10: AP Command	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	15: Wind mode	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP Nav

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	10
Event	10

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 12 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	10: AP Command	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	10: Nav mode	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	



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10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP Heading

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	10
Event	9

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 11 bytes long and contains 12 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	10: AP Command	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	9: Heading mode	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP Tack

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	10
Event	17

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect



- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

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This fast-packet PGN is 12 bytes long and contains 13 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	10: AP Command	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	17: Tack	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown A		0 .. 252	8 bits unsigned NUMBER	
10	Unknown B		0 .. 252	8 bits unsigned NUMBER	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	
13	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP Follow Up

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Command Type	2
Event	14

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 13 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	



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6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type	2: Follow Up	0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	14: Follow Up mode	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	
13	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Command AP Change Course

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255
Event	26

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 12 fields. The default priority (as observed in the field) is 2.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type		0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event	26: Change course	0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Unknown		0 .. 252	8 bits unsigned NUMBER	
10	Direction		0 .. 252	8 bits lookup SIMNET_DIRECTION	
11	Angle	Absolute change in desired attitude, generally 1 or 10 degrees	0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
12	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Alarm

This PGN description applies when the following field(s) match:



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Manufacturer Code	1857
Industry Code	4
Proprietary ID	1

There may follow a PGN 130856 'Simnet: Alarm Text' message with a textual explanation of the alarm

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 12 bytes long and contains 11 fields. The default priority (as observed in the field) is 2. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	1: Alarm	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Reserved			8 bits RESERVED	
8	Alarm		0 .. 65532	16 bits lookup SIMNET_ALARM	
9	Message ID		0 .. 65532	16 bits unsigned NUMBER	
10	F		0 .. 252	8 bits unsigned NUMBER	
11	G		0 .. 252	8 bits unsigned NUMBER	

0x1FF22: PGN 130850 - Simnet: AP command

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	255

Other Simnet AP command

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 14 bytes long and contains 15 fields. The default priority (as observed in the field) is 2. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	



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4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID	255: Autopilot	0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Command Type		0 .. 252	8 bits lookup SIMNET_AP_COMMAND_TYPE	
8	Event		0 .. 252	8 bits lookup SIMNET_AP_EVENTS	
9	Reserved			8 bits RESERVED	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	
13	Reserved			8 bits RESERVED	
14	Reserved			8 bits RESERVED	
15	Reserved			8 bits RESERVED	

0x1FF22: PGN 130850 - Simnet: Event

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

Other Simnet Event

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 12 bytes long and contains 13 fields. The default priority (as observed in the field) is 2. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Address	NMEA 2000 address of commanded device	0 .. 252	8 bits unsigned NUMBER	
5	Reserved			8 bits RESERVED	
6	Proprietary ID		0 .. 255	8 bits lookup SIMNET_EVENT_COMMAND	true
7	Reserved			8 bits RESERVED	
8	Reserved			8 bits RESERVED	
9	Reserved			8 bits RESERVED	
10	Reserved			8 bits RESERVED	
11	Reserved			8 bits RESERVED	
12	Reserved			8 bits RESERVED	
13	Reserved			8 bits RESERVED	



0x1FF23: PGN 130851 - Simnet: Event Reply: AP command

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4
Proprietary ID	2

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The default transmission interval is not known

This fast-packet PGN is 12 bytes long and contains 11 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	Proprietary ID	2: AP command	0 .. 255	8 bits lookup SIMNET EVENT COMMAND	true
5	B		0 .. 65532	16 bits unsigned NUMBER	
6	Address	NMEA 2000 address of controlling device	0 .. 252	8 bits unsigned NUMBER	
7	Event		0 .. 252	8 bits lookup SIMNET AP EVENTS	
8	C		0 .. 252	8 bits unsigned NUMBER	
9	Direction		0 .. 252	8 bits lookup SIMNET DIRECTION	
10	Angle		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
11	G		0 .. 252	8 bits unsigned NUMBER	

0x1FF28: PGN 130856 - Simnet: Alarm Message

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

Usually accompanied by a PGN 130850 'Simnet: Alarm' message with the same information in binary form.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 229 bytes long and contains 7 fields. It is not transmitted at a regular interval, but when data is available or it is requested.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	

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4	Message ID		0 .. 65532	16 bits unsigned NUMBER	
5	B		0 .. 252	8 bits unsigned NUMBER	
6	C		0 .. 252	8 bits unsigned NUMBER	
7	Text			1784 bits STRING FIX	

0x1FF2C: PGN 130860 - Simnet: AP Unknown 4

This PGN description applies when the following field(s) match:

Manufacturer Code	1857
Industry Code	4

Seen as sent by AC-42 and H5000 AP only so far.

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect

This fast-packet PGN is 23 bytes long and contains 9 fields. The default priority (as observed in the field) is 7. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1857: Simrad	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	A		0 .. 252	8 bits unsigned NUMBER	
5	B		2147483647 .. 2147483644	32 bits signed NUMBER	
6	C		2147483647 .. 2147483644	32 bits signed NUMBER	
7	D		0 .. 4294967292	32 bits unsigned NUMBER	
8	E		2147483647 .. 2147483644	32 bits signed NUMBER	
9	F		0 .. 4294967292	32 bits unsigned NUMBER	

0x1FF40: PGN 130880 - Airmar: Additional Weather Data

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY CODE	
4	C		0 .. 252	8 bits unsigned NUMBER	



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5	Apparent Windchill Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	True Windchill Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
7	Dewpoint		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	

Source: <http://www.airmartechonology.com/uploads/installguide/PB2000UserManual.pdf>

0x1FF41: PGN 130881 - Airmar: Heater Control

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 9 bytes long and contains 7 fields.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	C		0 .. 252	8 bits unsigned NUMBER	
5	Plate Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
6	Air Temperature		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	
7	Dewpoint		0.01 K 0 .. 655.32	16 bits unsigned NUMBER	

Source: <http://www.airmartechonology.com/uploads/installguide/PB2000UserManual.pdf>

0x1FF66: PGN 130918 - SeaTalk: Route Information

This PGN description applies when the following field(s) match:

Manufacturer Code	1851
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet

This fast-packet PGN is 47 bytes long and contains 11 fields. The default priority (as observed in the field) is 7. It is normally transmitted every 1000 milliseconds.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	1851: Raymarine	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	



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3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Current Waypoint Sequence		0 .. 65532	16 bits unsigned NUMBER	
5	Current Waypoint Name			128 bits STRING_FIX	
6	Next Waypoint Sequence		0 .. 65532	16 bits unsigned NUMBER	
7	Next Waypoint Name			128 bits STRING_FIX	
8	Unknown		0 .. 252	8 bits unsigned NUMBER	
9	Distance, Position to Next Waypoint		m 0 .. 4294967292	32 bits unsigned NUMBER	
10	Bearing, Position to Next Waypoint, True		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	
11	Bearing, Current Waypoint to Next Waypoint, True		0.0001 rad 0 .. 6.2831852	16 bits unsigned NUMBER	

0x1FF80: PGN 130944 - Airmar: POST

This PGN description applies when the following field(s) match:

Manufacturer Code	135
Industry Code	4

This PGN is not fully reverse engineered. Some aspects that are known to be missing/incorrect:

- The list of fields is incomplete; some fields maybe be missing or their attributes may be incorrect
- The length of one or more fields is likely incorrect
- The resolution of one or more fields is likely incorrect
- The PGN has not been seen in any logfiles yet
- The default transmission interval is not known

This fast-packet PGN is 6 bytes long and contains 8 fields. The default priority (as observed in the field) is 7.

Field #	Field Name	Description	Unit	Type	PartOfPrimaryKey
1	Manufacturer Code	135: Airmar	0 .. 2044	11 bits lookup MANUFACTURER_CODE	
2	Reserved			2 bits RESERVED	
3	Industry Code	4: Marine Industry	0 .. 6	3 bits lookup INDUSTRY_CODE	
4	Control		0 .. 1	1 bits lookup AIRMAR_POST_CONTROL	
5	Reserved			7 bits RESERVED	
6	Number of ID/test result pairs to follow		0 .. 252	8 bits unsigned NUMBER	
7	Test ID	See Airmar docs for table of IDs and failure codes; these lookup values are for DST200	0 .. 252	8 bits lookup AIRMAR_POST_ID	
8	Test result	Values other than 0 are failure codes	0 .. 252	8 bits unsigned NUMBER	

Source: <http://www.airmartechonology.com/uploads/installguide/DST200UserManual.pdf>

Physical quantities

A lot of fields represent a physical, observable, quantity.

ELECTRICAL_CURRENT - Electrical current

In the CAN data these are expressed in Ampere, abbreviated as A.



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For more information see: https://en.wikipedia.org/wiki/Electric_current

ELECTRICAL_CHARGE - Electrical charge

In the CAN data these are expressed in Coulomb, abbreviated as C.

For more information see: https://en.wikipedia.org/wiki/Electric_charge

ELECTRICAL_ENERGY - Electrical energy

The amount of electricity used or stored. The base unit used in NMEA 2000 is the Kilo Watt Hour (kWh) which is equivalent to 3.6e6 J (Joules).

In the CAN data these are expressed in Kilo Watt Hour, abbreviated as kWh.

For more information see: https://en.wikipedia.org/wiki/Electrical_energy

ELECTRICAL_POWER - Electrical power

The amount of energy transferred or converted per unit time.

In the CAN data these are expressed in Watt, abbreviated as W.

For more information see: https://en.wikipedia.org/wiki/Electrical_power

ELECTRICAL_APPARENT_POWER - AC apparent power

The amount of power transferred where the current and voltage are in phase.

In the CAN data these are expressed in Volt Ampere, abbreviated as VA.

For more information see: <https://en.wikipedia.org/wiki/Volt-ampere>

ELECTRICAL_REACTIVE_POWER - AC reactive power

The amount of power transferred where the current and voltage are not in phase.

In the CAN data these are expressed in Volt Ampere Reactive, abbreviated as VAR.

For more information see: <https://en.wikipedia.org/wiki/Volt-ampere#Reactive>

POTENTIAL_DIFFERENCE - Potential difference

In the CAN data these are expressed in Volt, abbreviated as V.

For more information see: <https://en.wikipedia.org/wiki/Voltage>

POWER_FACTOR - Power Factor

Used in AC circuits only, the ratio of the real power absorbed by the load to the apparent power flowing in the circuit. If less than one, the voltage and current are not in phase.

In the CAN data these are expressed in Cos(Phi), abbreviated as Cos Phi.

For more information see: https://en.wikipedia.org/wiki/Power_factor

LENGTH - Length

The physical size in one dimension of an object.

In the CAN data these are expressed in Meter, abbreviated as m.

For more information see: <https://en.wikipedia.org/wiki/Length>

DISTANCE - Distance

The amount of separation between two objects.

In the CAN data these are expressed in meter, abbreviated as m.

For more information see: <https://en.wikipedia.org/wiki/Distance>



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SPEED - Speed

The velocity, or length per unit of time.

In the CAN data these are expressed in meter per second, abbreviated as m/s.

For more information see: <https://en.wikipedia.org/wiki/Speed>

ANGLE - Angle

All standardized PGNs seen so far all use radians, but some manufacturer specific PGNs use degrees (deg).

In the CAN data these are expressed in radian, abbreviated as rad.

For more information see: <https://en.wikipedia.org/wiki/Angle>

ANGULAR_VELOCITY - Angular velocity

The speed at which a measured angle changes

In the CAN data these are expressed in radians per second, abbreviated as rad/s.

For more information see: https://en.wikipedia.org/wiki/Angular_velocity

VOLUME - Volume

A measure of occupied three-dimensional space.

In the CAN data these are expressed in liter, abbreviated as L.

For more information see: <https://en.wikipedia.org/wiki/Volume>

VOLUMETRIC_FLOW - Volumetric flow

The volume of fluid which passes per unit time.

In the CAN data these are expressed in liter per hour, abbreviated as L/h.

For more information see: https://en.wikipedia.org/wiki/Volumetric_flow_rate

MAGNETIC_FIELD - Magnetic field

In the CAN data these are expressed in Tesla, abbreviated as T.

For more information see: https://en.wikipedia.org/wiki/Magnetic_field

FREQUENCY - Frequency

In the CAN data these are expressed in Hertz, abbreviated as Hz.

For more information see: https://en.wikipedia.org/wiki/Radio_frequency

DATE - Date

A calendar date is a reference to a particular day in time, in NMEA 2000 expressed as the number of days since 1970-01-01 (UNIX epoch).

In the CAN data these are expressed in days, abbreviated as d.

For more information see: https://en.wikipedia.org/wiki/Calendar_date

TIME - Time

Time is what clocks measure. We use time to place events in sequence one after the other, and we use time to compare how long events last. Absolute times in NMEA2000 are expressed as seconds since midnight(in an undefined timezone)

In the CAN data these are expressed in Second, abbreviated as s.

For more information see: <https://en.wikipedia.org/wiki/Time>

DURATION - Duration of an event or interval between two points in time



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In the CAN data these are expressed in Second, abbreviated as s.

GEOGRAPHICAL_LATITUDE - Geographical latitude

Combined with a geographical longitude the two fields form a unique point on earth, when height is disregarded.

In the CAN data these are expressed in degree, abbreviated as deg.

For more information see: https://en.wikipedia.org/wiki/Geographic_coordinate_system

GEOGRAPHICAL_LONGITUDE - Geographical longitude

Combined with a geographical latitude the two fields form a unique point on earth, when height is disregarded.

In the CAN data these are expressed in degree, abbreviated as deg.

For more information see: https://en.wikipedia.org/wiki/Geographic_coordinate_system

TEMPERATURE - Temperature

In the CAN data these are expressed in Kelvin, abbreviated as K.

For more information see: <https://en.wikipedia.org/wiki/Temperature>

PRESSURE - Pressure

In the CAN data these are expressed in Pascal, abbreviated as Pa.

For more information see: <https://en.wikipedia.org/wiki/Pressure>

PRESSURE_RATE - Pressure rate

How the pressure changes over time.

In the CAN data these are expressed in Pascal per hour, abbreviated as Pa/hr.

For more information see: <https://en.wikipedia.org/wiki/Pressure>

CONCENTRATION - Concentration of one substance in another, in this marine context usually the amount of salts in water

In the CAN data these are expressed in parts per million, abbreviated as ppm.

For more information see: https://www.engineeringtoolbox.com/water-salinity-d_1251.html

SIGNAL_STRENGTH - Signal strength expressed in dB with respect to 1 uV/m

In the CAN data these are expressed in decibel, abbreviated as dB.

For more information see: https://en.wikipedia.org/wiki/Signal_strength_in_telecommunications

SIGNAL_TO_NOISE_RATIO - Signal-to-noise ratio

In the CAN data these are expressed in decibel, abbreviated as dB.

For more information see: https://en.wikipedia.org/wiki/Signal-to-noise_ratio

Field types

All fields are of one of the following field types.

NUMBER - Number

Encoding: Binary numbers are little endian. Number fields that are at least two bits in length use the highest positive value to represent unknown. Number fields with at least 7 as maximum (3 bits unsigned, 4 bits signed) use the highest value minus one as an error indicator. This is likely also true for numbers where 3 is the maximum value, but there are few fields that have this length -- certainly as a number, there are a lot of lookup fields of two bits length. For signed numbers the maximum values are the maximum positive value and that minus 1, not the all-ones bit encoding which is the maximum negative value.

For more information see: https://en.wikipedia.org/wiki/Binary_number



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FLOAT - 32 bit IEEE-754 floating point number

32 bits signed number

For more information see: https://en.wikipedia.org/wiki/IEEE_754

DECIMAL - A unsigned numeric value represented with 2 decimal digits per byte

Encoding: Each byte represent 2 digits, so 1234 is represented by 2 bytes containing 0x12 and 0x34. A number with an odd number of digits will have 0 as the first digit in the first byte.

For more information see: https://en.wikipedia.org/wiki/Binary-coded_decimal

LOOKUP - Number value where each value encodes for a distinct meaning

For almost all lookups the list of values is known with some precision, but it is quite possible that a value occurs that has no corresponding textual explanation. Generally the same special values for Error (max value - 1) and Unknown (max value) that are used for plain numbers are also valid for lookups, but some lookups define a lookup key/value pair for the maximum possible value in that field. In fact, recent additions to PGNs such as alerts have two bit lookup fields that specify four actual valid values. This makes it impossible to send an unknown value for these fields.

Encoding: Each lookup has a LookupEnumeration defining what the possible values mean

INDIRECT_LOOKUP - Number value where each value encodes for a distinct meaning but the meaning also depends on the value in another field

For almost all lookups the list of values is known with some precision, but it is quite possible that a value occurs that has no corresponding textual explanation.

Encoding: Each lookup has a LookupIndirectEnumeration defining what the possible values mean

BITLOOKUP - Number value where each bit value encodes for a distinct meaning

For almost all lookups the list of values is known with some precision, but it is quite possible that a value occurs that has no corresponding textual explanation.

Encoding: Each LookupBit has a LookupBitEnumeration defining what the possible values mean. A bitfield can have any combination of bits set.

DYNAMIC_FIELD_KEY - Number value where each value encodes for a distinct meaning including a fieldtype of the next variable field; generally followed by an optional **DYNAMIC_FIELD_LENGTH** and a **DYNAMIC_FIELD_VALUE** field; when there is no **DYNAMIC_FIELD_LENGTH** field the length is contained in the lookup table

These values have been determined by reverse engineering, given the known values it is anticipated that there are unknown enumeration values and some known values have incorrect datatypes

Encoding: Each lookup has a LookupFieldTypeEnumeration defining what the possible values mean

DYNAMIC_FIELD_LENGTH - Number value that indicates the length of the following **DYNAMIC_FIELD_VALUE** field

DYNAMIC_FIELD_VALUE - Variable field whose type and length is dynamic

Encoding: The type definition of the field is defined by an earlier LookupFieldTypeEnumeration field. The length is defined by the preceding length field or the length determined by the lookup value.

TIME - Time

For more information see: <https://en.wikipedia.org/wiki/Time>

DURATION - Duration

DATE - Date

16 bits unsigned number

Encoding: The date, in days since 1 January 1970.

For more information see: https://en.wikipedia.org/wiki/Calendar_date



PGN - Parameter Group Number

Encoding: A 24 bit number referring to a PGN

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ISO_NAME - ISO NAME field

64 bits unsigned number

Encoding: A 64 bit field containing the ISO name, e.g. all fields produced by PGN 60928. Use the definition of PGN 60928 to explain the subfields.

STRING_FIX - A fixed length string containing single byte codepoints.

It is unclear what character sets are allowed/supported. Possibly UTF-8 but it could also be that only ASCII values are supported.

Encoding: The length of the string is determined by the PGN field definition. Trailing bytes have been observed as '@', ' ', 0x0 or 0xff.

STRING_LZ - A varying length string containing single byte codepoints encoded with a length byte and terminating zero.

This type of string is found only in SonicHub and Fusion brand specific PGNs. SonicHub is no longer produced. The use of the Fusion specific PGNs is no longer very prevalent now that there are generic PGNs for audio/media devices.

Encoding: The length of the string is determined by a starting length byte. It also contains a terminating zero byte. The length byte includes neither the zero byte or itself. The character encoding is UTF-8.

STRING_LAU - A varying length string containing double or single byte codepoints encoded with a length byte and terminating zero.

Encoding: The length of the string is determined by a starting length byte. This count includes the length and type bytes, so any empty string contains count 2. The 2nd byte contains 0 for UNICODE or 1 for ASCII.

BINARY - Binary field

Encoding: Unspecified content consisting of any number of bits.

RESERVED - Reserved field

NMEA reserved for future expansion and/or to align next data on byte boundary

Encoding: All reserved bits shall be 1

SPARE - Spare field

This is like a reserved field but originates from other sources where unused fields shall be 0, like the AIS ITU-1371 standard.

Encoding: All spare bits shall be 0

MMSI - MMSI

32 bits unsigned number

Encoding: The MMSI is encoded as a 32 bit number, but is always printed as a 9 digit number and should be considered as a string. The first three or four digits are special, see the USCG link for a detailed explanation.

For more information see: <https://navcen.uscg.gov/maritime-mobile-service-identity>

VARIABLE - Variable

Encoding: The definition of the field is that of the reference PGN and reference field, this is totally variable.

FIELD_INDEX - Field Index

8 bits unsigned number

Encoding: Index of the specified field in the PGN referenced.

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LIGHTING_COMMAND (0 - 7)

Value	Description
0	Idle
1	Detect Devices
2	Reboot
3	Factory Reset
4	Powering Up

INDUSTRY_CODE (0 - 7)

Value	Description
0	Global
1	Highway
2	Agriculture
3	Construction
4	Marine Industry
5	Industrial

MANUFACTURER_CODE (0 - 2047)

Value	Description
69	ARKS Enterprises, Inc.
78	FW Murphy/Enovation Controls
80	Twin Disc
85	Kohler Power Systems
88	Hemisphere GPS Inc
116	BEP Marine
135	Airmar
137	Maretron
140	Lowrance
144	Mercury Marine
147	Nautibus Electronic GmbH
148	Blue Water Data
154	Westerbeke
157	ISSPRO Inc
161	Offshore Systems (UK) Ltd.
163	Evinrude/BRP
165	CPAC Systems AB
168	Xantrex Technology Inc.
169	Marlin Technologies, Inc.
172	Yanmar Marine
174	Volvo Penta
175	Honda Marine
176	Carling Technologies Inc. (Moritz Aerospace)
185	Beede Instruments



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193	Nobletec
198	Mystic Valley Communications
199	Actia
200	Honda Marine
201	Disenos Y Tecnologia
211	Digital Switching Systems
215	Xintex/Atena
224	EMMI NETWORK S.L.
225	Honda Marine
228	ZF
229	Garmin
233	Yacht Monitoring Solutions
235	Sailormade Marine Telemetry/Tetra Technology LTD
243	Eride
250	Honda Marine
257	Honda Motor Company LTD
272	Groco
273	Actisense
274	Amphenol LTW Technology
275	Navico
283	Hamilton Jet
285	Sea Recovery
286	Coelmo SRL Italy
295	BEP Marine
304	Empir Bus
305	NovAtel
306	Sleipner Motor AS
307	MBW Technologies
311	Fischer Panda
315	ICOM
328	Qwerty
329	Dief
341	Boeing Automationstechnologie GmbH & Co. KG
345	Korean Maritime University
351	Thrane and Thrane
355	Mastervolt
356	Fischer Panda Generators
358	Victron Energy
370	Rolls Royce Marine
373	Electronic Design
374	Northern Lights
378	Glendinning



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384	Rose Point Navigation Systems
385	Johnson Outdoors Marine Electronics Inc Geonav
394	Capi 2
396	Beyond Measure
400	Livorsi Marine
404	ComNav
409	Chetco
419	Fusion Electronics
421	Standard Horizon
422	True Heading AB
426	Egersund Marine Electronics AS
427	em-trak Marine Electronics
431	Tohatsu Co, JP
437	Digital Yacht
438	Comar Systems Limited
440	Cummins
443	VDO (aka Continental-Corporation)
451	Parker Hannifin aka Village Marine Tech
459	Alltek Marine Electronics Corp
460	SAN GIORGIO S.E.I.N
466	Veethree Electronics & Marine
467	Humminbird Marine Electronics
470	SI-TEX Marine Electronics
471	Sea Cross Marine AB
475	GME aka Standard Communications Pty LTD
476	Humminbird Marine Electronics
478	Ocean Sat BV
481	Chetco Digital Instruments
493	Watcheye
499	Lcj Capteurs
502	Attwood Marine
503	Naviop S.R.L.
504	Vesper Marine Ltd
510	Marinesoft Co. LTD
513	Simarine
517	NoLand Engineering
518	Transas USA
529	National Instruments Korea
530	National Marine Electronics Association
532	Onwa Marine
540	Webasto
571	Marinecraft (South Korea)



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578	Advanseae
579	KVH
580	San Jose Technology
583	Yacht Control
586	Suzuki Motor Corporation
591	US Coast Guard
595	Ship Module aka Customware
600	Aquatic AV
605	Aventics GmbH
606	Intellian
612	SamwonIT
614	Arlt Technologies
637	Bavaria Yachts
641	Diverse Yacht Services
644	Wema U.S.A dba KUS
645	Garmin
658	Shenzhen Jiuzhou Himunication
688	Rockford Corp
699	Harman International
704	JL Audio
708	Lars Thrane
715	Autonnic
717	Yacht Devices
734	REAP Systems
735	Au Electronics Group
739	LxNav
741	Littelfuse, Inc (formerly Carling Technologies)
743	DaeMyung
744	Woosung
748	ISOTTA IFRA srl
773	Clarion US
776	HMI Systems
777	Ocean Signal
778	Seekeeper
781	Poly Planar
785	Fischer Panda DE
795	Broyda Industries
796	Canadian Automotive
797	Tides Marine
798	Lumishore
799	Still Water Designs and Audio
802	BJ Technologies (Beneteau)



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811	Blue Water Desalination
815	FLIR
824	Undheim Systems
826	Lewmar Inc
838	TeamSurv
844	Fell Marine
847	Oceanvolt
862	Prospec
868	Data Panel Corp
890	L3 Technologies
894	Rhodan Marine Systems
896	Nexfour Solutions
905	ASA Electronics
909	Marines Co (South Korea)
911	Nautic-on
917	Sentinel
929	JL Marine ystems
930	Ecotronix
944	Zontisa Marine
951	EXOR International
962	Timbolier Industries
963	TJC Micro
968	Cox Powertrain
969	Blue Seas
981	Kobelt Manufacturing Co. Ltd
992	Blue Ocean IOT
997	Xenta Systems
1004	Ultraflex SpA
1008	Lintest SmartBoat
1011	Soundmax
1020	Team Italia Marine (Onyx Marine Automation s.r.l)
1021	Entratech
1022	ITC Inc.
1029	The Marine Guardian LLC
1047	Sonic Corporation
1051	ProNav
1053	Vetus Maxwell INC.
1056	Lithium Pros
1059	Boatrax
1062	Marol Co ltd
1065	CALYPSO Instruments
1066	Spot Zero Water



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1070	Quick-teck Electronics Ltd
1075	Uniden America
1083	Nauticoncept
1084	Shadow-Caster LED lighting LLC
1085	Wet Sounds, LLC
1088	E-T-A Circuit Breakers
1092	Scheiber
1100	Smart Yachts International Limited
1109	Dockmate
1114	Bobs Machine
1118	L3Harris ASV
1119	Balmar LLC
1120	Elettromedia spa
1127	Electromaax
1140	Across Oceans Systems Ltd.
1145	Kiwi Yachting
1150	BSB Artificial Intelligence GmbH
1151	Orca Technologoes AS
1154	TBS Electronics BV
1158	Technoton Electroics
1160	MG Energy Systems B.V.
1169	Sea Macine Robotics Inc.
1171	Vista Manufacturing
1183	Zipwake
1186	Sailmon BV
1192	Airmoniq Pro Kft
1194	Sierra Marine
1200	Xinuo Information Technology (Xiamen)
1218	Septentrio
1233	NKE Marine Elecronics
1238	SuperTrack Aps
1239	Honda Electronics Co., LTD
1245	Raritan Engineering Company, Inc
1249	Integrated Power Solutions AG
1260	Interactive Technologies, Inc.
1283	LTG-Tech
1299	Energy Solutions (UK) LTD.
1300	WATT Fuel Cell Corp
1302	Pro Mainer
1305	Dragonfly Energy
1306	Koden Electronics Co., Ltd
1311	Humphree AB



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1317	Global Marine Management GmbH (GMM)
1320	Triskel Marine Ltd
1330	Warwick Control Technologies
1331	Dolphin Charger
1337	Barnacle Systems Inc
1348	Radian IoT, Inc.
1353	Ocean LED Marine Ltd
1359	BluNav
1361	OVA (Nantong Saiyang Electronics Co., Ltd)
1368	RAD Propulsion
1369	Electric Yacht
1372	Elco Motor Yachts
1384	Tecnoseal Foundry S.r.l
1385	Pro Charging Systems, LLC
1389	EVEX Co., LTD
1398	Gobius Sensor Technology AB
1403	Arco Marine
1408	Lenco Marine Inc.
1413	Naocontrol S.L.
1417	Revatek
1438	Aeolionics
1439	PredictWind Ltd
1440	Egis Mobile Electric
1445	Starboard Yacht Group
1446	Roswell Marine
1451	ePropulsion (Guangdong ePropulsion Technology Ltd.)
1452	Micro-Air LLC
1453	Vital Battery
1458	Ride Controller LLC
1460	Tocaro Blue
1461	Vanquish Yachts
1471	FT Technologies
1478	Alps Alpine Co., Ltd.
1481	E-Force Marine
1482	CMC Marine
1483	Nanjing Sandemarine Information Technology Co., Ltd.
1850	Teleflex Marine (SeaStar Solutions)
1851	Raymarine
1852	Navionics
1853	Japan Radio Co
1854	Northstar Technologies
1855	Furuno



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1857	Simrad
1858	Litton
1859	Kvasar AB
1860	MMP
1861	Vector Cantech
1862	Yamaha Marine
1863	Faria Instruments

AIS_MESSAGE_ID (0 - 63)

Value	Description
1	Scheduled Class A position report
2	Assigned scheduled Class A position report
3	Interrogated Class A position report
4	Base station report
5	Static and voyage related data
6	Binary addressed message
7	Binary acknowledgement
8	Binary broadcast message
9	Standard SAR aircraft position report
10	UTC/date inquiry
11	UTC/date response
12	Safety related addressed message
13	Safety related acknowledgement
14	Safety related broadcast message
15	Interrogation
16	Assignment mode command
17	DGNSS broadcast binary message
18	Standard Class B position report
19	Extended Class B position report
20	Data link management message
21	ATON report
22	Channel management
23	Group assignment command
24	Static data report
25	Single slot binary message
26	Multiple slot binary message
27	Position report for long range applications

SHIP_TYPE (0 - 255)

Value	Description
0	Unavailable
20	Wing In Ground



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22	Wing In Ground (hazard cat Y)
23	Wing In Ground (hazard cat Z)
24	Wing In Ground (hazard cat OS)
29	Wing In Ground (no additional information)
30	Fishing
31	Towing
32	Towing exceeds 200m or wider than 25m
33	Engaged in dredging or underwater operations
34	Engaged in diving operations
35	Engaged in military operations
36	Sailing
37	Pleasure
40	High speed craft
41	High speed craft (hazard cat X)
42	High speed craft (hazard cat Y)
43	High speed craft (hazard cat Z)
44	High speed craft (hazard cat OS)
49	High speed craft (no additional information)
50	Pilot vessel
51	SAR
52	Tug
53	Port tender
54	Anti-pollution
55	Law enforcement
56	Spare
57	Spare #2
58	Medical
59	Ships and aircraft of States not parties to an armed conflict
60	Passenger ship
61	Passenger ship (hazard cat X)
62	Passenger ship (hazard cat Y)
63	Passenger ship (hazard cat Z)
64	Passenger ship (hazard cat OS)
69	Passenger ship (no additional information)
70	Cargo ship
71	Cargo ship (hazard cat X)
72	Cargo ship (hazard cat Y)
73	Cargo ship (hazard cat Z)
74	Cargo ship (hazard cat OS)
79	Cargo ship (no additional information)
80	Tanker
81	Tanker (hazard cat X)



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83	Tanker (hazard cat Z)
84	Tanker (hazard cat OS)
89	Tanker (no additional information)
90	Other
91	Other (hazard cat X)
92	Other (hazard cat Y)
93	Other (hazard cat Z)
94	Other (hazard cat OS)
99	Other (no additional information)

DEVICE_CLASS (0 - 127)

Value	Description
0	Reserved for 2000 Use
10	System tools
20	Safety systems
25	Internetwork device
30	Electrical Distribution
35	Electrical Generation
40	Steering and Control surfaces
50	Propulsion
60	Navigation
70	Communication
75	Sensor Communication Interface
80	Instrumentation/general systems
85	External Environment
90	Internal Environment
100	Deck + cargo + fishing equipment systems
110	Human Interface
120	Display
125	Entertainment

REPEAT_INDICATOR (0 - 3)

Value	Description
0	Initial
1	First retransmission
2	Second retransmission
3	Final retransmission

TX_RX_MODE (0 - 15)

Value	Description
0	Tx A/Tx B, Rx A/Rx B
1	Tx A, Rx A/Rx B



2	Tx B, Rx A/Rx B
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STATION_TYPE (0 - 15)

Value	Description
0	All types of mobile station
2	All types of Class B mobile station
3	SAR airborne mobile station
4	AtoN station
5	Class B CS shipborne mobile station
6	Inland waterways
7	Regional use 7
8	Regional use 8
9	Regional use 9

REPORTING_INTERVAL (0 - 15)

Value	Description
0	As given by the autonomous mode
1	10 min
2	6 min
3	3 min
4	1 min
5	30 sec
6	15 sec
7	10 sec
8	5 sec
9	2 sec (not applicable to Class B CS)
10	Next shorter reporting interval
11	Next longer reporting interval

AIS_TRANSCEIVER (0 - 31)

Value	Description
0	Channel A VDL reception
1	Channel B VDL reception
2	Channel A VDL transmission
3	Channel B VDL transmission
4	Own information not broadcast
5	Reserved

AIS_ASSIGNED_MODE (0 - 1)

Value	Description
0	Autonomous and continuous
1	Assigned mode

ATON_TYPE (0 - 31)



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Value	Description
0	Default: Type of AtoN not specified
1	Reference point
2	RACON
3	Fixed structure off-shore
4	Reserved for future use
5	Fixed light: without sectors
6	Fixed light: with sectors
7	Fixed leading light front
8	Fixed leading light rear
9	Fixed beacon: cardinal N
10	Fixed beacon: cardinal E
11	Fixed beacon: cardinal S
12	Fixed beacon: cardinal W
13	Fixed beacon: port hand
14	Fixed beacon: starboard hand
15	Fixed beacon: preferred channel port hand
16	Fixed beacon: preferred channel starboard hand
17	Fixed beacon: isolated danger
18	Fixed beacon: safe water
19	Fixed beacon: special mark
20	Floating AtoN: cardinal N
21	Floating AtoN: cardinal E
22	Floating AtoN: cardinal S
23	Floating AtoN: cardinal W
24	Floating AtoN: port hand mark
25	Floating AtoN: starboard hand mark
26	Floating AtoN: preferred channel port hand
27	Floating AtoN: preferred channel starboard hand
28	Floating AtoN: isolated danger
29	Floating AtoN: safe water
30	Floating AtoN: special mark
31	Floating AtoN: light vessel/LANBY/rigs

AIS_SPECIAL_MANEUVER (0 - 3)

Value	Description
0	Not available
1	Not engaged in special maneuver
2	Engaged in special maneuver
3	Reserved

POSITION_FIX_DEVICE (0 - 15)

Value	Description
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Notes

0	Default: undefined
1	GPS
2	GLONASS
3	Combined GPS/GLONASS
4	Loran-C
5	Chayka
6	Integrated navigation system
7	Surveyed
8	Galileo

GNS (0 - 15)

Value	Description
0	GPS
1	GLONASS
2	GPS+GLONASS
3	GPS+SBAS/WAAS
4	GPS+SBAS/WAAS+GLONASS
5	Chayka
6	integrated
7	surveyed
8	Galileo

ENGINE_INSTANCE (0 - 255)

Value	Description
0	Single Engine or Dual Engine Port
1	Dual Engine Starboard

GEAR_STATUS (0 - 3)

Value	Description
0	Forward
1	Neutral
2	Reverse

DIRECTION (0 - 15)

Value	Description
0	Forward
1	Reverse

POSITION_ACCURACY (0 - 1)

Value	Description
0	Low
1	High

RAIM_FLAG (0 - 1)



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Notes

Value	Description
0	not in use
1	in use

TIME_STAMP (0 - 63)

Value	Description
60	Not available
61	Manual input mode
62	Dead reckoning mode
63	Positioning system is inoperative

GNS_METHOD (0 - 15)

Value	Description
0	no GNSS
1	GNSS fix
2	DGNSS fix
3	Precise GNSS
4	RTK Fixed Integer
5	RTK float
6	Estimated (DR) mode
7	Manual Input
8	Simulate mode

GNS_INTEGRITY (0 - 3)

Value	Description
0	No integrity checking
1	Safe
2	Caution
3	Unsafe

SYSTEM_TIME (0 - 15)

Value	Description
0	GPS
1	GLONASS
2	Radio Station
3	Local Cesium clock
4	Local Rubidium clock
5	Local Crystal clock

MAGNETIC_VARIATION (0 - 15)

Value	Description
0	Manual
1	Automatic Chart



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2	Automatic Table
3	Automatic Calculation
4	WMM 2000
5	WMM 2005
6	WMM 2010
7	WMM 2015
8	WMM 2020

RESIDUAL_MODE (0 - 15)

Value	Description
0	Autonomous
1	Differential enhanced
2	Estimated
3	Simulator
4	Manual

WIND_REFERENCE (0 - 7)

Value	Description
0	True (ground referenced to North)
1	Magnetic (ground referenced to Magnetic North)
2	Apparent
3	True (boat referenced)
4	True (water referenced)

WATER_REFERENCE (0 - 255)

Value	Description
0	Paddle wheel
1	Pitot tube
2	Doppler
3	Correlation (ultra sound)
4	Electro Magnetic

YES_NO (0 - 3)

Value	Description
0	No
1	Yes

OK_WARNING (0 - 3)

Value	Description
0	OK
1	Warning

OFF_ON (0 - 3)



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Value	Description
0	Off
1	On

OFF_ON_CONTROL (0 - 3)

Value	Description
0	Off
1	On
2	Reserved
3	Take no action (no change)

DIRECTION_REFERENCE (0 - 3)

Value	Description
0	True
1	Magnetic
2	Error

DIRECTION_RUDDER (0 - 7)

Value	Description
0	No Order
1	Move to starboard
2	Move to port

NAV_STATUS (0 - 15)

Value	Description
0	Under way using engine
1	At anchor
2	Not under command
3	Restricted maneuverability
4	Constrained by her draught
5	Moored
6	Aground
7	Engaged in Fishing
8	Under way sailing
9	Hazardous material - High Speed
10	Hazardous material - Wing in Ground
11	Power-driven vessel towing astern
12	Power-driven vessel pushing ahead or towing alongside
14	AIS-SART

POWER_FACTOR (0 - 3)

Value	Description
0	Leading



1	Lagging
2	Error

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TEMPERATURE_SOURCE (0 - 255)

Value	Description
0	Sea Temperature
1	Outside Temperature
2	Inside Temperature
3	Engine Room Temperature
4	Main Cabin Temperature
5	Live Well Temperature
6	Bait Well Temperature
7	Refrigeration Temperature
8	Heating System Temperature
9	Dew Point Temperature
10	Apparent Wind Chill Temperature
11	Theoretical Wind Chill Temperature
12	Heat Index Temperature
13	Freezer Temperature
14	Exhaust Gas Temperature
15	Shaft Seal Temperature

HUMIDITY_SOURCE (0 - 255)

Value	Description
0	Inside
1	Outside

PRESSURE_SOURCE (0 - 255)

Value	Description
0	Atmospheric
1	Water
2	Steam
3	Compressed Air
4	Hydraulic
5	Filter
6	AltimeterSetting
7	Oil
8	Fuel

DSC_FORMAT (0 - 255)

Value	Description
102	Geographical area
112	Distress



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114	Common interest
116	All ships
120	Individual stations
121	Non-calling purpose
123	Individual station automatic

DSC_CATEGORY (0 - 255)

Value	Description
100	Routine
108	Safety
110	Urgency
112	Distress

DSC_NATURE (0 - 255)

Value	Description
100	Fire
101	Flooding
102	Collision
103	Grounding
104	Listing
105	Sinking
106	Disabled and adrift
107	Undesignated
108	Abandoning ship
109	Piracy
110	Man overboard
112	EPIRB emission

DSC_FIRST_TELECOMMAND (0 - 255)

Value	Description
100	F3E/G3E All modes TP
101	F3E/G3E duplex TP
103	Polling
104	Unable to comply
105	End of call
106	Data
109	J3E TP
110	Distress acknowledgement
112	Distress relay
113	F1B/J2B TTY-FEC
115	F1B/J2B TTY-ARQ
118	Test
121	Ship position or location registration updating



126	No information
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DSC_SECOND_TELECOMMAND (0 - 255)

Value	Description
100	No reason given
101	Congestion at MSC
102	Busy
103	Queue indication
104	Station barred
105	No operator available
106	Operator temporarily unavailable
107	Equipment disabled
108	Unable to use proposed channel
109	Unable to use proposed mode
110	Ships and aircraft of States not parties to an armed conflict
111	Medical transports
112	Pay phone/public call office
113	Fax/data
126	No information

DSC_EXPANSION_DATA (0 - 255)

Value	Description
100	Enhanced position
101	Source and datum of position
102	SOG
103	COG
104	Additional station identification
105	Enhanced geographic area
106	Number of persons on board

SEATALK_MESSAGE_ID (0 - 255)

Value	Description
240	Seatalk 1 Encoded
140	Display
108	Pilot Configuration

SEATALK_COMMAND (0 - 255)

Value	Description
129	Seatalk1
22	Hull Type
38	Auto Turn
12	Settings

SEATALK1_COMMAND (0 - 255)



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Value	Description
0	Depth Below Transducer
1	Equipment ID
5	Engine RPM and PITCH
16	Apparent Wind Angle
17	Apparent Wind Speed
32	Speed through water
33	Trip Mileage
34	Total Mileage
35	Water temperature (ST50)
36	Display units for Mileage & Speed
37	Total & Trip Log
38	Speed through water (with average)
39	Water temperature
48	Set lamp Intensity
54	Cancel MOB (Man Over Board) condition
56	Codelock data
80	LAT position
81	LON position
82	Speed over Ground
83	Course over Ground (COG)
84	GMT-time
85	TRACK keystroke on GPS unit
86	Date
87	Sat Info
88	LAT/LON (raw unfiltered)
89	Set Count Down Timer
97	Issued by E-80 multifunction display at initialization
101	Select Fathom display units for depth display
102	Wind alarm
104	Alarm acknowledgment keystroke
108	Second equipment-ID datagram
110	MOB (Man Over Board)
112	Keystroke on Raymarine A25006 ST60 Maxiview Remote Control
128	Set Lamp Intensity
129	Sent by course computer during setup
130	Target waypoint name
131	Sent by course computer
132	Compass heading Autopilot course and Rudder position
133	Navigation to waypoint information
134	Keystroke
135	Set Response level
136	Autopilot Parameter



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137	Compass heading sent by ST40 compass instrument
144	Device Identification
145	Set Rudder gain
146	Set Autopilot Parameter
147	Enter AP-Setup
149	Replaces command 84 while autopilot is in value setting mode
153	Compass variation
154	Version String
156	Compass heading and Rudder position
158	Waypoint definition
161	Destination Waypoint Info
162	Arrival Info
164	Broadcast query/response to identify devices
165	GPS and DGPS Info
167	Unknown meaning
168	Alarm ON/OFF for Guard
171	Alarm ON/OFF for Guard

SEATALK1_ATT (0 - 255)

Value	Description
0	Depth Below Transducer

SEATALK_ALARM_STATUS (0 - 255)

Value	Description
0	Alarm condition not met
1	Alarm condition met and not silenced
2	Alarm condition met and silenced

SEATALK_ALARM_ID (0 - 255)

Value	Description
0	No Alarm
1	Shallow Depth
2	Deep Depth
3	Shallow Anchor
4	Deep Anchor
5	Off Course
6	AWA High
7	AWA Low
8	AWS High
9	AWS Low
10	TWA High
11	TWA Low
12	TWS High



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13	TWS Low
14	WP Arrival
15	Boat Speed High
16	Boat Speed Low
17	Sea Temperature High
18	Sea Temperature Low
19	Pilot Watch
20	Pilot Off Course
21	Pilot Wind Shift
22	Pilot Low Battery
23	Pilot Last Minute Of Watch
24	Pilot No NMEA Data
25	Pilot Large XTE
26	Pilot NMEA DataError
27	Pilot CU Disconnected
28	Pilot Auto Release
29	Pilot Way Point Advance
30	Pilot Drive Stopped
31	Pilot Type Unspecified
32	Pilot Calibration Required
33	Pilot Last Heading
34	Pilot No Pilot
35	Pilot Route Complete
36	Pilot Variable Text
37	GPS Failure
38	MOB
39	Seataalk1 Anchor
40	Pilot Swapped Motor Power
41	Pilot Standby Too Fast To Fish
42	Pilot No GPS Fix
43	Pilot No GPS COG
44	Pilot Start Up
45	Pilot Too Slow
46	Pilot No Compass
47	Pilot Rate Gyro Fault
48	Pilot Current Limit
49	Pilot Way Point Advance Port
50	Pilot Way Point Advance Stbd
51	Pilot No Wind Data
52	Pilot No Speed Data
53	Pilot Seataalk Fail1
54	Pilot Seataalk Fail2
55	Pilot Warning Too Fast To Fish



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56	Pilot Auto Dockside Fail
57	Pilot Turn Too Fast
58	Pilot No Nav Data
59	Pilot Lost Waypoint Data
60	Pilot EEPROM Corrupt
61	Pilot Rudder Feedback Fail
62	Pilot Autolearn Fail1
63	Pilot Autolearn Fail2
64	Pilot Autolearn Fail3
65	Pilot Autolearn Fail4
66	Pilot Autolearn Fail5
67	Pilot Autolearn Fail6
68	Pilot Warning Cal Required
69	Pilot Warning OffCourse
70	Pilot Warning XTE
71	Pilot Warning Wind Shift
72	Pilot Warning Drive Short
73	Pilot Warning Clutch Short
74	Pilot Warning Solenoid Short
75	Pilot Joystick Fault
76	Pilot No Joystick Data
80	Pilot Invalid Command
81	AIS TX Malfunction
82	AIS Antenna VSWR fault
83	AIS Rx channel 1 malfunction
84	AIS Rx channel 2 malfunction
85	AIS No sensor position in use
86	AIS No valid SOG information
87	AIS No valid COG information
88	AIS 12V alarm
89	AIS 6V alarm
90	AIS Noise threshold exceeded channel A
91	AIS Noise threshold exceeded channel B
92	AIS Transmitter PA fault
93	AIS 3V3 alarm
94	AIS Rx channel 70 malfunction
95	AIS Heading lost/invalid
96	AIS internal GPS lost
97	AIS No sensor position
98	AIS Lock failure
99	AIS Internal GGA timeout
100	AIS Protocol stack restart
101	Pilot No IPS communications



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102	Pilot Power-On or Sleep-Switch Reset While Engaged
103	Pilot Unexpected Reset While Engaged
104	AIS Dangerous Target
105	AIS Lost Target
106	AIS Safety Related Message (used to silence)
107	AIS Connection Lost
108	No Fix

SEATALK_ALARM_GROUP (0 - 255)

Value	Description
0	Instrument
1	Autopilot
2	Radar
3	Chart Plotter
4	AIS

SEATALK_PILOT_MODE (0 - 255)

Value	Description
64	Standby
66	Auto
70	Wind
74	Track

SEATALK_PILOT_HULL_TYPE (0 - 255)

Value	Description
0	Sail
1	Sail (slow turn)
2	Sail Catamaran
3	Power (slow turn)
4	Power (fast turn)
8	Power

SEATALK_SHARED (0 - 255)

Value	Description
1	Shared
2	Not Shared

ENTERTAINMENT_ZONE (0 - 255)

Value	Description
0	All zones
1	Zone 1
2	Zone 2
3	Zone 3



4	Zone 4
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ENTERTAINMENT_SOURCE (0 - 255)

Value	Description
0	Vessel alarm
1	AM
2	FM
3	Weather
4	DAB
5	Aux
6	USB
7	CD
8	MP3
9	Apple iOS
10	Android
11	Bluetooth
12	Sirius XM
13	Pandora
14	Spotify
15	Slacker
16	Songza
17	Apple Radio
18	Last FM
19	Ethernet
20	Video MP4
21	Video DVD
22	Video BluRay
23	HDMI
24	Video

ENTERTAINMENT_PLAY_STATUS (0 - 65535)

Value	Description
0	Play
1	Pause
2	Stop
3	FF 1x
4	FF 2x
5	FF 3x
6	FF 4x
7	RW 1x
8	RW 2x
9	RW 3x
10	RW 4x



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11	Skip ahead
12	Skip back
13	Jog ahead
14	Jog back
15	Seek up
16	Seek down
17	Scan up
18	Scan down
19	Tune up
20	Tune down
21	Slow motion .75x
22	Slow motion .5x
23	Slow motion .25x
24	Slow motion .125x

ENTERTAINMENT_REPEAT_STATUS (0 - 15)

Value	Description
0	Off
1	One
2	All

ENTERTAINMENT_SHUFFLE_STATUS (0 - 15)

Value	Description
0	Off
1	Play queue
2	All

ENTERTAINMENT_LIKE_STATUS (0 - 255)

Value	Description
0	None
1	Thumbs up
2	Thumbs down

ENTERTAINMENT_TYPE (0 - 255)

Value	Description
0	File
1	Playlist Name
2	Genre Name
3	Album Name
4	Artist Name
5	Track Name
6	Station Name
7	Station Number



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8	Favourite Number
9	Play Queue
10	Content Info

ENTERTAINMENT_GROUP (0 - 255)

Value	Description
0	File
1	Playlist Name
2	Genre Name
3	Album Name
4	Artist Name
5	Track Name
6	Station Name
7	Station Number
8	Favourite Number
9	Play Queue
10	Content Info

ENTERTAINMENT_CHANNEL (0 - 255)

Value	Description
0	All channels
1	Stereo full range
2	Stereo front
3	Stereo back
4	Stereo surround
5	Center
6	Subwoofer
7	Front left
8	Front right
9	Back left
10	Back right
11	Surround left
12	Surround right

ENTERTAINMENT_EQ (0 - 255)

Value	Description
0	Flat
1	Rock
2	Hall
3	Jazz
4	Pop
5	Live
6	Classic



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7	Vocal
8	Arena
9	Cinema
10	Custom

ENTERTAINMENT_FILTER (0 - 255)

Value	Description
0	Full range
1	High pass
2	Low pass
3	Band pass
4	Notch filter

ALERT_TYPE (0 - 15)

Value	Description
1	Emergency Alarm
2	Alarm
5	Warning
8	Caution

ALERT_CATEGORY (0 - 15)

Value	Description
0	Navigational
1	Technical

ALERT_TRIGGER_CONDITION (0 - 15)

Value	Description
0	Manual
1	Auto
2	Test
3	Disabled

ALERT_THRESHOLD_STATUS (0 - 15)

Value	Description
0	Normal
1	Threshold Exceeded
2	Extreme Threshold Exceeded
3	Low Threshold Exceeded
4	Acknowledged
5	Awaiting Acknowledge

ALERT_STATE (0 - 255)

Value	Description
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Notes

0	Disabled
1	Normal
2	Active
3	Silenced
4	Acknowledged
5	Awaiting Acknowledge

ALERT_LANGUAGE_ID (0 - 255)

Value	Description
0	English (US)
1	English (UK)
2	Arabic
3	Chinese (simplified)
4	Croatian
5	Danish
6	Dutch
7	Finnish
8	French
9	German
10	Greek
11	Italian
12	Japanese
13	Korean
14	Norwegian
15	Polish
16	Portuguese
17	Russian
18	Spanish
19	Swedish

ALERT_RESPONSE_COMMAND (0 - 3)

Value	Description
0	Acknowledge
1	Temporary Silence
2	Test Command off
3	Test Command on

CONVERTER_STATE (0 - 255)

Value	Description
0	Off
1	Low Power Mode
2	Fault
3	Bulk



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4	Absorption
5	Float
6	Storage
7	Equalize
8	Pass thru
9	Inverting
10	Assisting

THRUSTER_DIRECTION_CONTROL (0 - 15)

Value	Description
0	Off
1	Ready
2	To Port
3	To Starboard

THRUSTER_RETRACT_CONTROL (0 - 3)

Value	Description
0	Off
1	Extend
2	Retract

THRUSTER_MOTOR_TYPE (0 - 15)

Value	Description
0	12VDC
1	24VDC
2	48VDC
3	24VAC
4	Hydraulic

BOOT_STATE (0 - 15)

Value	Description
0	in Startup Monitor
1	running Bootloader
2	running Application

ACCESS_LEVEL (0 - 7)

Value	Description
0	Locked
1	unlocked level 1
2	unlocked level 2

TRANSMISSION_INTERVAL (0 - 15)

Value	Description
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Notes

0	Acknowledge
1	Transmit Interval/Priority not supported
2	Transmit Interval too low
3	Access denied
4	Not supported

PARAMETER_FIELD (0 - 15)

Value	Description
0	Acknowledge
1	Invalid parameter field
2	Temporary error
3	Parameter out of range
4	Access denied
5	Not supported
6	Read or Write not supported

PGN_LIST_FUNCTION (0 - 255)

Value	Description
0	Transmit PGN list
1	Receive PGN list

FUSION_COMMAND (0 - 255)

Value	Description
1	Play
2	Pause
4	Next
6	Prev

FUSION_SIRIUS_COMMAND (0 - 255)

Value	Description
1	Next
2	Prev

FUSION_MUTE_COMMAND (0 - 255)

Value	Description
1	Mute On
2	Mute Off

SEATALK_KEYSTROKE (0 - 255)

Value	Description
1	Auto
2	Standby
3	Wind



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5	-1
6	-10
7	+1
8	+10
33	-1 and -10
34	+1 and +10
35	Track

SEATALK_DEVICE_ID (0 - 255)

Value	Description
3	S100
5	Course Computer

SEATALK_NETWORK_GROUP (0 - 255)

Value	Description
0	None
1	Helm 1
2	Helm 2
3	Cockpit
4	Flybridge
5	Mast
6	Group 1
7	Group 2
8	Group 3
9	Group 4
10	Group 5

SEATALK_DISPLAY_COLOR (0 - 255)

Value	Description
0	Day 1
2	Day 2
3	Red/Black
4	Inverse

AIRMAR_CALIBRATE_FUNCTION (0 - 255)

Value	Description
0	Normal/cancel calibration
1	Enter calibration mode
2	Reset calibration to 0
3	Verify
4	Reset compass to defaults
5	Reset damping to defaults

AIRMAR_CALIBRATE_STATUS (0 - 255)



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Value	Description
0	Queried
1	Passed
2	Failed - timeout
3	Failed - tilt error
4	Failed - other
5	In progress

AIRMAR_TEMPERATURE_INSTANCE (0 - 3)

Value	Description
0	Device Sensor
1	Onboard Water Sensor
2	Optional Water Sensor

AIRMAR_FILTER (0 - 15)

Value	Description
0	No filter
1	Basic IIR filter

CONTROLLER_STATE (0 - 3)

Value	Description
0	Error Active
1	Error Passive
2	Bus Off

EQUIPMENT_STATUS (0 - 3)

Value	Description
0	Operational
1	Fault

MOB_STATUS (0 - 7)

Value	Description
0	MOB Emitter Activated
1	Manual on-board MOB Button Activation
2	Test mode

LOW_BATTERY (0 - 7)

Value	Description
0	Good
1	Low

TURN_MODE (0 - 7)

Value	Description
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Notes

0	Rudder limit controlled
1	Turn rate controlled
2	Radius controlled

ACCEPTABILITY (0 - 3)

Value	Description
0	Bad level
1	Bad frequency
2	Being qualified
3	Good

LINE (0 - 3)

Value	Description
0	Line 1
1	Line 2
2	Line 3

WAVEFORM (0 - 7)

Value	Description
0	Sine wave
1	Modified sine wave

TANK_TYPE (0 - 15)

Value	Description
0	Fuel
1	Water
2	Gray water
3	Live well
4	Oil
5	Black water

DC_SOURCE (0 - 255)

Value	Description
0	Battery
1	Alternator
2	Convertor
3	Solar cell
4	Wind generator

CHARGER_STATE (0 - 15)

Value	Description
0	Not charging
1	Bulk



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2	Absorption
3	Overcharge
4	Equalise
5	Float
6	No float
7	Constant VI
8	Disabled
9	Fault

CHARGING_ALGORITHM (0 - 15)

Value	Description
0	Trickle
1	Constant voltage / Constant current
2	2 stage (no float)
3	3 stage

CHARGER_MODE (0 - 15)

Value	Description
0	Standalone
1	Primary
2	Secondary
3	Echo

INVERTER_STATE (0 - 15)

Value	Description
0	Invert
1	AC passthru
2	Load sense
3	Fault
4	Disabled

BATTERY_TYPE (0 - 15)

Value	Description
0	Flooded
1	Gel
2	AGM

BATTERY_VOLTAGE (0 - 15)

Value	Description
0	6V
1	12V
2	24V
3	32V



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4	36V
5	42V
6	48V

BATTERY_CHEMISTRY (0 - 15)

Value	Description
0	Pb (Lead)
1	Li
2	NiCd
3	ZnO
4	NiMH

GOOD_WARNING_ERROR (0 - 3)

Value	Description
0	Good
1	Warning
2	Error

TRACKING (0 - 3)

Value	Description
0	Cancelled
1	Acquiring
2	Tracking
3	Lost

TARGET_ACQUISITION (0 - 1)

Value	Description
0	Manual
1	Automatic

WINDLASS_DIRECTION (0 - 3)

Value	Description
0	Off
1	Down
2	Up

SPEED_TYPE (0 - 3)

Value	Description
0	Single speed
1	Dual speed
2	Proportional speed

WINDLASS_MOTION (0 - 3)



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Notes

Value	Description
0	Windlass stopped
1	Deployment occurring
2	Retrieval occurring

RODE_TYPE (0 - 3)

Value	Description
0	Chain presently detected
1	Rope presently detected

DOCKING_STATUS (0 - 3)

Value	Description
0	Not docked
1	Fully docked

AIS_TYPE (0 - 1)

Value	Description
0	SOTDMA
1	CS

AIS_BAND (0 - 1)

Value	Description
0	Top 525 kHz of marine band
1	Entire marine band

AIS_MODE (0 - 1)

Value	Description
0	Autonomous
1	Assigned

AIS_COMMUNICATION_STATE (0 - 1)

Value	Description
0	SOTDMA
1	ITDMA

AVAILABLE (0 - 1)

Value	Description
0	Available
1	Not available

BEARING_MODE (0 - 3)

Value	Description
0	Great Circle



1	Rhumbline
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MARK_TYPE (0 - 15)

Value	Description
0	Collision
1	Turning point
2	Reference
3	Wheelover
4	Waypoint

GNSS_MODE (0 - 7)

Value	Description
0	1D
1	2D
2	3D
3	Auto

RANGE_RESIDUAL_MODE (0 - 3)

Value	Description
0	Range residuals were used to calculate data
1	Range residuals were calculated after the position

DGNSS_MODE (0 - 7)

Value	Description
0	None
1	SBAS if available
3	SBAS

SATELLITE_STATUS (0 - 15)

Value	Description
0	Not tracked
1	Tracked
2	Used
3	Not tracked+Diff
4	Tracked+Diff
5	Used+Diff

AIS_VERSION (0 - 3)

Value	Description
0	ITU-R M.1371-1
1	ITU-R M.1371-3
2	ITU-R M.1371-5
3	ITU-R M.1371 future edition



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TIDE (0 - 3)

Value	Description
0	Falling
1	Rising

WATERMAKER_STATE (0 - 63)

Value	Description
0	Stopped
1	Starting
2	Running
3	Stopping
4	Flushing
5	Rinsing
6	Initiating
7	Manual

ENTERTAINMENT_ID_TYPE (0 - 255)

Value	Description
0	Group
1	File
2	Encrypted group
3	Encrypted file

ENTERTAINMENT_DEFAULT_SETTINGS (0 - 3)

Value	Description
0	Save current settings as user default
1	Load user default
2	Load manufacturer default

ENTERTAINMENT_REGIONS (0 - 15)

Value	Description
0	USA
1	Europe
2	Asia
3	Middle East
4	Latin America
5	Australia
6	Russia
7	Japan

VIDEO_PROTOCOLS (0 - 15)

Value	Description
0	PAL



1	NTSC
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ENTERTAINMENT_VOLUME_CONTROL (0 - 3)

Value	Description
0	Up
1	Down

BLUETOOTH_STATUS (0 - 255)

Value	Description
0	Connected
1	Not connected
2	Not paired

BLUETOOTH_SOURCE_STATUS (0 - 15)

Value	Description
0	Reserved
1	Connected
2	Connecting
3	Not connected

SONICHUB_COMMAND (0 - 255)

Value	Description
1	Init #2
4	AM Radio
5	Zone Info
6	Source
8	Source List
9	Control
12	FM Radio
13	Playlist
14	Track
15	Artist
16	Album
19	Menu Item
20	Zones
23	Max Volume
24	Volume
25	Init #1
48	Position
50	Init #3

SIMNET_AP_MODE (0 - 255)

Value	Description
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2	Heading
3	Wind
10	Nav
11	No Drift

SIMNET_DEVICE_MODEL (0 - 255)

Value	Description
0	AC
1	Other device
100	NAC

SIMNET_DEVICE_REPORT (0 - 255)

Value	Description
2	Status
3	Send Status
10	Mode
11	Send Mode
23	Sailing Processor Status

SIMNET_AP_STATUS (0 - 255)

Value	Description
2	Manual
16	Automatic

SIMNET_COMMAND (0 - 255)

Value	Description
50	Text

SIMNET_EVENT_COMMAND (0 - 255)

Value	Description
1	Alarm
2	AP command
255	Autopilot

SIMNET_NIGHT_MODE (0 - 255)

Value	Description
2	Day
4	Night

SIMNET_NIGHT_MODE_COLOR (0 - 255)

Value	Description
0	Red
1	Green



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2	Blue
3	White
4	Magenta

SIMNET_DISPLAY_GROUP (0 - 255)

Value	Description
1	Default
2	Group 1
3	Group 2
4	Group 3
5	Group 4
6	Group 5
7	Group 6

SIMNET_HOUR_DISPLAY (0 - 255)

Value	Description
0	24 hour
1	12 hour

SIMNET_TIME_FORMAT (0 - 255)

Value	Description
1	MM/dd/yyyy
2	dd/MM/yyyy

SIMNET_BACKLIGHT_LEVEL (0 - 255)

Value	Description
0	10% (Min)
1	Day mode
4	Night mode
11	20%
22	30%
33	40%
44	50%
55	60%
66	70%
77	80%
88	90%
99	100% (Max)

SIMNET_AP_EVENTS (0 - 255)

Value	Description
2	Follow/Non Follow
6	Standby



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9	Heading mode
10	Nav mode
12	No Drift mode
13	Non Follow Up mode
14	Follow Up mode
15	Wind mode
17	Tack
18	Square (Turn)
19	C-Turn
20	U-Turn
21	Spiral (Turn)
22	Zig Zag (Turn)
23	Lazy-S (Turn)
24	Depth (Turn)
26	Change course
61	Timer sync
112	Ping port end
113	Ping starboard end

SIMNET_DIRECTION (0 - 255)

Value	Description
2	Port
3	Starboard
4	Left rudder (port)
5	Right rudder (starboard)

SIMNET_AP_COMMAND_TYPE (0 - 255)

Value	Description
2	Follow Up
10	AP Command

SIMNET_ALARM (0 - 255)

Value	Description
57	Low boat speed
56	No autopilot computer
58	Nav data missing

FUSION_MESSAGE_ID (0 - 65535)

Value	Description
1	Request Status
2	Set Source
3	Media Command
5	Tuner Command



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6	Marine Tuner Command
7	Set Marine Tuner Squelch
8	Set Marine Tuner Scan Mode
9	Menu Action
10	Request Menu Count
11	Request Menu Item
12	Request Menu Lock ID
13	Set Aux Gain
15	Set Settings
16	DAB Updtate Command
17	Set Mute
18	Set Balance
19	Set Low Pass Filer
20	Set Sublevel
22	Set Equalizer
23	Set Volume Limit
24	Set Zone Volume
25	Set All Volumes
27	Set Line Level Control
28	Power
29	Set Device Name
30	Send Sirius Command
31	Set Sirius Parental
33	Send Factory Reset Command
34	Set Zone Name
35	Send Dvd Command
36	Dvd Press Ir Key
39	Send Select Sirius Team
40	Send Select Sirius Artist
41	Send Sirius Sport Alert User Action
45	Send Sirius Artist Song User Action
50	Send Multiroom Command
51	Get Multiroom Device Record
52	Scan Multirooom Devices
53	Send File Transfer
54	Set Loud
56	Fapi Set Source Multiroom Enabled
57	Request Head Unit Dsp Settings
64	Send Transfer Status
65	Fapi Get Server Info
69	Fapi Set Source Enabled
70	Fapi Set Source Name
73	Send External Amp Gain



74	Send Internal Amp Gain
75	Send Mono

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FUSION_PLAY_STATUS (0 - 65535)

Value	Description
0	Invalid
1	Playing
2	Paused
3	Stopped
4	Skip Forward
5	Skip Rewind

FUSION_SOURCE_TYPE (0 - 255)

Value	Description
0	AM
1	FM
2	Aux
3	Sirius
4	Ipod
5	USB
6	DVD
7	VHF
8	Invalid
9	MTP
10	Bluetooth
11	ARC
12	Android
13	Pandora
14	DAB
15	AirPlay
16	UPNP
17	Unknown

FUSION_SIRIUS_COM_STATE (0 - 255)

Value	Description
255	Unknown
1	Off
2	Initialising
3	On

FUSION_SIRIUS_ALERT (0 - 255)

Value	Description
255	Unknown



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1	None
2	Antenna
3	NoSignal
4	Subscription Update

FUSION_SIRIUS_TUNING_MODE (0 - 255)

Value	Description
1	Normal
2	Category
3	Preset

FUSION_STATUS_MESSAGE_ID (0 - 65535)

Value	Description
0	Unknown
32769	API Version
32770	Source
32771	Source Count
32772	Track Info
32773	Track Title
32774	Track Artist
32775	Track Album
32776	Cover Art
32777	Track Progress
32778	Tuner Align
32779	Tuner
32780	Marine Tuner
32781	Marine Squelch
32782	Marine Scan Mode
32783	Menu Action
32784	Menu Count
32785	Menu Item
32786	Menu Lock ID
32787	Aux Gain
32788	Setting
32789	Settings
32790	Update Firmware Result
32791	Mute
32792	Balance
32793	Low Pass Filter
32794	Sublevels
32795	Tone
32796	Volume Limits
32797	Volume



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32798	Capabilities
32799	Line Level Control
32800	Power
32801	Unit Name
32802	Sirius
32803	SiriusXM Preset Event
32804	SiriusXM Channel
32805	SiriusXM Title
32806	SiriusXM Artist
32807	SiriusXM Genre
32808	SiriusXM Category
32809	SiriusXm Signal
32810	SiriusXM Parental Request
32811	SiriusXM Diagnostics
32812	SiriusXM Presets
32813	Zone Name

SONICHUB_CONTROL (0 - 255)

Value	Description
0	Set
128	Ack

SONICHUB_SOURCE (0 - 255)

Value	Description
0	AM
1	FM
2	iPod
3	USB
4	AUX
5	AUX 2
6	Mic

ISO_CONTROL (0 - 255)

Value	Description
0	ACK
1	NAK
2	Access Denied
3	Address Busy

ISO_COMMAND (0 - 255)

Value	Description
0	ACK
16	RTS



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17	CTS
19	EOM
32	BAM
255	Abort

GROUP_FUNCTION (0 - 255)

Value	Description
0	Request
1	Command
2	Acknowledge
3	Read Fields
4	Read Fields Reply
5	Write Fields
6	Write Fields Reply

AIRMAR_COMMAND (0 - 255)

Value	Description
32	Attitude Offsets
33	Calibrate Compass
34	True Wind Options
35	Simulate Mode
40	Calibrate Depth
41	Calibrate Speed
42	Calibrate Temperature
43	Speed Filter
44	Temperature Filter
46	NMEA 2000 options

AIRMAR_DEPTH_QUALITY_FACTOR (0 - 15)

Value	Description
0	Depth unlocked
1	Quality 10%
2	Quality 20%
3	Quality 30%
4	Quality 40%
5	Quality 50%
6	Quality 60%
7	Quality 70%
8	Quality 80%
9	Quality 90%
10	Quality 100%

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Value	Description
0	Acknowledge
1	PGN not supported
2	PGN not available
3	Access denied
4	Not supported
5	Tag not supported
6	Read or Write not supported

AIRMAR_TRANSMISSION_INTERVAL (0 - 3)

Value	Description
0	Measure interval
1	Requested by user

MOB_POSITION_SOURCE (0 - 7)

Value	Description
0	Position estimated by the vessel
1	Position reported by MOB emitter

STEERING_MODE (0 - 7)

Value	Description
0	Main Steering
1	Non-Follow-Up Device
2	Follow-Up Device
3	Heading Control Standalone
4	Heading Control
5	Track Control

FUSION_RADIO_SOURCE (0 - 255)

Value	Description
0	AM
1	FM

FUSION_SETTING (0 - 4294967295)

Value	Description
0	Alpha Search Threshold
1	iPod Subtitles
2	Zone 2 Linked
3	Zone 2 Enabled
4	Zone 3 Enabled
5	Zone 4 Enabled
6	Telemute
7	Tuner Region



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8	Marine Zone
9	USB repeat
10	USB shuffle
11	iPod Album Artwork
12	iPod repeat
13	iPod shuffle
14	AM Preset 0
15	AM Preset 1
16	AM Preset 2
17	AM Preset 3
18	AM Preset 4
19	AM Preset 5
20	AM Preset 6
21	AM Preset 7
22	AM Preset 8
23	AM Preset 9
24	AM Preset 10
25	AM Preset 11
26	AM Preset 12
27	AM Preset 13
28	AM Preset 14
29	FM Preset 0
30	FM Preset 1
31	FM Preset 2
32	FM Preset 3
33	FM Preset 4
34	FM Preset 5
35	FM Preset 6
36	FM Preset 7
37	FM Preset 8
38	FM Preset 9
39	FM Preset 10
40	FM Preset 11
41	FM Preset 12
42	FM Preset 13
43	FM Preset 14
44	VHF Preset 0
45	VHF Preset 1
46	VHF Preset 2
47	VHF Preset 3
48	VHF Preset 4
49	VHF Preset 5
50	VHF Preset 6



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52	VHF Preset 8
53	VHF Preset 9
54	VHF Preset 10
55	VHF Preset 11
56	VHF Preset 12
57	VHF Preset 13
58	VHF Preset 14
59	Clock Time
60	Clock Alarm
61	iPod Video Signal
62	iPod Monitor Aspect
63	Aux Name Index
64	AM Enabled
65	VHF Enabled
66	Language
67	Internal Amps On
68	MTP Repeat
69	MTP Shuffle
70	Id Accessory Source
71	NMEA Power
72	Low Power Mode
73	DVD region
74	Volume Zone Sync
75	Max Volume Start
76	BT Auto Connect
77	Null Setting

FUSION_REPEAT_STATUS (0 - 4294967295)

Value	Description
0	Off
1	One/track
2	All/album

AIRMAR_POST_CONTROL (0 - 1)

Value	Description
0	Report previous values
1	Generate new values

AIRMAR_POST_ID (0 - 255)

Value	Description
1	Format Code
2	Factory EEPROM



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3	User EEPROM
4	Water Temperature Sensor
5	Sonar Transceiver
6	Speed sensor
7	Internal temperature sensor
8	Battery voltage sensor

SONICHUB_TUNING (0 - 255)

Value	Description
1	Seeking up
2	Tuned
3	Seeking down

SONICHUB_PLAYLIST (0 - 255)

Value	Description
1	Report
4	Next song
6	Previous song

FUSION_POWER_STATE (0 - 255)

Value	Description
1	On
2	Off

PRIORITY (0 - 15)

Value	Description
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	Leave unchanged
9	Reset to default

DEVICE_TEMP_STATE (0 - 15)

Value	Description
0	Cold
1	Warm
2	Hot

BANDG_DECIMALS (0 - 255)



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Value	Description
0	0
1	1
2	2
3	3
4	4
254	Auto

GARMIN_COLOR_MODE (0 - 255)

Value	Description
0	Day
1	Night
13	Color

GARMIN_COLOR (0 - 255)

Value	Description
0	Day full color
1	Day high contrast
2	Night full color
3	Night red/black
4	Night green/black

GARMIN_BACKLIGHT_LEVEL (0 - 255)

Value	Description
0	0%
1	5%
2	10%
3	15%
4	20%
5	25%
6	30%
7	35%
8	40%
9	45%
10	50%
11	55%
12	60%
13	65%
14	70%
15	75%
16	80%
17	85%
18	90%



19	95%
20	100%

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SEATALK_PILOT_MODE_16 (0 - 65535)

Value	Description
0	Standby
64	Auto, compass commanded
256	Vane, Wind Mode
384	Track Mode
385	No Drift, COG referenced (In track, course changes)

STATION_HEALTH (0 - 15)

Value	Description
0	Not Working
1	Unmonitored
2	Healthy Operational
3	Healthy Test Mode
4	Test Mode

SERIAL_BIT_RATE (0 - 31)

Value	Description
0	25
1	50
2	100
3	200
4	300
5	600
6	1200
7	2400
8	4800
9	9600
10	19200
11	38400
12	57600

SERIAL_DETECTION_MODE (0 - 7)

Value	Description
0	Auto bit rate
1	Manual bit rate

DIFFERENTIAL_SOURCE (0 - 15)

Value	Description
0	Auto



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1	Loran
2	MSK Beacon
3	FM Subcarrier
4	AIS
5	Ground based radio
6	SBAS
7	Satellite

DIFFERENTIAL_MODE (0 - 15)

Value	Description
0	Manual
1	Auto Power
2	Auto Range

WP_POSITION_RESOLUTION (0 - 15)

Value	Description
0	more than 0.1 min
1	<0.01 .. 0.1] min
2	<0.001 .. 0.01] min
3	<0.0001 .. 0.001] min
4	<0 .. 0.0001] min

WP_IDENTIFICATION_METHOD (0 - 3)

Value	Description
0	Waypoints in WP list
1	Waypoints embedded in route

WP_ROUTE_STATUS (0 - 15)

Value	Description
0	Active
1	Inactive
2	Deleted

WP_NAVIGATION_METHOD (0 - 3)

Value	Description
0	Great Circle
1	Rhumb Line

INVERTER_MODE (0 - 15)

Value	Description
0	Standalone
1	Series Master
2	Series Slave



3	Parallel Master
4	Parallel Slave

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CERTIFICATION_LEVEL (0 - 255)

Value	Description
0	Level A
1	Level B

AGS_MODE (0 - 15)

Value	Description
0	Off
1	On
2	Automatic

AGS_OPERATING_STATE (0 - 15)

Value	Description
0	Quiet time
1	Auto on
2	Auto off
3	Manual On
4	Manual Off
5	Generator shutdown
6	External shutdown
7	Fault
8	Suspend
9	Not operating

AGS_GENERATING_STATE (0 - 15)

Value	Description
0	Preheating
1	Start delay
2	Cranking
3	Starter cooling
4	Warming up
5	Cooling down
6	Spinning up
7	Shutdown bypass
8	Stopping
9	Running
10	Stopped
11	Crank delay

AGS_ON_REASON (0 - 255)



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Value	Description
0	Not on
1	DC voltage low
2	Battery state of charge low
3	AC current high
4	Contact closed
5	Manual on
6	Exercise
7	Non Quiet time
8	External on via AGS
9	External on via generator
10	Unable to stop

AGS_OFF_REASON (0 - 255)

Value	Description
0	Not off
1	DC voltage high
2	Battery state of charge high
3	AC current low
4	Contact opened
5	Reached absorption
6	Reached float
7	Manual off
8	Max run time
9	Max auto cycle
10	Exercise done
11	Quiet time
12	External off via AGS
13	Safe mode
14	External off via generator
15	External shutdown
16	Auto off
17	Fault
18	Unable to start

TELEPHONE_MODE (0 - 255)

Value	Description
0	F3E/G3E simplex, telephone
1	F3E/G3E duplex, telephone
2	J3E, telephone
3	H3E, telephone
4	F1B/J2B FEC NBDP, telex/teleprinter
5	F1B/J2B ARQ NBDP, telex/teleprinter



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6	F1B/J2B receive only, teleprinter/DSC
7	F1B/J2B, teleprinter/DSC
8	A1A Morse, tape recorder
9	A1A Morse, Morse key/head set
10	F1C/F2C/F3C, FAX machine

POWER_MODE (0 - 1)

Value	Description
0	High
1	Low

BROADCAST_INDICATOR (0 - 15)

Value	Description
0	Broadcast geo area message
1	Addressed message

BANDWIDTH (0 - 1)

Value	Description
0	Default
1	12.5 kHz

FLOOD_STATE (0 - 7)

Value	Description
0	Flood
1	Slack
2	Ebb

AC_LINE (0 - 3)

Value	Description
0	Line 1
1	Line 2
2	Line 3

ZONE_SIZE (0 - 7)

Value	Description
0	1 nm
1	2 nm
2	3 nm
3	4 nm
4	5 nm
5	6 nm

MARETRON_PRODUCT_CODE (0 - 65535)



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Value	Description
434	SSC200
2686	SSC300

MARETRON_SOFTWARE_CODE (0 - 65535)

Value	Description
1	Version 1

MARETRON_COMMAND (0 - 255)

Value	Description
80	Deviation calibration

MARETRON_STATUS_DEVIATION (0 - 255)

Value	Description
1	Started
2	Completed successfully
3	Failed to complete
4	Turning too fast
5	Turning too slow
6	Invalid movement

AUTOMATIC_MANUAL (0 - 255)

Value	Description
0	Automatic
1	Manual

SBAS_SV (0 - 255)

Value	Description
0	120
1	121
2	122
3	123
4	124
5	125
6	126
7	127
8	128
9	129
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15	135
16	136
17	137
18	138

Indirect Lookup enumerations

DEVICE_FUNCTION (0 - 255)

Value in other field	Value	Description
10	130	Diagnostic
10	140	Bus Traffic Logger
20	110	Alarm Enunciator
20	130	Emergency Position Indicating Radio Beacon (EPIRB)
20	135	Man Overboard
20	140	Voyage Data Recorder
20	150	Camera
25	130	PC Gateway
25	131	NMEA 2000 to Analog Gateway
25	132	Analog to NMEA 2000 Gateway
25	133	NMEA 2000 to Serial Gateway
25	135	NMEA 0183 Gateway
25	136	NMEA Network Gateway
25	137	NMEA 2000 Wireless Gateway
25	140	Router
25	150	Bridge
25	160	Repeater
30	130	Binary Event Monitor
30	140	Load Controller
30	141	AC/DC Input
30	150	Function Controller
35	140	Engine
35	141	DC Generator/Alternator
35	142	Solar Panel (Solar Array)
35	143	Wind Generator (DC)
35	144	Fuel Cell
35	145	Network Power Supply
35	151	AC Generator
35	152	AC Bus
35	153	AC Mains (Utility/Shore)
35	154	AC Output
35	160	Power Converter - Battery Charger
35	161	Power Converter - Battery Charger+Inverter
35	162	Power Converter - Inverter
35	163	Power Converter - DC



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35	170	Battery
35	180	Engine Gateway
40	130	Follow-up Controller
40	140	Mode Controller
40	150	Autopilot
40	155	Rudder
40	160	Heading Sensors
40	170	Trim (Tabs)/Interceptors
40	180	Attitude (Pitch, Roll, Yaw) Control
50	130	Engineroom Monitoring
50	140	Engine
50	141	DC Generator/Alternator
50	150	Engine Controller
50	151	AC Generator
50	155	Motor
50	160	Engine Gateway
50	165	Transmission
50	170	Throttle/Shift Control
50	180	Actuator
50	190	Gauge Interface
50	200	Gauge Large
50	210	Gauge Small
60	130	Bottom Depth
60	135	Bottom Depth/Speed
60	136	Bottom Depth/Speed/Temperature
60	140	Ownship Attitude
60	145	Ownship Position (GNSS)
60	150	Ownship Position (Loran C)
60	155	Speed
60	160	Turn Rate Indicator
60	170	Integrated Navigation
60	175	Integrated Navigation System
60	190	Navigation Management
60	195	Automatic Identification System (AIS)
60	200	Radar
60	201	Infrared Imaging
60	205	ECDIS
60	210	ECS
60	220	Direction Finder
60	230	Voyage Status
70	130	EPIRB
70	140	AIS
70	150	DSC



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70	160	Data Receiver/Transceiver
70	170	Satellite
70	180	Radio-telephone (MF/HF)
70	190	Radiotelephone
75	130	Temperature
75	140	Pressure
75	150	Fluid Level
75	160	Flow
75	170	Humidity
80	130	Time/Date Systems
80	140	VDR
80	150	Integrated Instrumentation
80	160	General Purpose Displays
80	170	General Sensor Box
80	180	Weather Instruments
80	190	Transducer/General
80	200	NMEA 0183 Converter
85	130	Atmospheric
85	160	Aquatic
90	130	HVAC
100	130	Scale (Catch)
110	130	Button Interface
110	135	Switch Interface
110	140	Analog Interface
120	130	Display
120	140	Alarm Enunciator
125	130	Multimedia Player
125	140	Multimedia Controller

Bitfield Lookup enumerations

STATION_STATUS (0 - 2³-1)

Bit	Description
0	Station in use
1	Low SNR
2	Cycle Error
3	Blink

ENGINE_STATUS_1 (0 - 2⁴-1)

Bit	Description
0	Check Engine
1	Over Temperature
2	Low Oil Pressure
3	Low Oil Level



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4	Low Fuel Pressure
5	Low System Voltage
6	Low Coolant Level
7	Water Flow
8	Water In Fuel
9	Charge Indicator
10	Preheat Indicator
11	High Boost Pressure
12	Rev Limit Exceeded
13	EGR System
14	Throttle Position Sensor
15	Emergency Stop

ENGINE_STATUS_2 (0 - 2^15-1)

Bit	Description
0	Warning Level 1
1	Warning Level 2
2	Power Reduction
3	Maintenance Needed
4	Engine Comm Error
5	Sub or Secondary Throttle
6	Neutral Start Protect
7	Engine Shutting Down

ENTERTAINMENT_PLAY_STATUS_BITFIELD (0 - 2^31-1)

Bit	Description
0	Play
1	Pause
2	Stop
3	FF 1x
4	FF 2x
5	FF 3x
6	FF 4x
7	RW 1x
8	RW 2x
9	RW 3x
10	RW 4x
11	Skip ahead
12	Skip back
13	Jog ahead
14	Jog back
15	Seek up
16	Seek down



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17	Scan up
18	Scan down
19	Tune up
20	Tune down
21	Slow motion .75x
22	Slow motion .5x
23	Slow motion .25x
24	Slow motion .125x
25	Source renaming

ENTERTAINMENT_GROUP_BITFIELD (0 - 2¹⁵-1)

Bit	Description
0	File
1	Playlist Name
2	Genre Name
3	Album Name
4	Artist Name
5	Track Name
6	Station Name
7	Station Number
8	Favourite Number
9	Play Queue
10	Content Info

THRUSTER_CONTROL_EVENTS (0 - 2⁷-1)

Bit	Description
0	Another device controlling thruster
1	Boat speed too fast to safely use thruster

THRUSTER_MOTOR_EVENTS (0 - 2⁷-1)

Bit	Description
0	Motor over temperature cutout
1	Motor over current cutout
2	Low oil level warning
3	Oil over temperature warning
4	Controller under voltage cutout
5	Manufacturer defined

WINDLASS_CONTROL (0 - 2³-1)

Bit	Description
0	Another device controlling windlass

WINDLASS_OPERATION (0 - 2⁵-1)



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Bit	Description
0	System error
1	Sensor error
2	No windlass motion detected
3	Retrieval docking distance reached
4	End of rode reached

WINDLASS_MONITORING (0 - 2⁷-1)

Bit	Description
0	Controller under voltage cut-out
1	Controller over current cut-out
2	Controller over temperature cut-out
3	Manufacturer defined

SIMNET_AP_MODE_BITFIELD (0 - 2¹⁵-1)

Bit	Description
3	Standby
4	Heading
6	Nav
8	No Drift
10	Wind

SIMNET_ALERT_BITFIELD (0 - 2⁶³-1)

Bit	Description
0	No GPS fix
2	No active autopilot control unit
4	No autopilot computer
6	AP clutch overload
8	AP clutch disengaged
10	Rudder controller fault
12	No rudder response
14	Rudder drive overload
16	High drive supply
18	Low drive supply
20	Memory fail
22	AP position data missing
24	AP speed data missing
26	AP depth data missing
28	AP heading data missing
30	AP nav data missing
32	AP rudder data missing
34	AP wind data missing
36	AP off course



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38	High drive temperature
40	Drive inhibit
42	Rudder limit
44	Drive computer missing
46	Drive ready missing
48	EVC com error
50	EVC override
52	Low CAN bus voltage
54	CAN bus supply overload
56	Wind sensor battery low

ENTERTAINMENT_REPEAT_BITFIELD (0 - 2¹-1)

Bit	Description
0	Song
1	Play queue

ENTERTAINMENT_SHUFFLE_BITFIELD (0 - 2¹-1)

Bit	Description
0	Play queue
1	All

WP_CHANGE (0 - 2⁷-1)

Bit	Description
0	Change in main data (Position, Name)
1	Change in supplementary parameters (or new added)
2	Changed number of WPs in Route/WP-List, and/or name changed/added
3	Route: Change supplementary parameters (or new added)
6	Other not specified changed

WP_CRITICAL_PARAMETERS (0 - 2⁷-1)

Bit	Description
0	Navigation Method
1	XTE Limit

DISABLED_SATELLITES (0 - 2³⁹-1)

Bit	Description
0	Disable SV #1
1	Disable SV #2
2	Disable SV #3
3	Disable SV #4
4	Disable SV #5
5	Disable SV #6
6	Disable SV #7

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7	Disable SV #8
8	Disable SV #9
9	Disable SV #10
10	Disable SV #11
11	Disable SV #12
12	Disable SV #13
13	Disable SV #14
14	Disable SV #15
15	Disable SV #16
16	Disable SV #17
17	Disable SV #18
18	Disable SV #19
19	Disable SV #20
20	Disable SV #21
21	Disable SV #22
22	Disable SV #23
23	Disable SV #24
24	Disable SV #25
25	Disable SV #26
26	Disable SV #27
27	Disable SV #28
28	Disable SV #29
29	Disable SV #30
30	Disable SV #31
31	Disable SV #32
32	Disable SV #33
33	Disable SV #34
34	Disable SV #35
35	Disable SV #36
36	Disable SV #37
37	Disable SV #38
38	Disable SV #39
39	Disable SV #40

Lookup with FieldType enumerations

SIMNET_KEY_VALUE (0 - 65535)

Value	Description	Unit	Type
0	Heading Offset	0.0001 rad	16 bits signed NUMBER
41	Timezone offset	60 s	16 bits signed DURATION
260	True wind high	0.01 m/s	16 bits unsigned NUMBER
264	Deep water	0.01 m	16 bits unsigned NUMBER
516	True wind low	0.01 m/s	16 bits unsigned NUMBER
517	Low boat speed	0.01 m/s	16 bits unsigned NUMBER



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520	Shallow water	0.01 m	16 bits unsigned NUMBER
768	Local field	0.0001 rad	16 bits signed NUMBER
1024	Field angle	0.0001 rad	16 bits signed NUMBER
1800	Anchor depth	0.01 m	16 bits unsigned NUMBER
4863	Backlight level		8 bits lookup SIMNET BACKLIGHT LEVEL
5160	Time format		8 bits lookup SIMNET TIME FORMAT
5161	Time hour display		8 bits lookup SIMNET HOUR DISPLAY
9983	Night mode		8 bits lookup SIMNET NIGHT MODE
11524	True wind shift	0.0001 rad	16 bits signed NUMBER
22296	AP low boat speed	0.01 m/s	16 bits unsigned NUMBER
32789	Alert bits		16 bits bitfield SIMNET ALERT BITFIELD
44079	Night mode color		8 bits lookup SIMNET NIGHT MODE COLOR
55087	Day mode invert		8 bits unsigned NUMBER

BANDG_KEY_VALUE (0 - 4095)

Value	Description	Unit	Type
0	Altitude		16 bits signed NUMBER
11	Rudder Angle	0.0001 rad	16 bits signed NUMBER
16	User 5	0.01	32 bits signed NUMBER
17	User 6	0.01	32 bits signed NUMBER
18	User 7	0.01	32 bits signed NUMBER
19	User 8	0.01	32 bits signed NUMBER
20	User 9	0.01	32 bits signed NUMBER
21	User 10	0.01	32 bits signed NUMBER
22	User 11	0.01	32 bits signed NUMBER
23	User 12	0.01	32 bits signed NUMBER
24	User 13	0.01	32 bits signed NUMBER
25	User 14	0.01	32 bits signed NUMBER
26	User 15	0.01	32 bits signed NUMBER
27	User 16	0.01	32 bits signed NUMBER
28	Outside Temperature	0.01 K	16 bits unsigned NUMBER
29	Outside Temperature	0.01 K	16 bits unsigned NUMBER
30	Water Temperature	0.01 K	16 bits unsigned NUMBER
31	Water Temperature	0.01 K	16 bits unsigned NUMBER
50	Tacking Performance	0.1 %	16 bits signed NUMBER
52	Attitude Roll	0.0001 rad	16 bits signed NUMBER
53	Optimum Wind Angle	0.0001 rad	16 bits signed NUMBER
56	User 1	0.01	32 bits signed NUMBER
57	User 2	0.01	32 bits signed NUMBER
58	User 3	0.01	32 bits signed NUMBER
59	User 4	0.01	32 bits signed NUMBER
60	Roll Rate	0.0001 rad	16 bits signed NUMBER
64	Forestay	0.001	32 bits unsigned NUMBER



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65	Water Speed	0.01 m/s	16 bits unsigned NUMBER
77	Wind Speed Apparent	0.01 m/s	16 bits unsigned NUMBER
79	Wind Speed Apparent	0.01 m/s	16 bits unsigned NUMBER
80	Average True Wind Direction	0.0001 rad	16 bits signed NUMBER
81	Wind Angle Apparent	0.0001 rad	16 bits signed NUMBER
83	Target TWA	0.0001 rad	16 bits signed NUMBER
85	Wind Speed True	0.01 m/s	16 bits unsigned NUMBER
86	Wind Speed True	0.01 m/s	16 bits unsigned NUMBER
89	Wind Angle True	0.0001 rad	16 bits signed NUMBER
100	Unknown		16 bits signed NUMBER
102	Keel Angle	0.0001 rad	16 bits signed NUMBER
103	Canard Angle	0.0001 rad	16 bits signed NUMBER
104	Keel Trim Tab Angle	0.0001 rad	16 bits signed NUMBER
105	Course	0.0001 rad	16 bits signed NUMBER
109	Wind Direction	0.0001 rad	16 bits signed NUMBER
111	Next Leg AWA	0.0001 rad	16 bits signed NUMBER
113	Next Leg AWS	0.01 m/s	16 bits unsigned NUMBER
117	Race Timer	0.001 s	32 bits signed DURATION
124	Polar Performance	0.1 %	16 bits signed NUMBER
125	Target Boat Speed	0.01 m/s	16 bits signed NUMBER
126	Polar Speed	0.01 m/s	16 bits unsigned NUMBER
127	VMG to Wind	0.01 m/s	16 bits unsigned NUMBER
129	DR Distance	0.01 m	32 bits unsigned NUMBER
130	Leeway Angle	0.0001 rad	16 bits signed NUMBER
131	Current Drift	0.01 m/s	16 bits unsigned NUMBER
132	Current Set	0.0001 rad	16 bits signed NUMBER
135	Barometric Pressure	100 Pa	16 bits unsigned NUMBER
152	Distance to Start Line	0.01 m	32 bits signed NUMBER
154	Heading on Opposite Tack	0.0001 rad	16 bits signed NUMBER
155	Attitude Pitch	0.0001 rad	16 bits signed NUMBER
156	Mast Angle	0.0001 rad	16 bits signed NUMBER
157	Wind Angle to Mast	0.0001 rad	16 bits signed NUMBER
158	Pitch Angle	0.0001 rad	16 bits signed NUMBER
163	Daggerboard Position		16 bits unsigned NUMBER
164	Boom Position		16 bits unsigned NUMBER
185	MOB DR Bearing	0.0001 rad	16 bits signed NUMBER
186	MOB DR Range	0.01 m	32 bits unsigned NUMBER
194	Depth	0.01 m	32 bits unsigned NUMBER
195	Depth	0.01 m	32 bits unsigned NUMBER
199	Aft Depth	0.01 m	32 bits unsigned NUMBER
205	Odometer	0.01 m	32 bits unsigned NUMBER
207	Trip Distance	0.01 m	32 bits unsigned NUMBER
211	DR Bearing	0.0001 rad	16 bits signed NUMBER



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233	Course Over Ground	0.0001 rad	16 bits signed NUMBER
235	Speed Over Ground	0.01 m/s	16 bits unsigned NUMBER
239	Remote 0	0.001	16 bits unsigned NUMBER
240	Remote 1	0.001	16 bits unsigned NUMBER
241	Remote 2	0.001	16 bits unsigned NUMBER
242	Remote 3	0.001	16 bits unsigned NUMBER
243	Remote 4	0.001	16 bits unsigned NUMBER
244	Remote 5	0.001	16 bits unsigned NUMBER
245	Remote 6	0.001	16 bits unsigned NUMBER
246	Remote 7	0.001	16 bits unsigned NUMBER
247	Remote 8	0.001	16 bits unsigned NUMBER
248	Remote 9	0.001	16 bits unsigned NUMBER
256	Layline Time	0.001 s	32 bits unsigned DURATION
258	Layline Distance	0.01 m	32 bits unsigned NUMBER
259	Layline Distance	0.01 m	32 bits unsigned NUMBER
260	Sailing Time to Waypoint	0.001 s	32 bits unsigned DURATION
261	Sailing Distance to Waypoint	0.01 m	32 bits unsigned NUMBER
262	Sailing ETA	0.001 s	32 bits unsigned DURATION
265	Trip Time	0.001 s	32 bits unsigned DURATION
270	Bow Latitude	1e-07 deg	32 bits signed NUMBER
271	Bow Longitude	1e-07 deg	32 bits signed NUMBER
272	Start Line Bearing	0.0001 rad	16 bits signed NUMBER
273	Start Line Bias	0.0001 rad	16 bits signed NUMBER
274	Distance to Start Line Port	0.01 m	32 bits unsigned NUMBER
275	Distance to Start Line Starboard	0.01 m	32 bits unsigned NUMBER
280	Bias Advantage in Boat Lengths	0.1	16 bits signed NUMBER
281	Distance to Start Line in Boat Lengths	0.1	16 bits signed NUMBER
282	Backstay	0.001	32 bits unsigned NUMBER
283	Boom Vang	0.001	32 bits unsigned NUMBER
284	Chain Length	0.01 m	32 bits unsigned NUMBER
285	VMG Performance	0.1 %	16 bits signed NUMBER
286	Inner Forestay Load	0.001	32 bits unsigned NUMBER
287	Inner Forestay Halyard Load	0.001	32 bits unsigned NUMBER
288	Jib Furl	0.01 m	32 bits unsigned NUMBER
289	Jib Halyard Load	0.001	32 bits unsigned NUMBER
290	Outhaul Load	0.001	32 bits unsigned NUMBER
291	Plow Angle	0.0001 rad	16 bits signed NUMBER
292	Cunningham	0.001	32 bits unsigned NUMBER
293	Jacuzzi Temperature	0.01 K	16 bits unsigned NUMBER
294	Pool Temperature	0.01 K	16 bits unsigned NUMBER
296	Keel Draught	0.01 m	16 bits signed NUMBER
297	Boom Angle	0.0001 rad	16 bits signed NUMBER
298	Code Zero Load	0.001	32 bits unsigned NUMBER



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301	Distance Behind Start Line	0.01 m	32 bits signed NUMBER
302	Distance Behind Start Line in Boat Lengths	0.1	16 bits signed NUMBER
305	Bias Advantage	0.01 m	32 bits unsigned NUMBER
306	Opposite Tack COG	0.0001 rad	16 bits signed NUMBER
307	Opposite Tack Target Heading	0.0001 rad	16 bits signed NUMBER
308	Mast Rake	0.0001 rad	16 bits signed NUMBER
309	Next Leg Bearing	0.0001 rad	16 bits signed NUMBER
310	Next Leg Target Speed	0.01 m/s	16 bits signed NUMBER
311	Ground Wind Direction	0.0001 rad	16 bits signed NUMBER
312	Ground Wind Speed	0.01 m/s	16 bits signed NUMBER
313	Mast Cant Angle	0.0001 rad	16 bits signed NUMBER
314	Rudder Toe In	0.0001 rad	16 bits signed NUMBER
315	Daggerboard Port		16 bits unsigned NUMBER
316	Daggerboard Starboard		16 bits unsigned NUMBER
317	User 17	0.01	32 bits signed NUMBER
318	User 18	0.01	32 bits signed NUMBER
319	User 19	0.01	32 bits signed NUMBER
320	User 20	0.01	32 bits signed NUMBER
321	User 21	0.01	32 bits signed NUMBER
322	User 22	0.01	32 bits signed NUMBER
323	User 23	0.01	32 bits signed NUMBER
324	User 24	0.01	32 bits signed NUMBER
325	User 25	0.01	32 bits signed NUMBER
326	User 26	0.01	32 bits signed NUMBER
327	User 27	0.01	32 bits signed NUMBER
328	User 28	0.01	32 bits signed NUMBER
329	User 29	0.01	32 bits signed NUMBER
330	User 30	0.01	32 bits signed NUMBER
331	User 31	0.01	32 bits signed NUMBER
332	User 32	0.01	32 bits signed NUMBER
336	Average True Wind Direction	0.0001 rad	16 bits signed NUMBER
337	Wind Phase	0.0001 rad	16 bits signed NUMBER
338	Wind Lift	0.0001 rad	16 bits signed NUMBER
380	Active Perf Mode		16 bits signed NUMBER
381	Gust Bear Away	0.0001 rad	16 bits signed NUMBER
382	TWS Bear Away	0.0001 rad	16 bits signed NUMBER
383	Heel Compensation	0.0001 rad	16 bits signed NUMBER
384	Pilot Net Course	0.0001 rad	16 bits signed NUMBER
385	Pilot Target Wind Angle	0.0001 rad	16 bits signed NUMBER
386	Pilot Weather Helm	0.0001 rad	16 bits signed NUMBER
387	Pilot Mean Heel	0.0001 rad	16 bits signed NUMBER

Notes



Excess-K offset

Some fields are not stored as a straight signed two's complement binary number, but as an binary number with an offset.

So in effect the value *is* signed, but it is stored as an unsigned number where the unsigned number n should be interpreted as $(n + K)$. As the offset in these definitions is negative, it results in a value that can be negative.

For more information see: https://en.wikipedia.org/wiki/Offset_binary.

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